# IDAES Grid Integration Code Notebook

# Contents


**1 grid_integration (root) 15**

1.1 __init__.py . . . 15

1.2 bidder.py . . . 15

1.2.1 AbstractBidder . . . 20

1.2.2 AbstractBidder.update_day_ahead_model . . . 20

1.2.3 AbstractBidder.update_real_time_model . . . 20

1.2.4 AbstractBidder.compute_day_ahead_bids . . . 21

1.2.5 AbstractBidder.compute_real_time_bids . . . 21

1.2.6 AbstractBidder.write_results . . . 21

1.2.7 AbstractBidder.formulate_DA_bidding_problem . . . 21

1.2.8 AbstractBidder.formulate_RT_bidding_problem . . . 21

1.2.9 AbstractBidder.record_bids . . . 22

1.2.10 AbstractBidder.generator . . . 22

1.2.11 AbstractBidder._check_inputs . . . 22

1.2.12 AbstractBidder._check_bidding_model_object . . . 22

1.2.13 AbstractBidder._check_n_scenario . . . 23

1.2.14 AbstractBidder._check_solver . . . 23

1.2.15 StochasticProgramBidder . . . 23

1.2.16 StochasticProgramBidder.__init__ . . . 24

1.2.17 StochasticProgramBidder._set_up_bidding_problem . . . 25

1.2.18 StochasticProgramBidder.formulate_DA_bidding_problem . . . 25

1.2.19 StochasticProgramBidder.formulate_RT_bidding_problem . . . 26

1.2.20 StochasticProgramBidder._save_power_outputs . . . 26

1.2.21 StochasticProgramBidder._add_bidding_params . . . 26

1.2.22 StochasticProgramBidder._add_bidding_vars . . . 27

1.2.23 StochasticProgramBidder._add_bidding_vars.relaxed_day_ahead_power_ub_rule 27

1.2.24 StochasticProgramBidder._add_bidding_objective . . . 27

1.2.25 StochasticProgramBidder._compute_bids . . . 28

1.2.26 StochasticProgramBidder.compute_day_ahead_bids . . . 28

1.2.27 StochasticProgramBidder.compute_real_time_bids . . . 28

1.2.28 StochasticProgramBidder._pass_realized_day_ahead_prices . . . 29

1.2.29 StochasticProgramBidder._pass_realized_day_ahead_dispatches . . . 29

1.2.30 StochasticProgramBidder.update_day_ahead_model . . . 30

1.2.31 StochasticProgramBidder.update_real_time_model . . . 30

1.2.32 StochasticProgramBidder._update_model . . . 30

1.2.33 StochasticProgramBidder.record_bids . . . 30

1.2.34 StochasticProgramBidder._pass_price_forecasts . . . 31

1.2.35 StochasticProgramBidder.write_results . . . 31
1.2.36 StochasticProgramBidder.generator . . . 31
1.2.37 StochasticProgramBidder.generator . . . 32
1.2.38 SelfScheduler . . . 32
1.2.39 SelfScheduler.__init__ . . . 33
1.2.40 SelfScheduler._add_DA_bidding_constraints . . . 33
1.2.41 SelfScheduler._add_DA_bidding_constraints.day_ahead_bidding_constraints_rule 33
1.2.42 SelfScheduler._add_RT_bidding_constraints . . . 33
1.2.43 SelfScheduler._add_RT_bidding_constraints.real_time_bidding_constraints_rule 34
1.2.44 SelfScheduler._assemble_bids . . . 34
1.2.45 SelfScheduler._record_bids . . . 34
1.2.46 Bidder . . . 35
1.2.47 Bidder.__init__ . . . 36
1.2.48 Bidder._add_DA_bidding_constraints . . . 36
1.2.49 Bidder._add_DA_bidding_constraints.day_ahead_bidding_constraints_rule 36
1.2.50 Bidder._add_RT_bidding_constraints . . . 37
1.2.51 Bidder._add_RT_bidding_constraints.real_time_bidding_constraints_rule 37
1.2.52 Bidder._assemble_bids . . . 37
1.2.53 Bidder._record_bids . . . 38
1.2.54 ParametrizedBidder . . . 38
1.2.55 ParametrizedBidder.__init__ . . . 39
1.2.56 ParametrizedBidder.generator . . . 39
1.2.57 ParametrizedBidder.generator . . . 40
1.2.58 ParametrizedBidder.formulate_DA_bidding_problem . . . 40
1.2.59 ParametrizedBidder.formulate_RT_bidding_problem . . . 40
1.2.60 ParametrizedBidder.compute_day_ahead_bids . . . 40
1.2.61 ParametrizedBidder.compute_real_time_bids . . . 40
1.2.62 ParametrizedBidder.update_day_ahead_model . . . 41
1.2.63 ParametrizedBidder.update_real_time_model . . . 41
1.2.64 ParametrizedBidder.record_bids . . . 41
1.2.65 ParametrizedBidder._record_bids . . . 41
1.2.66 ParametrizedBidder.write_results . . . 42
1.2.67 PEMParametrizedBidder . . . 42
1.2.68 PEMParametrizedBidder.__init__ . . . 43
1.2.69 PEMParametrizedBidder._check_power . . . 43
1.2.70 PEMParametrizedBidder.compute_day_ahead_bids . . . 44
1.2.71 PEMParametrizedBidder.compute_real_time_bids . . . 44
1.3 coordinator.py . . . 45
1.3.1 PrescientPluginModule . . . 48
1.3.2 PrescientPluginModule.__init__ . . . 48
1.3.3 DoubleLoopCoordinator . . . 48
1.3.4 DoubleLoopCoordinator.__init__ . . . 49
1.3.5 DoubleLoopCoordinator.register_plugins . . . 49
1.3.6 DoubleLoopCoordinator.get_configuration . . . 50
1.3.7 DoubleLoopCoordinator.prescient_plugin_module . . . 50
1.3.8 DoubleLoopCoordinator.initialize_customized_results . . . 51
1.3.9 DoubleLoopCoordinator.push_hourly_stats_to_forecaster . . . 51
1.3.10 DoubleLoopCoordinator.push_day_ahead_stats_to_forecaster . . . 51

1.3.11 DoubleLoopCoordinator._update_bids . . . 51
1.3.12 DoubleLoopCoordinator._update_bids._update_p_cost . . . 52
1.3.13 DoubleLoopCoordinator._update_bids._update_time_series_params . . . 52
1.3.14 DoubleLoopCoordinator._update_bids._update_non_time_series_params . 52
1.3.15 DoubleLoopCoordinator._pass_DA_bid_to_prescient . . . 53
1.3.16 DoubleLoopCoordinator.assemble_project_tracking_signal . . . 53
1.3.17 DoubleLoopCoordinator.project_tracking_trajectory . . . 53
1.3.18 DoubleLoopCoordinator._clone_tracking_model . . . 54
1.3.19 DoubleLoopCoordinator._update_static_params . . . 54
1.3.20 DoubleLoopCoordinator.update_static_params . . . 55
1.3.21 DoubleLoopCoordinator.bid_into_DAM . . . 55
1.3.22 DoubleLoopCoordinator.fetch_DA_prices . . . 55
1.3.23 DoubleLoopCoordinator.fetch_DA_dispatches . . . 56
1.3.24 DoubleLoopCoordinator._pass_RT_bid_to_prescient . . . 56
1.3.25 DoubleLoopCoordinator.bid_into_RTM . . . 56
1.3.26 DoubleLoopCoordinator.assemble_sced_tracking_market_signals . . . 56
1.3.27 DoubleLoopCoordinator._assemble_sced_tracking_market_signals . . . 57
1.3.28 DoubleLoopCoordinator.track_sced_signal . . . 57
1.3.29 DoubleLoopCoordinator.update_observed_dispatch . . . 58
1.3.30 DoubleLoopCoordinator.activate_pending_DA_data . . . 58
1.3.31 DoubleLoopCoordinator.write_plugin_results . . . 58
1.4 forecaster.py . . . 58
1.4.1 ForecastError . . . 61
1.4.2 AbstractPriceForecaster . . . 61
1.4.3 AbstractPriceForecaster.forecast_day_ahead_and_real_time_prices . . . 61
1.4.4 AbstractPriceForecaster.forecast_real_time_prices . . . 62
1.4.5 AbstractPriceForecaster.forecast_day_ahead_prices . . . 62
1.4.6 AbstractPrescientPriceForecaster . . . 62
1.4.7 AbstractPrescientPriceForecaster.fetch_hourly_stats_from_prescient . . . 62
1.4.8 AbstractPrescientPriceForecaster.fetch_day_ahead_stats_from_prescient . 62
1.4.9 PlaceHolderForecaster . . . 63
1.4.10 PlaceHolderForecaster.__init__ . . . 63
1.4.11 PlaceHolderForecaster.forecast_day_ahead_and_real_time_prices . . . 63
1.4.12 PlaceHolderForecaster.forecast_day_ahead_prices . . . 64
1.4.13 PlaceHolderForecaster.forecast_real_time_prices . . . 64
1.4.14 PlaceHolderForecaster._forecast . . . 64
1.4.15 PlaceHolderForecaster.fetch_hourly_stats_from_prescient . . . 64
1.4.16 PlaceHolderForecaster.fetch_day_ahead_stats_from_prescient . . . 65
1.4.17 Backcaster . . . 65
1.4.18 Backcaster.__init__ . . . 66
1.4.19 Backcaster._validate_input_historical_price . . . 66
1.4.20 Backcaster.max_historical_days . . . 66
1.4.21 Backcaster.max_historical_days . . . 67
1.4.22 Backcaster.historical_da_prices . . . 67
1.4.23 Backcaster.historical_da_prices . . . 67
1.4.24 Backcaster.historical_rt_prices . . . 67
1.4.25 Backcaster.historical_rt_prices . . . 68
1.4.26 Backcaster.forecast_day_ahead_and_real_time_prices . . . 68

1.4.27 Backcaster.forecast_real_time_prices . . . 68
1.4.28 Backcaster.forecast_day_ahead_prices . . . 69
1.4.29 Backcaster._forecast . . . 69
1.4.30 Backcaster.fetch_hourly_stats_from_prescient . . . 69
1.4.31 Backcaster.fetch_day_ahead_stats_from_prescient . . . 70
1.4.32 PerfectForecaster . . . 70
1.4.33 PerfectForecaster.__init__ . . . 71
1.4.34 PerfectForecaster.__getitem__ . . . 71
1.4.35 PerfectForecaster.fetch_hourly_stats_from_prescient . . . 71
1.4.36 PerfectForecaster.fetch_day_ahead_stats_from_prescient . . . 71
1.4.37 PerfectForecaster.forecast_day_ahead_and_real_time_prices . . . 72
1.4.38 PerfectForecaster.get_column_from_data . . . 72
1.4.39 PerfectForecaster.forecast_day_ahead_prices . . . 72
1.4.40 PerfectForecaster.forecast_real_time_prices . . . 73
1.4.41 PerfectForecaster.forecast_day_ahead_capacity_factor . . . 73
1.4.42 PerfectForecaster.forecast_real_time_capacity_factor . . . 73
1.5 model_data.py . . . 73
1.5.1 BaseValidator . . . 75
1.5.2 BaseValidator.__set_name__ . . . 76
1.5.3 BaseValidator.__set__ . . . 76
1.5.4 BaseValidator.__get__ . . . 76
1.5.5 BaseValidator._validate . . . 77
1.5.6 StrValidator . . . 77
1.5.7 StrValidator._validate . . . 77
1.5.8 RealValueValidator . . . 77
1.5.9 RealValueValidator.__init__ . . . 78
1.5.10 RealValueValidator._validate . . . 78
1.5.11 AtLeastPminValidator . . . 78
1.5.12 AtLeastPminValidator._validate . . . 79
1.5.13 GeneratorModelData . . . 79
1.5.14 GeneratorModelData.__init__ . . . 80
1.5.15 GeneratorModelData.__iter__ . . . 80
1.5.16 GeneratorModelData.__next__ . . . 81
1.5.17 ThermalGeneratorModelData . . . 81
1.5.18 ThermalGeneratorModelData.__init__ . . . 82
1.5.19 ThermalGeneratorModelData._check_empty_and_sort_cost_pairs . . . 82
1.5.20 ThermalGeneratorModelData._assemble_default_cost_bids . . . 83
1.5.21 ThermalGeneratorModelData._assemble_default_startup_cost_bids . . . 83
1.5.22 ThermalGeneratorModelData.generator_type . . . 83
1.5.23 ThermalGeneratorModelData.initial_status . . . 83
1.5.24 ThermalGeneratorModelData.initial_status . . . 84
1.5.25 ThermalGeneratorModelData.initial_p_output . . . 84
1.5.26 ThermalGeneratorModelData.initial_p_output . . . 84
1.5.27 RenewableGeneratorModelData . . . 85
1.5.28 RenewableGeneratorModelData.__init__ . . . 85
1.5.29 RenewableGeneratorModelData.generator_type . . . 85
1.6 tracker.py . . . 86
1.6.1 Tracker . . . 88

1.6.2 Tracker.__init__ . . . 89
1.6.3 Tracker._check_inputs . . . 90
1.6.4 Tracker._check_tracking_model_object . . . 90
1.6.5 Tracker._check_n_tracking_hour . . . 91
1.6.6 Tracker._check_solver . . . 91
1.6.7 Tracker.formulate_tracking_problem . . . 91
1.6.8 Tracker._add_tracking_vars . . . 92
1.6.9 Tracker._add_tracking_params . . . 92
1.6.10 Tracker._add_tracking_constraints . . . 92
1.6.11 Tracker._add_tracking_constraints.tracking_dispatch_constraint_rule . . . 93
1.6.12 Tracker._add_tracking_objective . . . 93
1.6.13 Tracker.update_model . . . 93
1.6.14 Tracker.track_market_dispatch . . . 93
1.6.15 Tracker._record_daily_stats . . . 94
1.6.16 Tracker._pass_market_dispatch . . . 94
1.6.17 Tracker.get_last_delivered_power . . . 95
1.6.18 Tracker._record_tracker_results . . . 95
1.6.19 Tracker.record_results . . . 95
1.6.20 Tracker.write_results . . . 96
1.7 utils.py . . . 96
1.7.1 convert_marginal_costs_to_actual_costs . . . 96

**2 examples** **98**
2.1 __init__.py . . . 98
2.2 thermal_generator.py . . . 98
2.2.1 ThermalGenerator . . . 101
2.2.2 ThermalGenerator.__init__ . . . 102
2.2.3 ThermalGenerator.assemble_model_data . . . 102
2.2.4 ThermalGenerator.model_data . . . 102
2.2.5 ThermalGenerator._add_UT_DT_constraints . . . 103
2.2.6 ThermalGenerator._add_UT_DT_constraints.pre_shut_down_trajectory_set_rule 103
2.2.7 ThermalGenerator._add_UT_DT_constraints.pre_start_up_trajectory_set_rule 103
2.2.8 ThermalGenerator._add_UT_DT_constraints.min_down_time_rule . . . 104
2.2.9 ThermalGenerator._add_UT_DT_constraints.min_up_time_rule . . . 104
2.2.10 ThermalGenerator.populate_model . . . 104
2.2.11 ThermalGenerator.populate_model.init_pre_pow_fun . . . 105
2.2.12 ThermalGenerator.populate_model.lhs_bnd_gen_pow_fun . . . 105
2.2.13 ThermalGenerator.populate_model.rhs_bnd_gen_pow_fun . . . 105
2.2.14 ThermalGenerator.populate_model.linear_power_fun . . . 106
2.2.15 ThermalGenerator.populate_model.seg_pow_bnd_fun . . . 106
2.2.16 ThermalGenerator.populate_model.start_up_shut_dw_fun . . . 106
2.2.17 ThermalGenerator.populate_model.start_up_or_shut_dw_fun . . . 106
2.2.18 ThermalGenerator.populate_model.ramp_up_fun . . . 107
2.2.19 ThermalGenerator.populate_model.ramp_shut_dw_fun . . . 107
2.2.20 ThermalGenerator.populate_model.ramp_dw_fun . . . 107
2.2.21 ThermalGenerator.populate_model.prod_cost_fun . . . 107
2.2.22 ThermalGenerator.populate_model.start_cost_fun . . . 108
2.2.23 ThermalGenerator.populate_model.tot_cost_fun . . . 108

2.2.24 ThermalGenerator._update_UT_DT . . . 108
2.2.25 ThermalGenerator._update_power . . . 108
2.2.26 ThermalGenerator.update_model . . . 109
2.2.27 ThermalGenerator.get_implemented_profile . . . 109
2.2.28 ThermalGenerator.get_last_delivered_power . . . 109
2.2.29 ThermalGenerator.record_results . . . 110
2.2.30 ThermalGenerator.write_results . . . 110
2.2.31 ThermalGenerator.power_output . . . 111
2.2.32 ThermalGenerator.total_cost . . . 111
2.3 thermal_generator_prescient_plugin.py . . . 111
2.4 utils.py . . . 111

**3 multiperiod** **113**
3.1 __init__.py . . . 113
3.2 multiperiod.py . . . 113
3.2.1 MultiPeriodModel . . . 115
3.2.2 MultiPeriodModel.__init__ . . . 117
3.2.3 MultiPeriodModel.build_multi_period_model . . . 117
3.2.4 MultiPeriodModel.advance_time . . . 118
3.2.5 MultiPeriodModel.pyomo_model . . . 118
3.2.6 MultiPeriodModel.current_time . . . 119
3.2.7 MultiPeriodModel.get_active_process_blocks . . . 119
3.2.8 MultiPeriodModel._create_linking_constraints . . . 119
3.2.9 MultiPeriodModel._create_periodic_constraints . . . 120
3.2.10 MultiPeriodModel._construct_flowsheet_instance . . . 120
3.2.11 MultiPeriodModel.build_stochastic_multi_period . . . 120
3.2.12 MultiPeriodModel.build_stochastic_multi_period._build_scenario_model . 121
3.2.13 MultiPeriodModel.plot_lmp_signal . . . 122
3.2.14 MultiPeriodModel.plot_lmp_and_schedule . . . 122

**4 multiperiod/tests** **124**
4.1 __init__.py . . . 124
4.2 test_multiperiod.py . . . 124
4.2.1 build_flowsheet . . . 126
4.2.2 fix_dof_and_initialize . . . 126
4.2.3 fix_dof_and_initialize_2 . . . 127
4.2.4 unfix_dof . . . 127
4.2.5 get_linking_variable_pairs . . . 127
4.2.6 build_multi_period_model . . . 127
4.2.7 build_multi_period_stochastic_model . . . 128
4.2.8 build_multi_day_model . . . 128
4.2.9 build_multi_day_stochastic_model . . . 128
4.2.10 build_multi_year_model . . . 129
4.2.11 build_multi_year_stochastic_model . . . 129
4.2.12 build_multi_day_year_model . . . 129
4.2.13 build_multi_day_year_stochastic_model . . . 130
4.2.14 build_multi_period_model_manual . . . 130
4.2.15 test_lmp_plot . . . 130

4.2.16 test_lmp_and_schedule_plot . . . 131
4.2.17 test_multi_period_model . . . 131
4.2.18 test_multi_period_stochastic_model . . . 132
4.2.19 test_multi_day_model . . . 132
4.2.20 test_multi_day_stochastic_model . . . 132
4.2.21 test_multi_year_model . . . 133
4.2.22 test_multi_year_stochastic_model . . . 133
4.2.23 test_multi_day_year_model . . . 133
4.2.24 test_multi_day_year_stochastic_model . . . 134
4.2.25 test_no_initialization . . . 134
4.2.26 test_data_kwargs . . . 134
4.2.27 test_data_kwargs_fail . . . 135
4.2.28 test_initialization_fail . . . 135
4.2.29 test_multi_period_model_manual . . . 135

**5 pricetaker** **136**
5.1 __init__.py . . . 136
5.2 clustering.py . . . 136
5.2.1 generate_daily_data . . . 138
5.2.2 cluster_lmp_data . . . 138
5.2.3 get_optimal_num_clusters . . . 139
5.2.4 _elbow_method . . . 139
5.2.5 _normalize_values . . . 140
5.3 design_and_operation_models.py . . . 140
5.3.1 _format_data . . . 142
5.3.2 is_valid_variable_design_data . . . 143
5.3.3 is_valid_polynomial_surrogate_data . . . 143
5.3.4 DesignModelData . . . 144
5.3.5 DesignModelData.build . . . 144
5.3.6 DesignModelData._build_variable_design_model . . . 145
5.3.7 OperationModelData . . . 146
5.3.8 OperationModelData.build . . . 147
5.3.9 OperationModelData.build_polynomial_surrogates . . . 147
5.3.10 OperationModelData.build_expressions . . . 148
5.3.11 _is_valid_data_type_for_storage_model . . . 148
5.3.12 StorageModelData . . . 148
5.3.13 StorageModelData.build . . . 149
5.3.14 StorageModelData.build._add_upper_bound . . . 150
5.4 price_taker_model.py . . . 150
5.4.1 PriceTakerModel . . . 154
5.4.2 PriceTakerModel.__init__ . . . 155
5.4.3 PriceTakerModel.num_representative_days . . . 155
5.4.4 PriceTakerModel.num_representative_days . . . 156
5.4.5 PriceTakerModel.horizon_length . . . 156
5.4.6 PriceTakerModel.horizon_length . . . 156
5.4.7 PriceTakerModel._assert_lmp_data_exists . . . 156
5.4.8 PriceTakerModel.append_lmp_data . . . 157
5.4.9 PriceTakerModel.get_optimal_representative_days . . . 157

5.4.10 PriceTakerModel._assert_mp_model_exists . . . 157
5.4.11 PriceTakerModel.build_multiperiod_model . . . 158
5.4.12 PriceTakerModel._get_operation_blocks . . . 159
5.4.13 PriceTakerModel._get_operation_vars . . . 159
5.4.14 PriceTakerModel.fix_operation_var . . . 159
5.4.15 PriceTakerModel.unfix_operation_var . . . 160
5.4.16 PriceTakerModel.update_operation_params . . . 160
5.4.17 PriceTakerModel.add_linking_constraints . . . 161
5.4.18 PriceTakerModel.add_linking_constraints._rule_linking_constraints . . . 161
5.4.19 PriceTakerModel.add_periodic_constraints . . . 161
5.4.20 PriceTakerModel.add_periodic_constraints._rule_linking_constraints . . . 162
5.4.21 PriceTakerModel._retrieve_uc_data . . . 162
5.4.22 PriceTakerModel._get_valid_block_name . . . 162
5.4.23 PriceTakerModel.add_capacity_limits . . . 163
5.4.24 PriceTakerModel.add_ramping_limits . . . 163
5.4.25 PriceTakerModel.add_startup_shutdown . . . 164
5.4.26 PriceTakerModel.add_hourly_cashflows . . . 165
5.4.27 PriceTakerModel.add_overall_cashflows . . . 165
5.4.28 PriceTakerModel.add_objective_function . . . 166
5.4.29 PriceTakerModel._get_num . . . 166
5.4.30 PriceTakerModel.get_num_startups . . . 166
5.4.31 PriceTakerModel.get_num_shutdowns . . . 167
5.4.32 PriceTakerModel._get_pyomo_obj_value . . . 167
5.4.33 PriceTakerModel.get_operation_var_values . . . 167
5.4.34 PriceTakerModel.get_design_var_values . . . 168
5.4.35 PriceTakerModel.plot_operation_profile . . . 168
5.4.36 PriceTakerModel.plot_lmp_histogram . . . 169
5.5 unit_commitment.py . . . 169
5.5.1 UnitCommitmentData . . . 170
5.5.2 UnitCommitmentData.__init__ . . . 171
5.5.3 UnitCommitmentData._get_config . . . 171
5.5.4 UnitCommitmentData.update . . . 171
5.5.5 UnitCommitmentData.assert_startup_rate_validity . . . 172
5.5.6 UnitCommitmentData.assert_shutdown_rate_validity . . . 172
5.5.7 UnitCommitmentData.assert_ramping_args_present . . . 172
5.5.8 startup_shutdown_constraints . . . 173
5.5.9 startup_shutdown_constraints.binary_relationship_con . . . 173
5.5.10 startup_shutdown_constraints.minimum_up_time_con . . . 173
5.5.11 startup_shutdown_constraints.minimum_down_time_con . . . 174
5.5.12 capacity_limits . . . 174
5.5.13 capacity_limits.capacity_low_limit_con . . . 174
5.5.14 capacity_limits.capacity_high_limit_con . . . 175
5.5.15 ramping_limits . . . 175
5.5.16 ramping_limits.ramp_up_con . . . 175
5.5.17 ramping_limits.ramp_down_con . . . 176

**6 pricetaker/tests** **177**
6.1 __init__.py . . . 177
6.2 lmp_data.csv . . . 177
6.3 test_clustering.py . . . 178
6.3.1 sample_data_fixture . . . 179
6.3.2 dummy_data_fixture . . . 179
6.3.3 test_daily_data_logger_message1 . . . 180
6.3.4 test_daily_data_logger_message2 . . . 180
6.3.5 test_generate_daily_data . . . 180
6.3.6 test_cluster_lmp_data . . . 181
6.3.7 test_optimal_num_clusters_elbow . . . 181
6.3.8 test_optimal_num_clusters_silhouette . . . 182
6.3.9 test_optimal_clusters_logger_message1 . . . 182
6.3.10 test_optimal_clusters_logger_message2 . . . 182
6.3.11 test_optimal_clusters_logger_message3 . . . 183
6.3.12 test_optimal_clusters_logger_message4 . . . 183
6.3.13 test_optimal_clusters_logger_message5 . . . 184
6.3.14 test_normalize_values . . . 184
6.4 test_design_and_operation_models.py . . . 184
6.4.1 test_format_data . . . 186
6.4.2 test_is_valid_variable_design_data . . . 186
6.4.3 test_is_valid_polynomial_surrogate_data . . . 186
6.4.4 test_design_model_class . . . 187
6.4.5 test_design_model_class._my_design_model . . . 187
6.4.6 test_design_model_class.dummy_func . . . 188
6.4.7 test_design_model_class_logger_message1 . . . 188
6.4.8 test_design_model_with_fixed_design . . . 188
6.4.9 test_design_model_with_variable_design . . . 188
6.4.10 test_operation_model_class . . . 189
6.4.11 test_operation_model_class.des_model . . . 189
6.4.12 test_operation_model_class.op_model . . . 190
6.4.13 test_operation_model_class_logger_message1 . . . 190
6.4.14 test_operation_model_class_logger_message1.des_model . . . 190
6.4.15 test_operation_model_with_surrogates . . . 191
6.4.16 test_is_valid_data_type_for_storage_model . . . 191
6.4.17 test_storage_model_fixed_design . . . 192
6.4.18 test_storage_model_variable_design . . . 192
6.5 test_price_taker_model.py . . . 192
6.5.1 simple_flowsheet_func . . . 196
6.5.2 foo_design_model . . . 197
6.5.3 foo_operation_model . . . 197
6.5.4 build_foo_flowsheet . . . 197
6.5.5 build_power_cycle_with_storage . . . 198
6.5.6 initialized_foo_flowsheet_fixture . . . 198
6.5.7 test_num_representative_days . . . 199
6.5.8 test_horizon_length . . . 199
6.5.9 test_append_lmp_data . . . 199
6.5.10 test_get_optimal_representative_days . . . 200

6.5.11 test_mp_model_full_year . . . 200
6.5.12 test_build_mp_model_w_period . . . 201
6.5.13 test_mp_model_rep_days . . . 201
6.5.14 test_get_operation_blocks . . . 201
6.5.15 test_get_operation_blocks_rep_days . . . 202
6.5.16 test_get_operation_vars . . . 202
6.5.17 test_get_operation_vars_rep_days . . . 203
6.5.18 test_update_operation_params . . . 204
6.5.19 test_update_operation_params_errors . . . 204
6.5.20 test_add_linking_constraints . . . 204
6.5.21 test_add_linking_constraints_rep_days . . . 205
6.5.22 test_add_periodic_constraints . . . 205
6.5.23 test_add_periodic_constraints_rep_days . . . 206
6.5.24 test_retrieve_uc_data . . . 206
6.5.25 test_add_capacity_limits . . . 207
6.5.26 test_add_capacity_limits_rep_days . . . 207
6.5.27 test_add_ramping_limits . . . 208
6.5.28 test_add_ramping_limits_rep_days . . . 208
6.5.29 test_start_shut_no_install . . . 209
6.5.30 test_start_shut_with_install . . . 209
6.5.31 test_start_shut_with_rep_days . . . 210
6.5.32 test_automatic_addition_operational_constr . . . 211
6.5.33 test_automatic_addition_operational_rep_days . . . 211
6.5.34 test_add_hourly_cashflows_warnings . . . 211
6.5.35 test_add_hourly_cashflows . . . 212
6.5.36 test_add_overall_cashflows_warnings . . . 212
6.5.37 test_add_overall_cashflows . . . 213
6.5.38 test_add_objective_function . . . 213
6.5.39 test_get_valid_block_name . . . 214
6.5.40 test_get_num_startups_shutdowns . . . 214
6.5.41 test_get_num_startups_shutdowns_rep_days . . . 215
6.5.42 test_get_num_startups_shutdowns_exception . . . 215
6.5.43 test_get_pyomo_obj_value . . . 216
6.5.44 test_get_operation_var_values . . . 216
6.5.45 test_get_design_var_values . . . 217
6.5.46 test_plot_operation_profile . . . 217
6.5.47 test_plot_lmp_histogram . . . 218
6.6 test_unit_commitment.py . . . 218
6.6.1 test_unit_commitment_data . . . 219
6.6.2 test_startup_shutdown_constraints . . . 220
6.6.3 test_startup_shutdown_constraints.op_blk . . . 220
6.6.4 test_capacity_limits_with_float . . . 220
6.6.5 test_capacity_limits_with_float.op_blk . . . 221
6.6.6 test_capacity_limits_with_var . . . 221
6.6.7 test_capacity_limits_with_var.op_blk . . . 222
6.6.8 test_ramping_limits . . . 222
6.6.9 test_ramping_limits.op_blk . . . 223

**7 tests** **224**
7.1 __init__.py . . . 224
7.2 self_scheduler_integration_test_plugin.py . . . 224
7.3 test_PEM_Parameterized_bidder.py . . . 224
7.3.1 test_creat_PEMParametrizedBidder_with_wrong_PEM_power . . . 225
7.3.2 wind_df . . . 226
7.3.3 bidder_object . . . 226
7.3.4 test_compute_DA_bids . . . 226
7.3.5 test_compute_RT_bids . . . 227
7.4 test_backcaster.py . . . 227
7.4.1 historical_da_prices . . . 229
7.4.2 historical_rt_prices . . . 229
7.4.3 base_backcaster . . . 230
7.4.4 test_create_backcaster . . . 230
7.4.5 test_create_backcaster_with_small_max_historical_days . . . 230
7.4.6 test_invalid_max_historical_days_type . . . 231
7.4.7 test_invalid_max_historical_days_value . . . 231
7.4.8 test_create_backcaster_with_non_dict . . . 231
7.4.9 test_create_backcaster_with_empty_dict . . . 232
7.4.10 test_create_backcaster_with_non_list . . . 232
7.4.11 test_create_backcaster_with_not_enough_entries . . . 232
7.4.12 test_create_backcaster_with_non_24_entries . . . 233
7.4.13 test_forecast_real_time_prices . . . 233
7.4.14 test_forecast_day_ahead_prices . . . 233
7.4.15 test_forecast_day_ahead_and_real_time_prices . . . 234
7.4.16 test_forecast_nonexistent_bus_prices . . . 234
7.4.17 MockPrescientHourlyStats . . . 234
7.4.18 MockPrescientHourlyStats.__init__ . . . 234
7.4.19 test_fetch_hourly_stats_from_prescient . . . 235
7.4.20 test_fetch_hourly_stats_from_prescient_greater_than_max_historical_days 235
7.4.21 DAPrices . . . 235
7.4.22 DAPrices.__init__ . . . 236
7.4.23 DAPrices.get . . . 236
7.4.24 MockRucMarket . . . 236
7.4.25 MockRucMarket.__init__ . . . 236
7.4.26 MockPrescientRucPlan . . . 237
7.4.27 MockPrescientRucPlan.__init__ . . . 237
7.4.28 test_fetch_day_ahead_stats_from_prescient . . . 237
7.5 test_bidder.py . . . 238
7.5.1 MissingModel . . . 239
7.5.2 MissingModel.__init__ . . . 239
7.5.3 test_model_object_missing_methods . . . 239
7.5.4 test_model_object_missing_attr . . . 240
7.5.5 test_n_scenario_checker . . . 240
7.5.6 test_solver_checker . . . 241
7.5.7 bidder_object . . . 241
7.5.8 test_compute_DA_bids . . . 242
7.5.9 test_compute_RT_bids . . . 242

7.5.10 bidder_object_pcost . . . 242
7.5.11 test_compute_RT_bids_pcost . . . 243
7.6 test_coordinator.py . . . 243
7.6.1 coordinator_object . . . 244
7.6.2 test_static_params . . . 244
7.6.3 test_assemble_sced_tracking_market_signals . . . 245
7.7 test_integration.py . . . 245
7.7.1 TestDoubleLoopIntegration . . . 246
7.7.2 TestDoubleLoopIntegration.data_path . . . 247
7.7.3 TestDoubleLoopIntegration.test_data_path_available . . . 247
7.7.4 TestDoubleLoopIntegration.output_dir . . . 248
7.7.5 TestDoubleLoopIntegration.self_scheduler_output_dir . . . 248
7.7.6 TestDoubleLoopIntegration.self_scheduler_plugin_path . . . 248
7.7.7 TestDoubleLoopIntegration.test_self_scheduler_plugin_path_is_existing_file 249
7.7.8 TestDoubleLoopIntegration.prescient_options . . . 249
7.7.9 TestDoubleLoopIntegration.bidder_sim_options . . . 249
7.7.10 TestDoubleLoopIntegration.self_scheduler_sim_options . . . 250
7.7.11 TestDoubleLoopIntegration.run_bidder_simulator . . . 250
7.7.12 TestDoubleLoopIntegration.run_self_scheduler_simulator . . . 250
7.7.13 TestDoubleLoopIntegration.simulation_results_dir . . . 251
7.7.14 TestDoubleLoopIntegration.self_scheduler_simulation_results_dir . . . 251
7.7.15 TestDoubleLoopIntegration.test_prescient_outputs_exist . . . 251
7.8 test_model_data.py . . . 251
7.8.1 generator_params . . . 253
7.8.2 renewable_generator_params . . . 253
7.8.3 generator_data_object . . . 253
7.8.4 renewable_generator_data_object . . . 253
7.8.5 test_create_thermal_model_data_object . . . 254
7.8.6 test_create_renewable_model_data_object . . . 254
7.8.7 test_create_model_data_with_non_real_numbers . . . 254
7.8.8 test_create_model_data_with_negative_values . . . 255
7.8.9 test_create_model_data_with_less_than_pmin_data . . . 255
7.8.10 test_invalid_fixed_commitment_value . . . 255
7.8.11 test_invalid_generator_name . . . 256
7.8.12 test_invalid_bus_name . . . 256
7.8.13 test_empty_bid_pairs . . . 256
7.8.14 test_bid_missing_pmin . . . 257
7.8.15 test_bid_missing_pmax . . . 257
7.8.16 test_invalid_start_up_bid . . . 257
7.8.17 test_invalid_pcost . . . 258
7.8.18 test_model_data_iterator . . . 258
7.8.19 test_invalid_initial_status . . . 258
7.8.20 test_invalid_initial_p_output . . . 259
7.9 test_perfectforecaster.py . . . 259
7.9.1 wind_df . . . 260
7.9.2 base_perfectforecaster . . . 260
7.9.3 test_create_perfectforecaster . . . 261
7.9.4 test_create_perfectforecaster_with_ndarray_and_list . . . 261

7.9.5 test_get_column_from_data . . . 261
7.9.6 test_forecast_day_ahead_prices . . . 262
7.9.7 test_forecast_real_time_prices . . . 262
7.9.8 test_forecast_day_ahead_capacity_factor . . . 263
7.9.9 test_forecast_real_time_capacity_factor . . . 263
7.10 test_tracker.py . . . 263
7.10.1 MissingModel . . . 264
7.10.2 MissingModel.__init__ . . . 265
7.10.3 test_model_object_missing_methods . . . 265
7.10.4 test_model_object_missing_attr . . . 265
7.10.5 test_n_tracking_hour_checker . . . 266
7.10.6 test_solver_checker . . . 266
7.10.7 tracker_object . . . 267
7.10.8 test_track_market_dispatch . . . 267
7.10.9 test_track_deviation_penalty . . . 268
7.11 util.py . . . 268
7.11.1 ExampleModel . . . 270
7.11.2 ExampleModel.__init__ . . . 271
7.11.3 ExampleModel.model_data . . . 271
7.11.4 ExampleModel.populate_model . . . 271
7.11.5 ExampleModel.populate_model.prod_cost_fun . . . 272
7.11.6 ExampleModel.populate_model.tot_cost_fun . . . 272
7.11.7 ExampleModel._update_power . . . 272
7.11.8 ExampleModel.update_model . . . 272
7.11.9 ExampleModel.get_implemented_profile . . . 273
7.11.10 ExampleModel.get_last_delivered_power . . . 273
7.11.11 ExampleModel.record_results . . . 273
7.11.12 ExampleModel.write_results . . . 274
7.11.13 ExampleModel.power_output . . . 274
7.11.14 ExampleModel.total_cost . . . 274
7.11.15 ExampleForecaster . . . 275
7.11.16 ExampleForecaster.__init__ . . . 275
7.11.17 ExampleForecaster.forecast_day_ahead_and_real_time_prices . . . 275
7.11.18 ExampleForecaster.forecast_real_time_prices . . . 276
7.11.19 ExampleForecaster.forecast_day_ahead_prices . . . 276
7.11.20 ExampleForecaster._forecast . . . 276
7.11.21 ExampleForecaster.fetch_hourly_stats_from_prescient . . . 276
7.11.22 ExampleForecaster.fetch_day_ahead_stats_from_prescient . . . 277
7.11.23 make_testing_forecaster . . . 277
7.11.24 make_testing_tracker . . . 277
7.11.25 make_testing_bidder . . . 277
7.11.26 make_testing_selfscheduler . . . 278

# Overview

This notebook is generated from files under `idaes/apps/grid_integration`. Chapters follow folders. Sections follow files. Function and method names are clickable and jump to where they are defined and described in this notebook.

# Chapter 1

# grid_integration (root)

## 1.1 __init__.py

**File path** `__init__.py`

## 1.2 bidder.py

**File path** `bidder.py`

**Definitions Index**

- **class**: `AbstractBidder` (line 30)
- **method**: `AbstractBidder.update_day_ahead_model` (line 36)
- **method**: `AbstractBidder.update_real_time_model` (line 48)
- **method**: `AbstractBidder.compute_day_ahead_bids` (line 60)
- **method**: `AbstractBidder.compute_real_time_bids` (line 78)
- **method**: `AbstractBidder.write_results` (line 96)
- **method**: `AbstractBidder.formulate_DA_bidding_problem` (line 108)
- **method**: `AbstractBidder.formulate_RT_bidding_problem` (line 121)
- **method**: `AbstractBidder.record_bids` (line 134)
- **method**: `AbstractBidder.generator` (line 155)
- **method**: `AbstractBidder._check_inputs` (line 158)
- **method**: `AbstractBidder._check_bidding_model_object` (line 167)
- **method**: `AbstractBidder._check_n_scenario` (line 192)
- **method**: `AbstractBidder._check_solver` (line 208)
- **class**: `StochasticProgramBidder` (line 219)

- **method**: `StochasticProgramBidder.__init__` (line 224)
- **method**: `StochasticProgramBidder._set_up_bidding_problem` (line 275)
- **method**: `StochasticProgramBidder.formulate_DA_bidding_problem` (line 303)
- **method**: `StochasticProgramBidder.formulate_RT_bidding_problem` (line 321)
- **method**: `StochasticProgramBidder._save_power_outputs` (line 339)
- **method**: `StochasticProgramBidder._add_bidding_params` (line 360)
- **method**: `StochasticProgramBidder._add_bidding_vars` (line 385)
- **method**: `StochasticProgramBidder._add_bidding_vars.relaxed_day_ahead_power_ub_rule` (line 396)
- **method**: `StochasticProgramBidder._add_bidding_objective` (line 418)
- **method**: `StochasticProgramBidder._compute_bids` (line 456)
- **method**: `StochasticProgramBidder.compute_day_ahead_bids` (line 508)
- **method**: `StochasticProgramBidder.compute_real_time_bids` (line 545)
- **method**: `StochasticProgramBidder._pass_realized_day_ahead_prices` (line 584)
- **method**: `StochasticProgramBidder._pass_realized_day_ahead_dispatches` (line 622)
- **method**: `StochasticProgramBidder.update_day_ahead_model` (line 649)
- **method**: `StochasticProgramBidder.update_real_time_model` (line 662)
- **method**: `StochasticProgramBidder._update_model` (line 675)
- **method**: `StochasticProgramBidder.record_bids` (line 695)
- **method**: `StochasticProgramBidder._pass_price_forecasts` (line 724)
- **method**: `StochasticProgramBidder.write_results` (line 750)
- **method**: `StochasticProgramBidder.generator` (line 772)
- **method**: `StochasticProgramBidder.generator` (line 776)
- **class**: `SelfScheduler` (line 780)
- **method**: `SelfScheduler.__init__` (line 785)
- **method**: `SelfScheduler._add_DA_bidding_constraints` (line 829)
- **method**: `SelfScheduler._add_DA_bidding_constraints.day_ahead_bidding_constraints_rule` (line 842)
- **method**: `SelfScheduler._add_RT_bidding_constraints` (line 858)
- **method**: `SelfScheduler._add_RT_bidding_constraints.real_time_bidding_constraints_rule` (line 871)

- **method**: `SelfScheduler._assemble_bids` (line 887)
- **method**: `SelfScheduler._record_bids` (line 957)
- **class**: `Bidder` (line 1000)
- **method**: `Bidder.__init__` (line 1005)
- **method**: `Bidder._add_DA_bidding_constraints` (line 1047)
- **method**: `Bidder._add_DA_bidding_constraints.day_ahead_bidding_constraints_rule` (line 1059)
- **method**: `Bidder._add_RT_bidding_constraints` (line 1080)
- **method**: `Bidder._add_RT_bidding_constraints.real_time_bidding_constraints_rule` (line 1092)
- **method**: `Bidder._assemble_bids` (line 1113)
- **method**: `Bidder._record_bids` (line 1242)
- **class**: `ParametrizedBidder` (line 1298)
- **method**: `ParametrizedBidder.__init__` (line 1304)
- **method**: `ParametrizedBidder.generator` (line 1342)
- **method**: `ParametrizedBidder.generator` (line 1346)
- **method**: `ParametrizedBidder.formulate_DA_bidding_problem` (line 1349)
- **method**: `ParametrizedBidder.formulate_RT_bidding_problem` (line 1361)
- **method**: `ParametrizedBidder.compute_day_ahead_bids` (line 1373)
- **method**: `ParametrizedBidder.compute_real_time_bids` (line 1376)
- **method**: `ParametrizedBidder.update_day_ahead_model` (line 1386)
- **method**: `ParametrizedBidder.update_real_time_model` (line 1398)
- **method**: `ParametrizedBidder.record_bids` (line 1410)
- **method**: `ParametrizedBidder._record_bids` (line 1439)
- **method**: `ParametrizedBidder.write_results` (line 1483)
- **class**: `PEMParametrizedBidder` (line 1502)
- **method**: `PEMParametrizedBidder.__init__` (line 1507)
- **method**: `PEMParametrizedBidder._check_power` (line 1555)
- **method**: `PEMParametrizedBidder.compute_day_ahead_bids` (line 1562)
- **method**: `PEMParametrizedBidder.compute_real_time_bids` (line 1625)

### Called functions/methods in this file

- line 23: `attempt_import`
- line 27: `getLogger`
- line 163: `_check_bidding_model_object`
- line 164: `_check_n_scenario`
- line 165: `_check_solver`
- line 177: `getattr`
- line 179: `AttributeError`
- line 198: `isinstance`
- line 199: `TypeError`
- line 200: `type`
- line 204: `ValueError`
- line 264: `_check_inputs`
- line 269: `formulate_DA_bidding_problem`
- line 270: `formulate_RT_bidding_problem`
- line 287: `ConcreteModel`
- line 289: `Set`
- line 289: `range`
- line 291: `Block`
- line 293: `populate_model`
- line 295: `_save_power_outputs`
- line 297: `_add_bidding_params`
- line 298: `_add_bidding_vars`
- line 299: `_add_bidding_objective`
- line 312: `_set_up_bidding_problem`
- line 313: `_add_DA_bidding_constraints`
- line 317: `fix`
- line 331: `_add_RT_bidding_constraints`
- line 335: `unfix`

- line 354: `Reference`
- line 372: `index_set`
- line 373: `Param`
- line 404: `Var`
- line 412: `Constraint`
- line 431: `Objective`
- line 493: `_pass_price_forecasts`
- line 495: `solve`
- line 497: `_assemble_bids`
- line 504: `record_bids`
- line 526: `forecast_day_ahead_and_real_time_prices`
- line 534: `_compute_bids`
- line 562: `forecast_real_time_prices`
- line 570: `_pass_realized_day_ahead_dispatches`
- line 571: `_pass_realized_day_ahead_prices`
- line 602: `len`
- line 603: `forecast_day_ahead_prices`
- line 604: `timedelta`
- line 660: `_update_model`
- line 691: `update_model`
- line 714: `_record_bids`
- line 718: `record_results`
- line 742: `zip`
- line 761: `info`
- line 762: `to_csv`
- line 762: `concat`
- line 763: `join`
- line 765: `write_results`
- line 818: `__init__`
- line 818: `super`
- line 847: `first`
- line 920: `round`

**Definition Notes**

### 1.2.1 AbstractBidder

**Kind** class

**Defined in file** `bidder.py` at line 30

**Signature** `class AbstractBidder(ABC):`

**Summary** The abstract class for all the bidder and self-schedulers.

**Calls**

- `_check_bidding_model_object`
- `_check_n_scenario`
- `_check_solver`
- `getattr`
- `isinstance`
- `TypeError`
- `ValueError`
- `AttributeError`
- `type`

### 1.2.2 AbstractBidder.update_day_ahead_model

**Kind** method

**Defined in file** `bidder.py` at line 36

**Signature** `def update_day_ahead_model(self, **kwargs):`

**Summary** Update the day-ahead model (advance timesteps) with necessary parameters in kwargs.

### 1.2.3 AbstractBidder.update_real_time_model

**Kind** method

**Defined in file** `bidder.py` at line 48

**Signature** `def update_real_time_model(self, **kwargs):`

**Summary** Update the real-time model (advance timesteps) with necessary parameters in kwargs.

### 1.2.4 AbstractBidder.compute_day_ahead_bids

**Kind** method

**Defined in file** `bidder.py` at line 60

**Signature** `def compute_day_ahead_bids(self, date, hour, **kwargs):`

**Summary** Solve the model to bid/self-schedule into the day-ahead market. After solving,

### 1.2.5 AbstractBidder.compute_real_time_bids

**Kind** method

**Defined in file** `bidder.py` at line 78

**Signature** `def compute_real_time_bids(self, date, hour, **kwargs):`

**Summary** Solve the model to bid/self-schedule into the real-time market. After solving,

### 1.2.6 AbstractBidder.write_results

**Kind** method

**Defined in file** `bidder.py` at line 96

**Signature** `def write_results(self, path):`

**Summary** This methods writes the saved results into an csv file.

### 1.2.7 AbstractBidder.formulate_DA_bidding_problem

**Kind** method

**Defined in file** `bidder.py` at line 108

**Signature** `def formulate_DA_bidding_problem(self):`

**Summary** Formulate the day-ahead bidding optimization problem by adding necessary

### 1.2.8 AbstractBidder.formulate_RT_bidding_problem

**Kind** method

**Defined in file** `bidder.py` at line 121

**Signature** `def formulate_RT_bidding_problem(self):`

**Summary** Formulate the real-time bidding optimization problem by adding necessary

### 1.2.9 AbstractBidder.record_bids

**Kind** method

**Defined in file** `bidder.py` at line 134

**Signature** `def record_bids(self, bids, model, date, hour):`

**Summary** This function records the bids (schedule) and the details in the

### 1.2.10 AbstractBidder.generator

**Kind** method

**Defined in file** `bidder.py` at line 155

**Signature** `def generator(self):`

**Summary** No docstring.

### 1.2.11 AbstractBidder._check_inputs

**Kind** method

**Defined in file** `bidder.py` at line 158

**Signature** `def _check_inputs(self):`

**Summary** Check if the inputs to construct the tracker is valid. If not raise errors.

**Calls**

- `_check_bidding_model_object`
- `_check_n_scenario`
- `_check_solver`

### 1.2.12 AbstractBidder._check_bidding_model_object

**Kind** method

**Defined in file** `bidder.py` at line 167

**Signature** `def _check_bidding_model_object(self):`

**Summary** Check if tracking model object has the necessary methods and attributes.

**Calls**

- `getattr`
- `AttributeError`

### 1.2.13 AbstractBidder._check_n_scenario

**Kind** method

**Defined in file** `bidder.py` at line 192

**Signature** `def _check_n_scenario(self):`

**Summary** Check if the number of LMP scenarios is an integer and greater than 0.

**Calls**

- `isinstance`
- `TypeError`
- `ValueError`
- `type`

### 1.2.14 AbstractBidder._check_solver

**Kind** method

**Defined in file** `bidder.py` at line 208

**Signature** `def _check_solver(self):`

**Summary** Check if provides solver is a valid Pyomo solver object.

**Calls**

- `isinstance`
- `TypeError`

### 1.2.15 StochasticProgramBidder

**Kind** class

**Defined in file** `bidder.py` at line 219

**Signature** `class StochasticProgramBidder(AbstractBidder):`

**Summary** Template class for bidders that use scenario-based stochastic programs.

**Calls**

- _check_inputs
- formulate_DA_bidding_problem
- formulate_RT_bidding_problem
- ConcreteModel
- Set
- Block
- _save_power_outputs
- _add_bidding_params
- _add_bidding_vars
- _add_bidding_objective
- _set_up_bidding_problem
- _add_DA_bidding_constraints
- _add_RT_bidding_constraints
- Objective
- _pass_price_forecasts
- solve
- _assemble_bids
- record_bids
- forecast_day_ahead_and_real_time_prices
- _compute_bids
- forecast_real_time_prices
- _pass_realized_day_ahead_dispatches
- _pass_realized_day_ahead_prices
- index_set
- _update_model

### 1.2.16 StochasticProgramBidder.__init__

**Kind** method

**Defined in file** bidder.py at line 224

**Signature** `def __init__(`

**Summary** Initializes the stochastic bidder object.

**Calls**

- `_check_inputs`
- `formulate_DA_bidding_problem`
- `formulate_RT_bidding_problem`

### 1.2.17 StochasticProgramBidder._set_up_bidding_problem

**Kind** method

**Defined in file** `bidder.py` at line 275

**Signature** `def _set_up_bidding_problem(self, horizon):`

**Summary** Set up the base stochastic programming bidding problems.

**Calls**

- `ConcreteModel`
- `Set`
- `Block`
- `_save_power_outputs`
- `_add_bidding_params`
- `_add_bidding_vars`
- `_add_bidding_objective`
- `populate_model`
- `range`

### 1.2.18 StochasticProgramBidder.formulate_DA_bidding_problem

**Kind** method

**Defined in file** `bidder.py` at line 303

**Signature** `def formulate_DA_bidding_problem(self):`

**Summary** Set up the day-ahead stochastic programming bidding problems.

**Calls**

- `_set_up_bidding_problem`
- `_add_DA_bidding_constraints`
- `fix`

### 1.2.19 StochasticProgramBidder.formulate_RT_bidding_problem

**Kind** method

**Defined in file** `bidder.py` at line 321

**Signature** `def formulate_RT_bidding_problem(self):`

**Summary** Set up the real-time stochastic programming bidding problems.

**Calls**

- `_set_up_bidding_problem`
- `_add_RT_bidding_constraints`
- `unfix`

### 1.2.20 StochasticProgramBidder._save_power_outputs

**Kind** method

**Defined in file** `bidder.py` at line 339

**Signature** `def _save_power_outputs(self, model):`

**Summary** Create references of the power output variable in each price scenario

**Calls**

- `Reference`
- `getattr`

### 1.2.21 StochasticProgramBidder._add_bidding_params

**Kind** method

**Defined in file** `bidder.py` at line 360

**Signature** `def _add_bidding_params(self, model):`

**Summary** Add necessary bidding parameters to the model, i.e., market energy price.

**Calls**

- `index_set`
- `Param`

### 1.2.22 StochasticProgramBidder._add_bidding_vars

**Kind** method

**Defined in file** `bidder.py` at line 385

**Signature** `def _add_bidding_vars(self, model):`

**Summary** Add necessary bidding parameters to the model, i.e., market energy price.

**Calls**

- `index_set`
- `Var`
- `Constraint`

### 1.2.23 StochasticProgramBidder._add_bidding_vars.relaxed_day_ahead_power_ub_ru

**Kind** method

**Defined in file** `bidder.py` at line 396

**Signature** `def relaxed_day_ahead_power_ub_rule(fs, t):`

**Summary** No docstring.

### 1.2.24 StochasticProgramBidder._add_bidding_objective

**Kind** method

**Defined in file** `bidder.py` at line 418

**Signature** `def _add_bidding_objective(self, model):`

**Summary** Add objective function to the model, i.e., maximizing the expected profit

**Calls**

- `Objective`
- `index_set`
- `getattr`

### 1.2.25 StochasticProgramBidder._compute_bids

**Kind** method

**Defined in file** `bidder.py` at line 456

**Signature** `def _compute_bids(`

**Summary** Solve the model to bid into the markets. After solving, record the bids from the solve.

**Calls**

- `_pass_price_forecasts`
- `solve`
- `_assemble_bids`
- `record_bids`

### 1.2.26 StochasticProgramBidder.compute_day_ahead_bids

**Kind** method

**Defined in file** `bidder.py` at line 508

**Signature** `def compute_day_ahead_bids(self, date, hour=0):`

**Summary** Solve the model to bid into the day-ahead market. After solving, record

**Calls**

- `forecast_day_ahead_and_real_time_prices`
- `_compute_bids`

### 1.2.27 StochasticProgramBidder.compute_real_time_bids

**Kind** method

**Defined in file** `bidder.py` at line 545

**Signature** `def compute_real_time_bids(`

**Summary** Solve the model to bid into the real-time market. After solving, record

**Calls**

- `forecast_real_time_prices`
- `_pass_realized_day_ahead_dispatches`
- `_pass_realized_day_ahead_prices`
- `_compute_bids`

### 1.2.28 StochasticProgramBidder._pass_realized_day_ahead_prices

**Kind** method

**Defined in file** `bidder.py` at line 584

**Signature** `def _pass_realized_day_ahead_prices(self, realized_day_ahead_prices, date, hour):`

**Summary** Pass the realized day-ahead prices into model parameters.

**Calls**

- `index_set`
- `len`
- `forecast_day_ahead_prices`
- `timedelta`

### 1.2.29 StochasticProgramBidder._pass_realized_day_ahead_dispatches

**Kind** method

**Defined in file** `bidder.py` at line 622

**Signature** `def _pass_realized_day_ahead_dispatches(self, realized_day_ahead_dispatches, hour):`

**Summary** Pass the realized day-ahead dispatches into model and fix the corresponding variables.

**Calls**

- `index_set`
- `fix`
- `unfix`

### 1.2.30 StochasticProgramBidder.update_day_ahead_model

**Kind** method

**Defined in file** `bidder.py` at line 649

**Signature** `def update_day_ahead_model(self, **kwargs):`

**Summary** This method updates the parameters in the day-ahead model based on the implemented profiles.

**Calls**

- `_update_model`

### 1.2.31 StochasticProgramBidder.update_real_time_model

**Kind** method

**Defined in file** `bidder.py` at line 662

**Signature** `def update_real_time_model(self, **kwargs):`

**Summary** This method updates the parameters in the real-time model based on the implemented profiles.

**Calls**

- `_update_model`

### 1.2.32 StochasticProgramBidder._update_model

**Kind** method

**Defined in file** `bidder.py` at line 675

**Signature** `def _update_model(self, model, **kwargs):`

**Summary** Update the flowsheets in all the price scenario blocks to advance time

**Calls**

- `update_model`

### 1.2.33 StochasticProgramBidder.record_bids

**Kind** method

**Defined in file** `bidder.py` at line 695

**Signature** `def record_bids(self, bids, model, date, hour, market):`

**Summary** This function records the bids and the details in the underlying bidding model.

**Calls**

- `_record_bids`
- `record_results`

### 1.2.34 StochasticProgramBidder._pass_price_forecasts

**Kind** method

**Defined in file** `bidder.py` at line 724

**Signature** `def _pass_price_forecasts(self, model, day_ahead_price, real_time_energy_price):`

**Summary** Pass the price forecasts into model parameters.

**Calls**

- `index_set`
- `zip`

### 1.2.35 StochasticProgramBidder.write_results

**Kind** method

**Defined in file** `bidder.py` at line 750

**Signature** `def write_results(self, path):`

**Summary** This methods writes the saved operation stats into an csv file.

**Calls**

- `info`
- `to_csv`
- `write_results`
- `join`
- `concat`

### 1.2.36 StochasticProgramBidder.generator

**Kind** method

**Defined in file** `bidder.py` at line 772

**Signature** `def generator(self):`

**Summary** No docstring.

### 1.2.37 StochasticProgramBidder.generator

**Kind** method

**Defined in file** `bidder.py` at line 776

**Signature** `def generator(self, name):`

**Summary** No docstring.

### 1.2.38 SelfScheduler

**Kind** class

**Defined in file** `bidder.py` at line 780

**Signature** `class SelfScheduler(StochasticProgramBidder):`

**Summary** Wrap a model object to self schedule into the market using stochastic programming.

**Calls**

- `__init__`
- `index_set`
- `Constraint`
- `getattr`
- `append`
- `concat`
- `super`
- `round`
- `get`
- `items`
- `from_dict`
- `value`
- `first`

### 1.2.39 SelfScheduler.__init__

**Kind** method

**Defined in file** `bidder.py` at line 785

**Signature** `def __init__(`

**Summary** Initializes the stochastic self-scheduler object.

**Calls**

- `__init__`
- `super`

### 1.2.40 SelfScheduler._add_DA_bidding_constraints

**Kind** method

**Defined in file** `bidder.py` at line 829

**Signature** `def _add_DA_bidding_constraints(self, model):`

**Summary** Add bidding constraints to the model, i.e., power outputs in the first

**Calls**

- `index_set`
- `Constraint`
- `first`

### 1.2.41 SelfScheduler._add_DA_bidding_constraints.day_ahead_bidding_constraints_ru

**Kind** method

**Defined in file** `bidder.py` at line 842

**Signature** `def day_ahead_bidding_constraints_rule(model, s1, s2, t):`

**Summary** No docstring.

### 1.2.42 SelfScheduler._add_RT_bidding_constraints

**Kind** method

**Defined in file** `bidder.py` at line 858

**Signature** `def _add_RT_bidding_constraints(self, model):`

**Summary** Add bidding constraints to the model, i.e., power outputs in the first

**Calls**

- `index_set`
- `Constraint`
- `first`

### 1.2.43 SelfScheduler._add_RT_bidding_constraints.real_time_bidding_constraints_rule

**Kind** method

**Defined in file** `bidder.py` at line 871

**Signature** `def real_time_bidding_constraints_rule(model, s1, s2, t):`

**Summary** No docstring.

### 1.2.44 SelfScheduler._assemble_bids

**Kind** method

**Defined in file** `bidder.py` at line 887

**Signature** `def _assemble_bids(self, model, power_var_name, energy_price_param_name, hour):`

**Summary** This methods extract the bids out of the stochastic programming model and

**Calls**

- `getattr`
- `index_set`
- `round`
- `value`

### 1.2.45 SelfScheduler._record_bids

**Kind** method

**Defined in file** `bidder.py` at line 957

**Signature** `def _record_bids(self, bids, date, hour, **kwargs):`

**Summary** This function records the bids (schedule) we computed for the given date into a

**Calls**

- `append`
- `concat`
- `get`
- `items`
- `from_dict`

### 1.2.46 Bidder

**Kind** class

**Defined in file** `bidder.py` at line 1000

**Signature** `class Bidder(StochasticProgramBidder):`

**Summary** Wrap a model object to bid into the market using stochastic programming.

**Calls**

- `__init__`
- `index_set`
- `Constraint`
- `append`
- `getattr`
- `dict`
- `items`
- `convert_marginal_costs_to_actual_costs`
- `concat`
- `super`
- `round`
- `sorted`
- `max`
- `list`
- `min`

- `len`
- `enumerate`
- `from_dict`
- `value`
- `validate_and_clean_cost_curve`
- `first`
- `RuntimeError`

### 1.2.47 Bidder.__init__

**Kind** method

**Defined in file** `bidder.py` at line 1005

**Signature** `def __init__(`

**Summary** Initializes the bidder object.

**Calls**

- `__init__`
- `super`

### 1.2.48 Bidder._add_DA_bidding_constraints

**Kind** method

**Defined in file** `bidder.py` at line 1047

**Signature** `def _add_DA_bidding_constraints(self, model):`

**Summary** Add bidding constraints to the model, i.e., the bid curves need to be

**Calls**

- `index_set`
- `Constraint`
- `first`

### 1.2.49 Bidder._add_DA_bidding_constraints.day_ahead_bidding_constraints_rule

**Kind** method

**Defined in file** `bidder.py` at line 1059

**Signature** `def day_ahead_bidding_constraints_rule(model, s1, s2, t):`

**Summary** No docstring.

### 1.2.50 Bidder._add_RT_bidding_constraints

**Kind** method

**Defined in file** `bidder.py` at line 1080

**Signature** `def _add_RT_bidding_constraints(self, model):`

**Summary** Add bidding constraints to the model, i.e., the bid curves need to be

**Calls**

- `index_set`
- `Constraint`
- `first`

### 1.2.51 Bidder._add_RT_bidding_constraints.real_time_bidding_constraints_rule

**Kind** method

**Defined in file** `bidder.py` at line 1092

**Signature** `def real_time_bidding_constraints_rule(model, s1, s2, t):`

**Summary** No docstring.

### 1.2.52 Bidder._assemble_bids

**Kind** method

**Defined in file** `bidder.py` at line 1113

**Signature** `def _assemble_bids(self, model, power_var_name, energy_price_param_name, hour):`

**Summary** This methods extract the bids out of the stochastic programming model and

**Calls**

- `getattr`
- `index_set`
- `dict`
- `items`
- `convert_marginal_costs_to_actual_costs`
- `round`
- `sorted`
- `max`
- `list`
- `min`
- `value`
- `validate_and_clean_cost_curve`
- `RuntimeError`

### 1.2.53 Bidder._record_bids

**Kind** method

**Defined in file** `bidder.py` at line 1242

**Signature** `def _record_bids(self, bids, date, hour, **kwargs):`

**Summary** This method records the bids we computed for the given date into a

**Calls**

- `append`
- `concat`
- `items`
- `len`
- `enumerate`
- `from_dict`

### 1.2.54 ParametrizedBidder

**Kind** class

**Defined in file** `bidder.py` at line 1298

**Signature** `class ParametrizedBidder(AbstractBidder):`

**Summary** Create a parameterized bidder.

**Calls**

- `_check_inputs`
- `_record_bids`
- `append`
- `info`
- `to_csv`
- `concat`
- `join`
- `items`
- `len`
- `enumerate`
- `from_dict`

### 1.2.55 ParametrizedBidder.__init__

**Kind** method

**Defined in file** `bidder.py` at line 1304

**Signature** `def __init__(`

**Summary** Arguments:

**Calls**

- `_check_inputs`

### 1.2.56 ParametrizedBidder.generator

**Kind** method

**Defined in file** `bidder.py` at line 1342

**Signature** `def generator(self):`

**Summary** No docstring.

### 1.2.57 ParametrizedBidder.generator

**Kind** method

**Defined in file** `bidder.py` at line 1346

**Signature** `def generator(self, name):`

**Summary** No docstring.

### 1.2.58 ParametrizedBidder.formulate_DA_bidding_problem

**Kind** method

**Defined in file** `bidder.py` at line 1349

**Signature** `def formulate_DA_bidding_problem(self):`

**Summary** No need to formulate a DA bidding problem here.

### 1.2.59 ParametrizedBidder.formulate_RT_bidding_problem

**Kind** method

**Defined in file** `bidder.py` at line 1361

**Signature** `def formulate_RT_bidding_problem(self):`

**Summary** No need to formulate a RT bidding problem here.

### 1.2.60 ParametrizedBidder.compute_day_ahead_bids

**Kind** method

**Defined in file** `bidder.py` at line 1373

**Signature** `def compute_day_ahead_bids(self, date, hour=0):`

**Summary** No docstring.

### 1.2.61 ParametrizedBidder.compute_real_time_bids

**Kind** method

**Defined in file** `bidder.py` at line 1376

**Signature** `def compute_real_time_bids(`

**Summary** No docstring.

### 1.2.62 ParametrizedBidder.update_day_ahead_model

**Kind** method

**Defined in file** `bidder.py` at line 1386

**Signature** `def update_day_ahead_model(self, **kwargs):`

**Summary** No need to update the RT bidding problem here.

### 1.2.63 ParametrizedBidder.update_real_time_model

**Kind** method

**Defined in file** `bidder.py` at line 1398

**Signature** `def update_real_time_model(self, **kwargs):`

**Summary** No need to update the RT bidding problem here.

### 1.2.64 ParametrizedBidder.record_bids

**Kind** method

**Defined in file** `bidder.py` at line 1410

**Signature** `def record_bids(self, bids: dict, model, date: str, hour: int, market):`

**Summary** This function records the bids and the details in the underlying bidding model.

**Calls**

- `_record_bids`

### 1.2.65 ParametrizedBidder._record_bids

**Kind** method

**Defined in file** `bidder.py` at line 1439

**Signature** `def _record_bids(self, bids: dict, date: str, hour: int, **kwargs):`

**Summary** Record the bis of each time period.

**Calls**

- `append`
- `concat`
- `items`
- `len`
- `enumerate`
- `from_dict`

### 1.2.66 ParametrizedBidder.write_results

**Kind** method

**Defined in file** `bidder.py` at line 1483

**Signature** `def write_results(self, path):`

**Summary** This methods writes the saved operation stats into an csv file.

**Calls**

- `info`
- `to_csv`
- `join`
- `concat`

### 1.2.67 PEMParametrizedBidder

**Kind** class

**Defined in file** `bidder.py` at line 1502

**Signature** `class PEMParametrizedBidder(ParametrizedBidder):`

**Summary** Renewable (PV or Wind) + PEM bidder that uses parameterized bid curves.

**Calls**

- `__init__`
- `_check_power`
- `forecast_day_ahead_capacity_factor`
- `range`
- `_record_bids`
- `forecast_real_time_capacity_factor`
- `ValueError`
- `max`
- `convert_marginal_costs_to_actual_costs`
- `validate_and_clean_cost_curve`
- `super`

### 1.2.68 PEMParametrizedBidder.__init__

**Kind** method

**Defined in file** `bidder.py` at line 1507

**Signature** `def __init__(`

**Summary** Arguments:

**Calls**

- `__init__`
- `_check_power`
- `super`

### 1.2.69 PEMParametrizedBidder._check_power

**Kind** method

**Defined in file** `bidder.py` at line 1555

**Signature** `def _check_power(self):`

**Summary** Check the power of PEM should not exceed the power of renewables

**Calls**

- `ValueError`

### 1.2.70 PEMParametrizedBidder.compute_day_ahead_bids

**Kind** method

**Defined in file** `bidder.py` at line 1562

**Signature** `def compute_day_ahead_bids(self, date: str, hour=0):`

**Summary** DA Bid: from 0 MW to (Wind Resource - PEM capacity) MW, bid $0/MWh.

**Calls**

- `forecast_day_ahead_capacity_factor`
- `range`
- `_record_bids`
- `max`
- `convert_marginal_costs_to_actual_costs`
- `validate_and_clean_cost_curve`

### 1.2.71 PEMParametrizedBidder.compute_real_time_bids

**Kind** method

**Defined in file** `bidder.py` at line 1625

**Signature** `def compute_real_time_bids(`

**Summary** RT Bid: from 0 MW to (Wind Resource - PEM capacity) MW, bid $0/MWh.

**Calls**

- `forecast_real_time_capacity_factor`
- `range`
- `max`
- `convert_marginal_costs_to_actual_costs`
- `validate_and_clean_cost_curve`
- `_record_bids`

## 1.3 coordinator.py

**File path** `coordinator.py`

**Definitions Index**

- **class**: `PrescientPluginModule` (line 24)
- **method**: `PrescientPluginModule.__init__` (line 25)
- **class**: `DoubleLoopCoordinator` (line 30)
- **method**: `DoubleLoopCoordinator.__init__` (line 35)
- **method**: `DoubleLoopCoordinator.register_plugins` (line 58)
- **method**: `DoubleLoopCoordinator.get_configuration` (line 100)
- **method**: `DoubleLoopCoordinator.prescient_plugin_module` (line 126)
- **method**: `DoubleLoopCoordinator.initialize_customized_results` (line 129)
- **method**: `DoubleLoopCoordinator.push_hourly_stats_to_forecaster` (line 147)
- **method**: `DoubleLoopCoordinator.push_day_ahead_stats_to_forecaster` (line 161)
- **method**: `DoubleLoopCoordinator._update_bids` (line 187)
- **method**: `DoubleLoopCoordinator._update_bids._update_p_cost` (line 208)
- **method**: `DoubleLoopCoordinator._update_bids._update_time_series_params` (line 229)
- **method**: `DoubleLoopCoordinator._update_bids._update_non_time_series_params` (line 243)
- **method**: `DoubleLoopCoordinator._pass_DA_bid_to_prescient` (line 272)
- **method**: `DoubleLoopCoordinator.assemble_project_tracking_signal` (line 296)
- **method**: `DoubleLoopCoordinator.project_tracking_trajectory` (line 321)
- **method**: `DoubleLoopCoordinator._clone_tracking_model` (line 366)
- **method**: `DoubleLoopCoordinator._update_static_params` (line 399)
- **method**: `DoubleLoopCoordinator.update_static_params` (line 448)
- **method**: `DoubleLoopCoordinator.bid_into_DAM` (line 468)
- **method**: `DoubleLoopCoordinator.fetch_DA_prices` (line 512)
- **method**: `DoubleLoopCoordinator.fetch_DA_dispatches` (line 545)
- **method**: `DoubleLoopCoordinator._pass_RT_bid_to_prescient` (line 587)
- **method**: `DoubleLoopCoordinator.bid_into_RTM` (line 616)
- **method**: `DoubleLoopCoordinator.assemble_sced_tracking_market_signals` (line 648)

- **method**: DoubleLoopCoordinator._assemble_sced_tracking_market_signals (line 683)
- **method**: DoubleLoopCoordinator.track_sced_signal (line 718)
- **method**: DoubleLoopCoordinator.update_observed_dispatch (line 759)
- **method**: DoubleLoopCoordinator.activate_pending_DA_data (line 791)
- **method**: DoubleLoopCoordinator.write_plugin_results (line 816)

**Called functions/methods in this file**

- line 21: attempt_import
- line 75: register_initialization_callback
- line 76: register_for_hourly_stats
- line 77: register_after_get_initial_actuals_model_for_sced_callback
- line 80: register_after_get_initial_actuals_model_for_simulation_actuals_callback
- line 83: register_after_get_initial_forecast_model_for_ruc_callback
- line 86: register_before_ruc_solve_callback
- line 87: register_after_ruc_generation_callback
- line 92: register_before_operations_solve_callback
- line 93: register_after_operations_callback
- line 94: register_update_operations_stats_callback
- line 95: register_after_ruc_activation_callback
- line 96: register_finalization_callback
- line 111: ConfigDict
- line 114: declare_as_argument
- line 114: declare
- line 116: ConfigValue
- line 127: PrescientPluginModule
- line 159: fetch_hourly_stats_from_prescient
- line 183: fetch_day_ahead_stats_from_prescient
- line 219: range
- line 225: pop
- line 232: get

- line 266: keys
- line 268: update_func
- line 292: _update_bids
- line 312: len
- line 314: _assemble_sced_tracking_market_signals
- line 341: _clone_tracking_model
- line 346: assemble_project_tracking_signal
- line 350: track_market_dispatch
- line 381: zip_longest
- line 382: component_objects
- line 390: ValueError
- line 393: index_set
- line 394: value
- line 395: round
- line 425: isinstance
- line 433: convert_marginal_costs_to_actual_costs
- line 442: NotImplementedError
- line 464: _update_static_params
- line 494: project_tracking_trajectory
- line 498: update_day_ahead_model
- line 501: compute_day_ahead_bids
- line 508: _pass_DA_bid_to_prescient
- line 636: compute_real_time_bids
- line 644: _pass_RT_bid_to_prescient
- line 709: append
- line 741: assemble_sced_tracking_market_signals
- line 754: update_model
- line 755: update_real_time_model
- line 787: get_last_delivered_power
- line 831: write_results

**Definition Notes**

### 1.3.1 PrescientPluginModule

**Kind** class

**Defined in file** `coordinator.py` at line 24

**Signature** `class PrescientPluginModule(ModuleType):`

**Summary** No docstring.

### 1.3.2 PrescientPluginModule.__init__

**Kind** method

**Defined in file** `coordinator.py` at line 25

**Signature** `def __init__(self, get_configuration, register_plugins):`

**Summary** No docstring.

### 1.3.3 DoubleLoopCoordinator

**Kind** class

**Defined in file** `coordinator.py` at line 30

**Signature** `class DoubleLoopCoordinator:`

**Summary** Coordinate Prescient, tracker and bidder.

**Calls**

- `register_initialization_callback`
- `register_for_hourly_stats`
- `register_after_get_initial_actuals_model_for_sced_callback`
- `register_after_get_initial_actuals_model_for_simulation_actuals_callback`
- `register_after_get_initial_forecast_model_for_ruc_callback`
- `register_before_ruc_solve_callback`
- `register_after_ruc_generation_callback`
- `register_before_operations_solve_callback`
- `register_after_operations_callback`

- register_update_operations_stats_callback
- register_after_ruc_activation_callback
- register_finalization_callback
- ConfigDict
- declare_as_argument
- PrescientPluginModule
- fetch_hourly_stats_from_prescient
- fetch_day_ahead_stats_from_prescient
- keys
- _update_bids
- len
- _assemble_sced_tracking_market_signals
- _clone_tracking_model
- range
- _update_static_params
- compute_day_ahead_bids

### 1.3.4 DoubleLoopCoordinator.__init__

**Kind** method

**Defined in file** `coordinator.py` at line 35

**Signature** `def __init__(self, bidder, tracker, projection_tracker):`

**Summary** Initializes the DoubleLoopCoordinator object and registers functionalities

### 1.3.5 DoubleLoopCoordinator.register_plugins

**Kind** method

**Defined in file** `coordinator.py` at line 58

**Signature** `def register_plugins(self, context, options, plugin_config):`

**Summary** Register functionalities in Prescient's plugin system.

**Calls**

- `register_initialization_callback`
- `register_for_hourly_stats`
- `register_after_get_initial_actuals_model_for_sced_callback`
- `register_after_get_initial_actuals_model_for_simulation_actuals_callback`
- `register_after_get_initial_forecast_model_for_ruc_callback`
- `register_before_ruc_solve_callback`
- `register_after_ruc_generation_callback`
- `register_before_operations_solve_callback`
- `register_after_operations_callback`
- `register_update_operations_stats_callback`
- `register_after_ruc_activation_callback`
- `register_finalization_callback`

### 1.3.6 DoubleLoopCoordinator.get_configuration

**Kind** method

**Defined in file** `coordinator.py` at line 100

**Signature** `def get_configuration(self, key):`

**Summary** Register customized commandline options.

**Calls**

- `ConfigDict`
- `declare_as_argument`
- `declare`
- `ConfigValue`

### 1.3.7 DoubleLoopCoordinator.prescient_plugin_module

**Kind** method

**Defined in file** `coordinator.py` at line 126

**Signature** `def prescient_plugin_module(self):`

**Summary** No docstring.

**Calls**

- PrescientPluginModule

### 1.3.8 DoubleLoopCoordinator.initialize_customized_results

**Kind** method

**Defined in file** `coordinator.py` at line 129

**Signature** `def initialize_customized_results(self, options, simulator):`

**Summary** This method is outdated.

### 1.3.9 DoubleLoopCoordinator.push_hourly_stats_to_forecaster

**Kind** method

**Defined in file** `coordinator.py` at line 147

**Signature** `def push_hourly_stats_to_forecaster(self, prescient_hourly_stats):`

**Summary** This method pushes the hourly stats from Prescient to the price forecaster

**Calls**

- `fetch_hourly_stats_from_prescient`

### 1.3.10 DoubleLoopCoordinator.push_day_ahead_stats_to_forecaster

**Kind** method

**Defined in file** `coordinator.py` at line 161

**Signature** `def push_day_ahead_stats_to_forecaster(`

**Summary** This method pushes the day-ahead market to the price forecaster after the

**Calls**

- `fetch_day_ahead_stats_from_prescient`

### 1.3.11 DoubleLoopCoordinator._update_bids

**Kind** method

**Defined in file** `coordinator.py` at line 187

**Signature** `def _update_bids(self, gen_dict, bids, start_hour, horizon):`

**Summary** This method takes bids and pass the information in the bids to generator

**Calls**

- `keys`
- `update_func`
- `pop`
- `get`
- `range`

### 1.3.12 DoubleLoopCoordinator._update_bids._update_p_cost

**Kind** method

**Defined in file** `coordinator.py` at line 208

**Signature** `def _update_p_cost(gen_dict, bids, param_name, start_hour, horizon):`

**Summary** No docstring.

**Calls**

- `pop`
- `range`

### 1.3.13 DoubleLoopCoordinator._update_bids._update_time_series_params

**Kind** method

**Defined in file** `coordinator.py` at line 229

**Signature** `def _update_time_series_params(gen_dict, bids, param_name, start_hour, horizon):`

**Summary** No docstring.

**Calls**

- `get`
- `range`

### 1.3.14 DoubleLoopCoordinator._update_bids._update_non_time_series_params

**Kind** method

**Defined in file** `coordinator.py` at line 243

**Signature** `def _update_non_time_series_params(`

**Summary** No docstring.

**Calls**

- `get`

### 1.3.15 DoubleLoopCoordinator._pass_DA_bid_to_prescient

**Kind** method

**Defined in file** `coordinator.py` at line 272

**Signature** `def _pass_DA_bid_to_prescient(self, options, ruc_instance, bids):`

**Summary** This method passes the bids into the RUC model for day-ahead market clearing.

**Calls**

- `_update_bids`

### 1.3.16 DoubleLoopCoordinator.assemble_project_tracking_signal

**Kind** method

**Defined in file** `coordinator.py` at line 296

**Signature** `def assemble_project_tracking_signal(self, options, simulator, hour):`

**Summary** This function assembles the signals for the tracking model to estimate the

**Calls**

- `len`
- `_assemble_sced_tracking_market_signals`

### 1.3.17 DoubleLoopCoordinator.project_tracking_trajectory

**Kind** method

**Defined in file** `coordinator.py` at line 321

**Signature** `def project_tracking_trajectory(self, options, simulator, ruc_hour):`

**Summary** This function projects the full power dispatch trajectory from the

**Calls**

- `_clone_tracking_model`
- `range`
- `assemble_project_tracking_signal`
- `track_market_dispatch`
- `get`

### 1.3.18 DoubleLoopCoordinator._clone_tracking_model

**Kind** method

**Defined in file** `coordinator.py` at line 366

**Signature** `def _clone_tracking_model(self):`

**Summary** Clone the model in tracker and replace that of projection tracker. In this

**Calls**

- `zip_longest`
- `component_objects`
- `index_set`
- `ValueError`
- `value`
- `round`

### 1.3.19 DoubleLoopCoordinator._update_static_params

**Kind** method

**Defined in file** `coordinator.py` at line 399

**Signature** `def _update_static_params(self, gen_dict):`

**Summary** Update static parameters in the Prescient generator parameter data dictionary depending on generator type.

**Calls**

- `isinstance`
- `convert_marginal_costs_to_actual_costs`
- `NotImplementedError`

### 1.3.20 DoubleLoopCoordinator.update_static_params

**Kind** method

**Defined in file** `coordinator.py` at line 448

**Signature** `def update_static_params(self, options, instance):`

**Summary** This method pass static generator parameters in Prescient

**Calls**

- `_update_static_params`

### 1.3.21 DoubleLoopCoordinator.bid_into_DAM

**Kind** method

**Defined in file** `coordinator.py` at line 468

**Signature** `def bid_into_DAM(self, options, simulator, ruc_instance, ruc_date, ruc_hour):`

**Summary** This function uses the bidding objects to bid into the day-ahead market

**Calls**

- `compute_day_ahead_bids`
- `_pass_DA_bid_to_prescient`
- `project_tracking_trajectory`
- `update_day_ahead_model`

### 1.3.22 DoubleLoopCoordinator.fetch_DA_prices

**Kind** method

**Defined in file** `coordinator.py` at line 512

**Signature** `def fetch_DA_prices(self, options, simulator, result, uc_date, uc_hour):`

**Summary** This method fetches the day-ahead market prices from unit commitment results,

**Calls**

- `get`
- `range`

### 1.3.23 DoubleLoopCoordinator.fetch_DA_dispatches

**Kind** method

**Defined in file** `coordinator.py` at line 545

**Signature** `def fetch_DA_dispatches(self, options, simulator, result, uc_date, uc_hour):`

**Summary** This method fetches the day-ahead dispatches from unit commitment results,

**Calls**

- `get`
- `range`

### 1.3.24 DoubleLoopCoordinator._pass_RT_bid_to_prescient

**Kind** method

**Defined in file** `coordinator.py` at line 587

**Signature** `def _pass_RT_bid_to_prescient(self, options, simulator, sced_instance, bids, hour):`

**Summary** This method passes the bids into the SCED model for real-time market

**Calls**

- `_update_bids`

### 1.3.25 DoubleLoopCoordinator.bid_into_RTM

**Kind** method

**Defined in file** `coordinator.py` at line 616

**Signature** `def bid_into_RTM(self, options, simulator, sced_instance):`

**Summary** This function bids into the real-time market. At this moment I just copy the

**Calls**

- `compute_real_time_bids`
- `_pass_RT_bid_to_prescient`

### 1.3.26 DoubleLoopCoordinator.assemble_sced_tracking_market_signals

**Kind** method

**Defined in file** `coordinator.py` at line 648

**Signature** `def assemble_sced_tracking_market_signals(`

**Summary** This function assembles the signals for the tracking model.

**Calls**

- `len`
- `_assemble_sced_tracking_market_signals`

### 1.3.27 DoubleLoopCoordinator._assemble_sced_tracking_market_signals

**Kind** method

**Defined in file** `coordinator.py` at line 683

**Signature** `def _assemble_sced_tracking_market_signals(`

**Summary** This function assembles the signals for the tracking model.

**Calls**

- `append`
- `range`
- `len`

### 1.3.28 DoubleLoopCoordinator.track_sced_signal

**Kind** method

**Defined in file** `coordinator.py` at line 718

**Signature** `def track_sced_signal(self, options, simulator, sced_instance, lmp_sced):`

**Summary** This methods uses the tracking object to track the current real-time market

**Calls**

- `assemble_sced_tracking_market_signals`
- `track_market_dispatch`
- `update_model`
- `update_real_time_model`

### 1.3.29 DoubleLoopCoordinator.update_observed_dispatch

**Kind** method

**Defined in file** `coordinator.py` at line 759

**Signature** `def update_observed_dispatch(self, options, simulator, ops_stats):`

**Summary** This methods extract the actual power delivered by the tracking model and

**Calls**

- `get_last_delivered_power`

### 1.3.30 DoubleLoopCoordinator.activate_pending_DA_data

**Kind** method

**Defined in file** `coordinator.py` at line 791

**Signature** `def activate_pending_DA_data(self, options, simulator):`

**Summary** This function puts the day-ahead data computed in the day before into effect,

### 1.3.31 DoubleLoopCoordinator.write_plugin_results

**Kind** method

**Defined in file** `coordinator.py` at line 816

**Signature** `def write_plugin_results(self, options, simulator):`

**Summary** After the simulation is completed, the plugins can write their own customized

**Calls**

- `write_results`

## 1.4 forecaster.py

**File path** `forecaster.py`

**Definitions Index**


- **class**: `ForecastError` (line 22)
- **class**: `AbstractPriceForecaster` (line 26)
- **method**: `AbstractPriceForecaster.forecast_day_ahead_and_real_time_prices` (line 32)
- **method**: `AbstractPriceForecaster.forecast_real_time_prices` (line 57)
- **method**: `AbstractPriceForecaster.forecast_day_ahead_prices` (line 78)
- **class**: `AbstractPrescientPriceForecaster` (line 99)
- **method**: `AbstractPrescientPriceForecaster.fetch_hourly_stats_from_prescient` (line 105)
- **method**: `AbstractPrescientPriceForecaster.fetch_day_ahead_stats_from_prescient` (line 118)
- **class**: `PlaceHolderForecaster` (line 135)
- **method**: `PlaceHolderForecaster.__init__` (line 142)
- **method**: `PlaceHolderForecaster.forecast_day_ahead_and_real_time_prices` (line 170)
- **method**: `PlaceHolderForecaster.forecast_day_ahead_prices` (line 203)
- **method**: `PlaceHolderForecaster.forecast_real_time_prices` (line 231)
- **method**: `PlaceHolderForecaster._forecast` (line 259)
- **method**: `PlaceHolderForecaster.fetch_hourly_stats_from_prescient` (line 288)
- **method**: `PlaceHolderForecaster.fetch_day_ahead_stats_from_prescient` (line 302)
- **class**: `Backcaster` (line 321)
- **method**: `Backcaster.__init__` (line 326)
- **method**: `Backcaster._validate_input_historical_price` (line 348)
- **method**: `Backcaster.max_historical_days` (line 397)
- **method**: `Backcaster.max_historical_days` (line 408)
- **method**: `Backcaster.historical_da_prices` (line 434)
- **method**: `Backcaster.historical_da_prices` (line 445)
- **method**: `Backcaster.historical_rt_prices` (line 460)
- **method**: `Backcaster.historical_rt_prices` (line 471)
- **method**: `Backcaster.forecast_day_ahead_and_real_time_prices` (line 485)
- **method**: `Backcaster.forecast_real_time_prices` (line 518)
- **method**: `Backcaster.forecast_day_ahead_prices` (line 548)

- **method**: `Backcaster._forecast` (line 578)
- **method**: `Backcaster.fetch_hourly_stats_from_prescient` (line 623)
- **method**: `Backcaster.fetch_day_ahead_stats_from_prescient` (line 654)
- **class**: `PerfectForecaster` (line 686)
- **method**: `PerfectForecaster.__init__` (line 692)
- **method**: `PerfectForecaster.__getitem__` (line 717)
- **method**: `PerfectForecaster.fetch_hourly_stats_from_prescient` (line 720)
- **method**: `PerfectForecaster.fetch_day_ahead_stats_from_prescient` (line 727)
- **method**: `PerfectForecaster.forecast_day_ahead_and_real_time_prices` (line 734)
- **method**: `PerfectForecaster.get_column_from_data` (line 762)
- **method**: `PerfectForecaster.forecast_day_ahead_prices` (line 784)
- **method**: `PerfectForecaster.forecast_real_time_prices` (line 787)
- **method**: `PerfectForecaster.forecast_day_ahead_capacity_factor` (line 790)
- **method**: `PerfectForecaster.forecast_real_time_capacity_factor` (line 793)

**Called functions/methods in this file**

- line 19: `getLogger`
- line 194: `forecast_real_time_prices`
- line 197: `forecast_day_ahead_prices`
- line 223: `_forecast`
- line 278: `range`
- line 281: `normal`
- line 286: `list`
- line 359: `isinstance`
- line 360: `TypeError`
- line 364: `len`
- line 365: `ValueError`
- line 367: `items`
- line 387: `warning`
- line 421: `type`

- line 424: `int`
- line 456: `_validate_input_historical_price`
- line 606: `ForecastError`
- line 638: `append`
- line 675: `get`
- line 707: `read_csv`
- line 777: `to_datetime`
- line 777: `Timedelta`
- line 778: `head`
- line 785: `get_column_from_data`

**Definition Notes**

### 1.4.1 ForecastError

**Kind** class

**Defined in file** `forecaster.py` at line 22

**Signature** `class ForecastError(Exception):`

**Summary** Error to indicate error with forecasters.

### 1.4.2 AbstractPriceForecaster

**Kind** class

**Defined in file** `forecaster.py` at line 26

**Signature** `class AbstractPriceForecaster(ABC):`

**Summary** The abstract class for price forecaster.

### 1.4.3 AbstractPriceForecaster.forecast_day_ahead_and_real_time_prices

**Kind** method

**Defined in file** `forecaster.py` at line 32

**Signature** `def forecast_day_ahead_and_real_time_prices(`

**Summary** Forecast both day-ahead and real-time market prices.

### 1.4.4 AbstractPriceForecaster.forecast_real_time_prices

**Kind** method

**Defined in file** `forecaster.py` at line 57

**Signature** `def forecast_real_time_prices(self, date, hour, bus, horizon, n_samples):`

**Summary** Forecast real-time market prices.

### 1.4.5 AbstractPriceForecaster.forecast_day_ahead_prices

**Kind** method

**Defined in file** `forecaster.py` at line 78

**Signature** `def forecast_day_ahead_prices(self, date, hour, bus, horizon, n_samples):`

**Summary** Forecast day-ahead market prices.

### 1.4.6 AbstractPrescientPriceForecaster

**Kind** class

**Defined in file** `forecaster.py` at line 99

**Signature** `class AbstractPrescientPriceForecaster(AbstractPriceForecaster):`

**Summary** Abstract class for price forecasters that will interface with Prescient.

### 1.4.7 AbstractPrescientPriceForecaster.fetch_hourly_stats_from_prescient

**Kind** method

**Defined in file** `forecaster.py` at line 105

**Signature** `def fetch_hourly_stats_from_prescient(self, prescient_hourly_stats):`

**Summary** This method fetches the hourly stats from Prescient to the price forecaster

### 1.4.8 AbstractPrescientPriceForecaster.fetch_day_ahead_stats_from_prescient

**Kind** method

**Defined in file** `forecaster.py` at line 118

**Signature** `def fetch_day_ahead_stats_from_prescient(self, uc_date, uc_hour, day_ahead_result):`

**Summary** This method fetches the day-ahead market to the price forecaster after the

### 1.4.9 PlaceHolderForecaster

**Kind** class

**Defined in file** `forecaster.py` at line 135

**Signature** `class PlaceHolderForecaster(AbstractPrescientPriceForecaster):`

**Summary** This a placeholder for a real price forecaster. This placeholder can takes

**Calls**

- `forecast_real_time_prices`
- `forecast_day_ahead_prices`
- `_forecast`
- `normal`
- `list`
- `range`

### 1.4.10 PlaceHolderForecaster.__init__

**Kind** method

**Defined in file** `forecaster.py` at line 142

**Signature** `def __init__(`

**Summary** Initialize the PlaceHolderForecaster.

### 1.4.11 PlaceHolderForecaster.forecast_day_ahead_and_real_time_prices

**Kind** method

**Defined in file** `forecaster.py` at line 170

**Signature** `def forecast_day_ahead_and_real_time_prices(`

**Summary** Forecast both day-ahead and real-time market prices.

**Calls**

- `forecast_real_time_prices`
- `forecast_day_ahead_prices`

### 1.4.12 PlaceHolderForecaster.forecast_day_ahead_prices

**Kind** method

**Defined in file** `forecaster.py` at line 203

**Signature** `def forecast_day_ahead_prices(self, date, hour, bus, horizon, n_samples):`

**Summary** Forecast day-ahead market prices.

**Calls**

- `_forecast`

### 1.4.13 PlaceHolderForecaster.forecast_real_time_prices

**Kind** method

**Defined in file** `forecaster.py` at line 231

**Signature** `def forecast_real_time_prices(self, date, hour, bus, horizon, n_samples):`

**Summary** Forecast real-time market prices.

**Calls**

- `_forecast`

### 1.4.14 PlaceHolderForecaster._forecast

**Kind** method

**Defined in file** `forecaster.py` at line 259

**Signature** `def _forecast(self, means, stds, hour, horizon, n_samples):`

**Summary** Generate price forecasts.

**Calls**

- `normal`
- `list`
- `range`

### 1.4.15 PlaceHolderForecaster.fetch_hourly_stats_from_prescient

**Kind** method

**Defined in file** `forecaster.py` at line 288

**Signature** `def fetch_hourly_stats_from_prescient(self, prescient_hourly_stats):`

**Summary** This method fetches the hourly stats from Prescient to the price forecaster

### 1.4.16 PlaceHolderForecaster.fetch_day_ahead_stats_from_prescient

**Kind** method

**Defined in file** `forecaster.py` at line 302

**Signature** `def fetch_day_ahead_stats_from_prescient(self, uc_date, uc_hour, day_ahead_result):`

**Summary** This method fetches the day-ahead market to the price forecaster after the

### 1.4.17 Backcaster

**Kind** class

**Defined in file** `forecaster.py` at line 321

**Signature** `class Backcaster(AbstractPrescientPriceForecaster):`

**Summary** Generate price forecasts by directly using historical prices.

**Calls**

- `items`
- `int`
- `_validate_input_historical_price`
- `forecast_real_time_prices`
- `forecast_day_ahead_prices`
- `_forecast`
- `len`
- `range`
- `isinstance`
- `TypeError`
- `ValueError`
- `ForecastError`

- `append`
- `warning`
- `get`
- `type`

### 1.4.18 Backcaster.__init__

**Kind** method

**Defined in file** `forecaster.py` at line 326

**Signature** `def __init__(`

**Summary** Initialize the Backcaster.

### 1.4.19 Backcaster._validate_input_historical_price

**Kind** method

**Defined in file** `forecaster.py` at line 348

**Signature** `def _validate_input_historical_price(self, historical_price):`

**Summary** Validate input historical prices.

**Calls**

- `items`
- `isinstance`
- `TypeError`
- `len`
- `ValueError`
- `warning`

### 1.4.20 Backcaster.max_historical_days

**Kind** method

**Defined in file** `forecaster.py` at line 397

**Signature** `def max_historical_days(self):`

**Summary** Property getter for max_historical_days.

### 1.4.21 Backcaster.max_historical_days

**Kind** method

**Defined in file** `forecaster.py` at line 408

**Signature** `def max_historical_days(self, value):`

**Summary** Property setter for max_historical_days (validate before setting).

**Calls**

- `int`
- `isinstance`
- `TypeError`
- `ValueError`
- `type`

### 1.4.22 Backcaster.historical_da_prices

**Kind** method

**Defined in file** `forecaster.py` at line 434

**Signature** `def historical_da_prices(self):`

**Summary** Property getter for historical_da_prices.

### 1.4.23 Backcaster.historical_da_prices

**Kind** method

**Defined in file** `forecaster.py` at line 445

**Signature** `def historical_da_prices(self, value):`

**Summary** Property setter for historical_da_prices (validate before setting).

**Calls**

- `_validate_input_historical_price`

### 1.4.24 Backcaster.historical_rt_prices

**Kind** method

**Defined in file** `forecaster.py` at line 460

**Signature** `def historical_rt_prices(self):`

**Summary** Property getter for historical_rt_prices.

### 1.4.25 Backcaster.historical_rt_prices

**Kind** method

**Defined in file** `forecaster.py` at line 471

**Signature** `def historical_rt_prices(self, value):`

**Summary** Property setter for historical_rt_prices (validate before setting).

**Calls**

- `_validate_input_historical_price`

### 1.4.26 Backcaster.forecast_day_ahead_and_real_time_prices

**Kind** method

**Defined in file** `forecaster.py` at line 485

**Signature** `def forecast_day_ahead_and_real_time_prices(`

**Summary** Forecast both day-ahead and real-time market prices.

**Calls**

- `forecast_real_time_prices`
- `forecast_day_ahead_prices`

### 1.4.27 Backcaster.forecast_real_time_prices

**Kind** method

**Defined in file** `forecaster.py` at line 518

**Signature** `def forecast_real_time_prices(self, date, hour, bus, horizon, n_samples):`

**Summary** Forecast real-time market prices.

**Calls**

- `_forecast`

### 1.4.28 Backcaster.forecast_day_ahead_prices

**Kind** method

**Defined in file** `forecaster.py` at line 548

**Signature** `def forecast_day_ahead_prices(self, date, hour, bus, horizon, n_samples):`

**Summary** Forecast day-ahead market prices.

**Calls**

- `_forecast`

### 1.4.29 Backcaster._forecast

**Kind** method

**Defined in file** `forecaster.py` at line 578

**Signature** `def _forecast(`

**Summary** Forecast energy market prices using historical prices.

**Calls**

- `len`
- `range`
- `ForecastError`

### 1.4.30 Backcaster.fetch_hourly_stats_from_prescient

**Kind** method

**Defined in file** `forecaster.py` at line 623

**Signature** `def fetch_hourly_stats_from_prescient(self, prescient_hourly_stats):`

**Summary** This method fetches the hourly real-time prices from Prescient and store

**Calls**

- `items`
- `append`
- `len`

### 1.4.31 Backcaster.fetch_day_ahead_stats_from_prescient

**Kind** method

**Defined in file** `forecaster.py` at line 654

**Signature** `def fetch_day_ahead_stats_from_prescient(self, uc_date, uc_hour, day_ahead_result):`

**Summary** This method fetches the hourly day-ahead prices from Prescient and store

**Calls**

- `get`
- `range`
- `len`

### 1.4.32 PerfectForecaster

**Kind** class

**Defined in file** `forecaster.py` at line 686

**Signature** `class PerfectForecaster(AbstractPrescientPriceForecaster):`

**Summary** PerfectForecaster is designed for the renewable-PEM Parameterized Bidder.

**Calls**

- `isinstance`
- `forecast_real_time_prices`
- `forecast_day_ahead_prices`
- `head`
- `get_column_from_data`
- `read_csv`
- `to_datetime`
- `Timedelta`
- `len`
- `append`
- `ValueError`

### 1.4.33 PerfectForecaster.__init__

**Kind** method

**Defined in file** `forecaster.py` at line 692

**Signature** `def __init__(self, data_path_or_df):`

**Summary** Perfect forecaster that reads the data from a Dataframe containing:

**Calls**

- `isinstance`
- `read_csv`
- `ValueError`

### 1.4.34 PerfectForecaster.__getitem__

**Kind** method

**Defined in file** `forecaster.py` at line 717

**Signature** `def __getitem__(self, index):`

**Summary** No docstring.

### 1.4.35 PerfectForecaster.fetch_hourly_stats_from_prescient

**Kind** method

**Defined in file** `forecaster.py` at line 720

**Signature** `def fetch_hourly_stats_from_prescient(self, prescient_hourly_stats):`

**Summary** Fetch the hourly stats from prescient.

### 1.4.36 PerfectForecaster.fetch_day_ahead_stats_from_prescient

**Kind** method

**Defined in file** `forecaster.py` at line 727

**Signature** `def fetch_day_ahead_stats_from_prescient(self, uc_date, uc_hour, day_ahead_result):`

**Summary** Fetch the day-ahead stats from prescient.

### 1.4.37 PerfectForecaster.forecast_day_ahead_and_real_time_prices

**Kind** method

**Defined in file** `forecaster.py` at line 734

**Signature** `def forecast_day_ahead_and_real_time_prices(`

**Summary** Forecast day-ahead and real-time prices.

**Calls**

- `forecast_real_time_prices`
- `forecast_day_ahead_prices`

### 1.4.38 PerfectForecaster.get_column_from_data

**Kind** method

**Defined in file** `forecaster.py` at line 762

**Signature** `def get_column_from_data(self, date: str, hour: int, horizon: int, col: str):`

**Summary** Get the data from the dataframe.

**Calls**

- `head`
- `to_datetime`
- `Timedelta`
- `len`
- `append`

### 1.4.39 PerfectForecaster.forecast_day_ahead_prices

**Kind** method

**Defined in file** `forecaster.py` at line 784

**Signature** `def forecast_day_ahead_prices(self, date, hour, bus, horizon, _):`

**Summary** No docstring.

**Calls**

- get_column_from_data

### 1.4.40 PerfectForecaster.forecast_real_time_prices

**Kind** method

**Defined in file** `forecaster.py` at line 787

**Signature** `def forecast_real_time_prices(self, date, hour, bus, horizon, _):`

**Summary** No docstring.

**Calls**

- get_column_from_data

### 1.4.41 PerfectForecaster.forecast_day_ahead_capacity_factor

**Kind** method

**Defined in file** `forecaster.py` at line 790

**Signature** `def forecast_day_ahead_capacity_factor(self, date, hour, gen, horizon):`

**Summary** No docstring.

**Calls**

- get_column_from_data

### 1.4.42 PerfectForecaster.forecast_real_time_capacity_factor

**Kind** method

**Defined in file** `forecaster.py` at line 793

**Signature** `def forecast_real_time_capacity_factor(self, date, hour, gen, horizon):`

**Summary** No docstring.

**Calls**

- get_column_from_data

## 1.5 model_data.py

**File path** `model_data.py`

**Definitions Index**


- **class**: `BaseValidator` (line 19)
- **method**: `BaseValidator.__set_name__` (line 24)
- **method**: `BaseValidator.__set__` (line 27)
- **method**: `BaseValidator.__get__` (line 32)
- **method**: `BaseValidator._validate` (line 39)
- **class**: `StrValidator` (line 49)
- **method**: `StrValidator._validate` (line 50)
- **class**: `RealValueValidator` (line 58)
- **method**: `RealValueValidator.__init__` (line 64)
- **method**: `RealValueValidator._validate` (line 68)
- **class**: `AtLeastPminValidator` (line 97)
- **method**: `AtLeastPminValidator._validate` (line 104)
- **class**: `GeneratorModelData` (line 129)
- **method**: `GeneratorModelData.__init__` (line 140)
- **method**: `GeneratorModelData.__iter__` (line 154)
- **method**: `GeneratorModelData.__next__` (line 171)
- **class**: `ThermalGeneratorModelData` (line 185)
- **method**: `ThermalGeneratorModelData.__init__` (line 203)
- **method**: `ThermalGeneratorModelData._check_empty_and_sort_cost_pairs` (line 245)
- **method**: `ThermalGeneratorModelData._assemble_default_cost_bids` (line 267)
- **method**: `ThermalGeneratorModelData._assemble_default_startup_cost_bids` (line 303)
- **method**: `ThermalGeneratorModelData.generator_type` (line 327)
- **method**: `ThermalGeneratorModelData.initial_status` (line 334)
- **method**: `ThermalGeneratorModelData.initial_status` (line 345)
- **method**: `ThermalGeneratorModelData.initial_p_output` (line 359)
- **method**: `ThermalGeneratorModelData.initial_p_output` (line 366)
- **class**: `RenewableGeneratorModelData` (line 391)
- **method**: `RenewableGeneratorModelData.__init__` (line 398)
- **method**: `RenewableGeneratorModelData.generator_type` (line 410)

**Called functions/methods in this file**

- line 29: `_validate`
- line 36: `get`
- line 52: `isinstance`
- line 53: `TypeError`
- line 54: `type`
- line 83: `isclose`
- line 85: `ValueError`
- line 113: `getattr`
- line 116: `RuntimeError`
- line 134: `StrValidator`
- line 137: `RealValueValidator`
- line 138: `AtLeastPminValidator`
- line 159: `hasattr`
- line 162: `dir`
- line 163: `startswith`
- line 165: `callable`
- line 177: `len`
- line 224: `__init__`
- line 224: `super`
- line 242: `_assemble_default_cost_bids`
- line 243: `_assemble_default_startup_cost_bids`
- line 263: `sort`
- line 279: `_check_empty_and_sort_cost_pairs`

**Definition Notes**

### 1.5.1 BaseValidator

**Kind** class

**Defined in file** `model_data.py` at line 19

**Signature** `class BaseValidator(ABC):`

**Summary** A base data descriptor that validate values before store them.

**Calls**

- `_validate`
- `get`

### 1.5.2 BaseValidator.__set_name__

**Kind** method

**Defined in file** `model_data.py` at line 24

**Signature** `def __set_name__(self, cls, prop_name):`

**Summary** No docstring.

### 1.5.3 BaseValidator.__set__

**Kind** method

**Defined in file** `model_data.py` at line 27

**Signature** `def __set__(self, instance, value):`

**Summary** No docstring.

**Calls**

- `_validate`

### 1.5.4 BaseValidator.__get__

**Kind** method

**Defined in file** `model_data.py` at line 32

**Signature** `def __get__(self, instance, cls):`

**Summary** No docstring.

**Calls**

- `get`

### 1.5.5 BaseValidator._validate

**Kind** method

**Defined in file** `model_data.py` at line 39

**Signature** `def _validate(self, instance, value):`

**Summary** Validate the provided value.

### 1.5.6 StrValidator

**Kind** class

**Defined in file** `model_data.py` at line 49

**Signature** `class StrValidator(BaseValidator):`

**Summary** No docstring.

**Calls**

- `isinstance`
- `TypeError`
- `type`

### 1.5.7 StrValidator._validate

**Kind** method

**Defined in file** `model_data.py` at line 50

**Signature** `def _validate(self, instance, value):`

**Summary** No docstring.

**Calls**

- `isinstance`
- `TypeError`
- `type`

### 1.5.8 RealValueValidator

**Kind** class

**Defined in file** `model_data.py` at line 58

**Signature** `class RealValueValidator(BaseValidator):`

**Summary** A data descriptor that validate whether a value is a real number and whether

**Calls**

- `isinstance`
- `TypeError`
- `ValueError`
- `isclose`

### 1.5.9 RealValueValidator.__init__

**Kind** method

**Defined in file** `model_data.py` at line 64

**Signature** `def __init__(self, min_val=None, max_val=None):`

**Summary** No docstring.

### 1.5.10 RealValueValidator._validate

**Kind** method

**Defined in file** `model_data.py` at line 68

**Signature** `def _validate(self, instance, value):`

**Summary** Validate whether the provided value is within bounds and is a real number.

**Calls**

- `isinstance`
- `TypeError`
- `ValueError`
- `isclose`

### 1.5.11 AtLeastPminValidator

**Kind** class

**Defined in file** `model_data.py` at line 97

**Signature** `class AtLeastPminValidator(BaseValidator):`

**Summary** A data descriptor that validate whether a value is greater or equal to the

**Calls**

- `getattr`
- `RuntimeError`
- `isinstance`
- `TypeError`
- `ValueError`
- `isclose`

### 1.5.12 AtLeastPminValidator._validate

**Kind** method

**Defined in file** `model_data.py` at line 104

**Signature** `def _validate(self, instance, value):`

**Summary** Validate whether the provided value is at least Pmin.

**Calls**

- `getattr`
- `RuntimeError`
- `isinstance`
- `TypeError`
- `ValueError`
- `isclose`

### 1.5.13 GeneratorModelData

**Kind** class

**Defined in file** `model_data.py` at line 129

**Signature** `class GeneratorModelData:`

**Summary** A class that holds data for generator parameters.

**Calls**

- `StrValidator`
- `RealValueValidator`
- `AtLeastPminValidator`
- `ValueError`
- `hasattr`
- `len`
- `getattr`
- `dir`
- `startswith`
- `callable`

### 1.5.14 GeneratorModelData.__init__

**Kind** method

**Defined in file** `model_data.py` at line 140

**Signature** `def __init__(self, gen_name, bus, p_min, p_max, fixed_commitment=None):`

**Summary** No docstring.

**Calls**

- `ValueError`

### 1.5.15 GeneratorModelData.__iter__

**Kind** method

**Defined in file** `model_data.py` at line 154

**Signature** `def __iter__(self):`

**Summary** Make it iteratble.

**Calls**

- `hasattr`
- `dir`
- `startswith`
- `callable`
- `getattr`

### 1.5.16 GeneratorModelData.__next__

**Kind** method

**Defined in file** `model_data.py` at line 171

**Signature** `def __next__(self):`

**Summary** Implement iterator method.

**Calls**

- `len`
- `getattr`

### 1.5.17 ThermalGeneratorModelData

**Kind** class

**Defined in file** `model_data.py` at line 185

**Signature** `class ThermalGeneratorModelData(GeneratorModelData):`

**Summary** Adds run time, ramping, shutdown and startup information for thermal generators.

**Calls**

- `RealValueValidator`
- `AtLeastPminValidator`
- `__init__`
- `_assemble_default_cost_bids`
- `_assemble_default_startup_cost_bids`
- `sort`

- _check_empty_and_sort_cost_pairs
- isclose
- ValueError
- len
- isinstance
- TypeError
- super

### 1.5.18 ThermalGeneratorModelData.__init__

**Kind** method

**Defined in file** model_data.py at line 203

**Signature** def __init__(

**Summary** No docstring.

**Calls**

- __init__
- _assemble_default_cost_bids
- _assemble_default_startup_cost_bids
- super

### 1.5.19 ThermalGeneratorModelData._check_empty_and_sort_cost_pairs

**Kind** method

**Defined in file** model_data.py at line 245

**Signature** def _check_empty_and_sort_cost_pairs(self, pair_description, pairs):

**Summary** Check whether a list of pairs for production and startup costs is empty,

**Calls**

- sort
- ValueError
- len

### 1.5.20 ThermalGeneratorModelData._assemble_default_cost_bids

**Kind** method

**Defined in file** `model_data.py` at line 267

**Signature** `def _assemble_default_cost_bids(self, production_cost_bid_pairs):`

**Summary** Assemble the default production cost bids.

**Calls**

- `_check_empty_and_sort_cost_pairs`
- `ValueError`
- `len`

### 1.5.21 ThermalGeneratorModelData._assemble_default_startup_cost_bids

**Kind** method

**Defined in file** `model_data.py` at line 303

**Signature** `def _assemble_default_startup_cost_bids(self, startup_cost_pairs):`

**Summary** Assemble the default startup cost bids.

**Calls**

- `_check_empty_and_sort_cost_pairs`
- `ValueError`

### 1.5.22 ThermalGeneratorModelData.generator_type

**Kind** method

**Defined in file** `model_data.py` at line 327

**Signature** `def generator_type(self):`

**Summary** Generator type property: fixed to thermal.

### 1.5.23 ThermalGeneratorModelData.initial_status

**Kind** method

**Defined in file** `model_data.py` at line 334

**Signature** `def initial_status(self):`

**Summary** Generator initial status property. If positive, the number of hours prior

### 1.5.24 ThermalGeneratorModelData.initial_status

**Kind** method

**Defined in file** `model_data.py` at line 345

**Signature** `def initial_status(self, value):`

**Summary** Generator initial status property setter. Validate the value before setting.

**Calls**

- `isclose`
- `isinstance`
- `TypeError`
- `ValueError`

### 1.5.25 ThermalGeneratorModelData.initial_p_output

**Kind** method

**Defined in file** `model_data.py` at line 359

**Signature** `def initial_p_output(self):`

**Summary** Generator initial power output property.

### 1.5.26 ThermalGeneratorModelData.initial_p_output

**Kind** method

**Defined in file** `model_data.py` at line 366

**Signature** `def initial_p_output(self, value):`

**Summary** Generator initial power output property setter.

**Calls**

- `isinstance`
- `TypeError`
- `ValueError`
- `isclose`

### 1.5.27 RenewableGeneratorModelData

**Kind** class

**Defined in file** `model_data.py` at line 391

**Signature** `class RenewableGeneratorModelData(GeneratorModelData):`

**Summary** Adds a single production cost for renewable generators.

**Calls**

- `RealValueValidator`
- `__init__`
- `super`

### 1.5.28 RenewableGeneratorModelData.__init__

**Kind** method

**Defined in file** `model_data.py` at line 398

**Signature** `def __init__(self, gen_name, bus, p_min, p_max, p_cost, fixed_commitment=None):`

**Summary** No docstring.

**Calls**

- `__init__`
- `super`

### 1.5.29 RenewableGeneratorModelData.generator_type

**Kind** method

**Defined in file** `model_data.py` at line 410

**Signature** `def generator_type(self):`

**Summary** Generator type property: fixed to renewable.

## 1.6 tracker.py

**File path** `tracker.py`

**Definitions Index**

- **class**: `Tracker` (line 19)
- **method**: `Tracker.__init__` (line 25)
- **method**: `Tracker._check_inputs` (line 70)
- **method**: `Tracker._check_tracking_model_object` (line 79)
- **method**: `Tracker._check_n_tracking_hour` (line 117)
- **method**: `Tracker._check_solver` (line 135)
- **method**: `Tracker.formulate_tracking_problem` (line 147)
- **method**: `Tracker._add_tracking_vars` (line 166)
- **method**: `Tracker._add_tracking_params` (line 187)
- **method**: `Tracker._add_tracking_constraints` (line 219)
- **method**: `Tracker._add_tracking_constraints.tracking_dispatch_constraint_rule` (line 232)
- **method**: `Tracker._add_tracking_objective` (line 244)
- **method**: `Tracker.update_model` (line 270)
- **method**: `Tracker.track_market_dispatch` (line 283)
- **method**: `Tracker._record_daily_stats` (line 315)
- **method**: `Tracker._pass_market_dispatch` (line 338)
- **method**: `Tracker.get_last_delivered_power` (line 361)
- **method**: `Tracker._record_tracker_results` (line 375)
- **method**: `Tracker.record_results` (line 414)
- **method**: `Tracker.write_results` (line 432)

### Called functions/methods in this file

- line 49: `_check_inputs`
- line 52: `ConcreteModel`
- line 53: `Block`
- line 54: `populate_model`
- line 58: `getattr`
- line 61: `index_set`
- line 63: `formulate_tracking_problem`
- line 75: `_check_tracking_model_object`
- line 76: `_check_n_tracking_hour`
- line 77: `_check_solver`
- line 98: `AttributeError`
- line 123: `isinstance`
- line 124: `TypeError`
- line 126: `format`
- line 126: `type`
- line 130: `ValueError`
- line 159: `_add_tracking_params`
- line 160: `_add_tracking_vars`
- line 161: `_add_tracking_constraints`
- line 162: `_add_tracking_objective`
- line 178: `Var`
- line 200: `Param`
- line 238: `Constraint`
- line 257: `Objective`
- line 281: `update_model`
- line 299: `_pass_market_dispatch`
- line 302: `solve`
- line 304: `record_results`

- line 307: `get_implemented_profile`
- line 311: `_record_daily_stats`
- line 332: `values`
- line 333: `len`
- line 334: `popleft`
- line 354: `deactivate`
- line 357: `activate`
- line 371: `get_last_delivered_power`
- line 395: `int`
- line 397: `round`
- line 398: `value`
- line 408: `from_dict`
- line 409: `append`
- line 412: `concat`
- line 427: `_record_tracker_results`
- line 443: `print`
- line 446: `to_csv`
- line 447: `join`
- line 449: `write_results`

**Definition Notes**

### 1.6.1 Tracker

**Kind** class

**Defined in file** `tracker.py` at line 19

**Signature** `class Tracker:`

**Summary** Wrap a model object to track the market dispatch signals. This class interfaces

**Calls**

- _check_inputs
- ConcreteModel
- Block
- populate_model
- getattr
- index_set
- formulate_tracking_problem
- _check_tracking_model_object
- _check_n_tracking_hour
- _check_solver
- _add_tracking_params
- _add_tracking_vars
- _add_tracking_constraints
- _add_tracking_objective
- Var
- Param
- Constraint
- Objective
- update_model
- _pass_market_dispatch
- solve
- record_results
- get_implemented_profile
- _record_daily_stats
- values

### 1.6.2 Tracker.__init__

**Kind** method

**Defined in file** tracker.py at line 25

**Signature** `def __init__(`

**Summary** Initializes the tracker object.

**Calls**

- `_check_inputs`
- `ConcreteModel`
- `Block`
- `populate_model`
- `getattr`
- `index_set`
- `formulate_tracking_problem`

### 1.6.3 Tracker._check_inputs

**Kind** method

**Defined in file** `tracker.py` at line 70

**Signature** `def _check_inputs(self):`

**Summary** Check if the inputs to construct the tracker is valid. If not raise errors.

**Calls**

- `_check_tracking_model_object`
- `_check_n_tracking_hour`
- `_check_solver`

### 1.6.4 Tracker._check_tracking_model_object

**Kind** method

**Defined in file** `tracker.py` at line 79

**Signature** `def _check_tracking_model_object(self):`

**Summary** Check if tracking model object has the necessary methods and attributes.

**Calls**

- `getattr`
- `AttributeError`

### 1.6.5 Tracker._check_n_tracking_hour

**Kind** method

**Defined in file** `tracker.py` at line 117

**Signature** `def _check_n_tracking_hour(self):`

**Summary** Check if the number of hour for tracking is an integer and greater than 0.

**Calls**

- `isinstance`
- `TypeError`
- `ValueError`
- `format`
- `type`

### 1.6.6 Tracker._check_solver

**Kind** method

**Defined in file** `tracker.py` at line 135

**Signature** `def _check_solver(self):`

**Summary** Check if provides solver is a valid Pyomo solver object.

**Calls**

- `isinstance`
- `TypeError`
- `format`

### 1.6.7 Tracker.formulate_tracking_problem

**Kind** method

**Defined in file** `tracker.py` at line 147

**Signature** `def formulate_tracking_problem(self):`

**Summary** Formulate the tracking optimization problem by adding necessary

**Calls**

- `_add_tracking_params`
- `_add_tracking_vars`
- `_add_tracking_constraints`
- `_add_tracking_objective`

### 1.6.8 Tracker._add_tracking_vars

**Kind** method

**Defined in file** `tracker.py` at line 166

**Signature** `def _add_tracking_vars(self):`

**Summary** Add necessary tracking variables to the model, i.e., power under and over

**Calls**

- `Var`

### 1.6.9 Tracker._add_tracking_params

**Kind** method

**Defined in file** `tracker.py` at line 187

**Signature** `def _add_tracking_params(self):`

**Summary** Add necessary tracking parameters to the model, i.e., market dispatch

**Calls**

- `Param`

### 1.6.10 Tracker._add_tracking_constraints

**Kind** method

**Defined in file** `tracker.py` at line 219

**Signature** `def _add_tracking_constraints(self):`

**Summary** Add necessary tracking constraints to the model, e.g., power output needs

**Calls**

- `Constraint`

### 1.6.11 Tracker._add_tracking_constraints.tracking_dispatch_constraint_rule

**Kind** method

**Defined in file** `tracker.py` at line 232

**Signature** `def tracking_dispatch_constraint_rule(m, t):`

**Summary** No docstring.

### 1.6.12 Tracker._add_tracking_objective

**Kind** method

**Defined in file** `tracker.py` at line 244

**Signature** `def _add_tracking_objective(self):`

**Summary** Add EMPC objective function to the model, i.e., minimizing different costs

**Calls**

- `Objective`
- `getattr`

### 1.6.13 Tracker.update_model

**Kind** method

**Defined in file** `tracker.py` at line 270

**Signature** `def update_model(self, **profiles):`

**Summary** This method updates the parameters in the model based on the implemented profiles.

**Calls**

- `update_model`

### 1.6.14 Tracker.track_market_dispatch

**Kind** method

**Defined in file** `tracker.py` at line 283

**Signature** `def track_market_dispatch(self, market_dispatch, date, hour):`

**Summary** Solve the model to track the market dispatch signals. After solving,

**Calls**

- `_pass_market_dispatch`
- `solve`
- `record_results`
- `get_implemented_profile`
- `_record_daily_stats`

### 1.6.15 Tracker._record_daily_stats

**Kind** method

**Defined in file** `tracker.py` at line 315

**Signature** `def _record_daily_stats(self, profiles):`

**Summary** Record the stats that are used to update the model in the past 24 hours.

**Calls**

- `values`
- `len`
- `popleft`

### 1.6.16 Tracker._pass_market_dispatch

**Kind** method

**Defined in file** `tracker.py` at line 338

**Signature** `def _pass_market_dispatch(self, market_dispatch):`

**Summary** Pass the received market signals into model parameters.

**Calls**

- `activate`
- `deactivate`

### 1.6.17 Tracker.get_last_delivered_power

**Kind** method

**Defined in file** `tracker.py` at line 361

**Signature** `def get_last_delivered_power(self):`

**Summary** Returns the last delivered power output.

**Calls**

- `get_last_delivered_power`

### 1.6.18 Tracker._record_tracker_results

**Kind** method

**Defined in file** `tracker.py` at line 375

**Signature** `def _record_tracker_results(self, **kwargs):`

**Summary** Record the tracker stats.

**Calls**

- `append`
- `int`
- `round`
- `from_dict`
- `concat`
- `value`

### 1.6.19 Tracker.record_results

**Kind** method

**Defined in file** `tracker.py` at line 414

**Signature** `def record_results(self, **kwargs):`

**Summary** Record the operations stats for the model.

**Calls**

- `_record_tracker_results`
- `record_results`

### 1.6.20 Tracker.write_results

**Kind** method

**Defined in file** `tracker.py` at line 432

**Signature** `def write_results(self, path):`

**Summary** This methods writes the saved operation stats into an csv file.

**Calls**

- `print`
- `to_csv`
- `write_results`
- `join`
- `concat`

## 1.7 utils.py

**File path** `utils.py`

**Definitions Index**

- **function**: `convert_marginal_costs_to_actual_costs` (line 15)

**Called functions/methods in this file**

- line 28: `sort`
- line 39: `append`

**Definition Notes**

### 1.7.1 convert_marginal_costs_to_actual_costs

**Kind** function

**Defined in file** `utils.py` at line 15

**Signature** `def convert_marginal_costs_to_actual_costs(power_marginal_cost_pairs):`

**Summary** Convert a list of power and marginal cost pairs to a list of power and actual

**Calls**

- `sort`
- `append`

# Chapter 2

# examples

## 2.1 __init__.py

**File path** `examples/__init__.py`

## 2.2 thermal_generator.py

**File path** `examples/thermal_generator.py`

**Definitions Index**

- **class**: `ThermalGenerator` (line 26)
- **method**: `ThermalGenerator.__init__` (line 36)
- **method**: `ThermalGenerator.assemble_model_data` (line 66)
- **method**: `ThermalGenerator.model_data` (line 170)
- **method**: `ThermalGenerator._add_UT_DT_constraints` (line 174)
- **method**: `ThermalGenerator._add_UT_DT_constraints.pre_shut_down_trajectory_set_rule` (line 188)
- **method**: `ThermalGenerator._add_UT_DT_constraints.pre_start_up_trajectory_set_rule` (line 195)
- **method**: `ThermalGenerator._add_UT_DT_constraints.min_down_time_rule` (line 209)
- **method**: `ThermalGenerator._add_UT_DT_constraints.min_up_time_rule` (line 230)
- **method**: `ThermalGenerator.populate_model` (line 253)
- **method**: `ThermalGenerator.populate_model.init_pre_pow_fun` (line 319)
- **method**: `ThermalGenerator.populate_model.lhs_bnd_gen_pow_fun` (line 350)
- **method**: `ThermalGenerator.populate_model.rhs_bnd_gen_pow_fun` (line 355)
- **method**: `ThermalGenerator.populate_model.linear_power_fun` (line 361)

- **method**: ThermalGenerator.populate_model.seg_pow_bnd_fun (line 370)
- **method**: ThermalGenerator.populate_model.start_up_shut_dw_fun (line 386)
- **method**: ThermalGenerator.populate_model.start_up_or_shut_dw_fun (line 395)
- **method**: ThermalGenerator.populate_model.ramp_up_fun (line 401)
- **method**: ThermalGenerator.populate_model.ramp_shut_dw_fun (line 423)
- **method**: ThermalGenerator.populate_model.ramp_dw_fun (line 437)
- **method**: ThermalGenerator.populate_model.prod_cost_fun (line 458)
- **method**: ThermalGenerator.populate_model.start_cost_fun (line 466)
- **method**: ThermalGenerator.populate_model.tot_cost_fun (line 472)
- **method**: ThermalGenerator._update_UT_DT (line 480)
- **method**: ThermalGenerator._update_power (line 527)
- **method**: ThermalGenerator.update_model (line 545)
- **method**: ThermalGenerator.get_implemented_profile (line 568)
- **method**: ThermalGenerator.get_last_delivered_power (line 600)
- **method**: ThermalGenerator.record_results (line 613)
- **method**: ThermalGenerator.write_results (line 678)
- **method**: ThermalGenerator.power_output (line 692)
- **method**: ThermalGenerator.total_cost (line 696)

### Called functions/methods in this file

- line 23: attempt_import
- line 55: merge
- line 61: assemble_model_data
- line 80: set_index
- line 100: to_dict
- line 104: min
- line 125: range
- line 127: format
- line 138: ThermalGeneratorModelData
- line 189: value

- line 191: `Set`
- line 202: `Param`
- line 212: `sum`
- line 228: `Constraint`
- line 267: `list`
- line 307: `int`
- line 329: `Var`
- line 455: `_add_UT_DT_constraints`
- line 463: `Expression`
- line 494: `deque`
- line 499: `append`
- line 500: `round`
- line 512: `len`
- line 513: `popleft`
- line 562: `_update_UT_DT`
- line 563: `_update_power`
- line 640: `float`
- line 659: `abs`
- line 669: `from_dict`
- line 674: `concat`
- line 689: `to_csv`
- line 714: `SolverFactory`
- line 723: `ThermalGenerator`
- line 729: `Tracker`
- line 738: `track_market_dispatch`
- line 741: `write_results`
- line 753: `PlaceHolderForecaster`
- line 760: `Bidder`
- line 771: `compute_day_ahead_bids`
- line 800: `simulate`
- line 800: `Prescient`

**Definition Notes**

### 2.2.1 ThermalGenerator

**Kind** class

**Defined in file** `examples/thermal_generator.py` at line 26

**Signature** `class ThermalGenerator:`

**Summary** Simple thermal generator model (MIP). Equations models are from Gao, Knueven,

**Calls**

- `merge`
- `assemble_model_data`
- `set_index`
- `to_dict`
- `min`
- `range`
- `ThermalGeneratorModelData`
- `Set`
- `Param`
- `Constraint`
- `Var`
- `_add_UT_DT_constraints`
- `Expression`
- `deque`
- `round`
- `_update_UT_DT`
- `_update_power`
- `value`
- `append`
- `to_csv`
- `len`

- `popleft`
- `int`
- `float`
- `from_dict`

### 2.2.2 ThermalGenerator.__init__

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 36

**Signature** `def __init__(`

**Summary** Initializes the class object by building the thermal generator model.

**Calls**

- `merge`
- `assemble_model_data`

### 2.2.3 ThermalGenerator.assemble_model_data

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 66

**Signature** `def assemble_model_data(self, generator_name, gen_params):`

**Summary** This function assembles the parameter data to build the thermal generator

**Calls**

- `set_index`
- `to_dict`
- `min`
- `range`
- `ThermalGeneratorModelData`
- `format`

### 2.2.4 ThermalGenerator.model_data

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 170

**Signature** `def model_data(self):`

**Summary** No docstring.

### 2.2.5 ThermalGenerator._add_UT_DT_constraints

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 174

**Signature** `def _add_UT_DT_constraints(b):`

**Summary** This function adds the minimum up/down time constraints using eq. 4 - 5

**Calls**

- `Set`
- `Param`
- `Constraint`
- `value`
- `range`
- `sum`

### 2.2.6 ThermalGenerator._add_UT_DT_constraints.pre_shut_down_trajectory_set_ru

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 188

**Signature** `def pre_shut_down_trajectory_set_rule(b):`

**Summary** No docstring.

**Calls**

- `range`
- `value`

### 2.2.7 ThermalGenerator._add_UT_DT_constraints.pre_start_up_trajectory_set_rule

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 195

**Signature** `def pre_start_up_trajectory_set_rule(b):`

**Summary** No docstring.

**Calls**

- `range`
- `value`

### 2.2.8 ThermalGenerator._add_UT_DT_constraints.min_down_time_rule

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 209

**Signature** `def min_down_time_rule(b, h):`

**Summary** No docstring.

**Calls**

- `value`
- `sum`
- `range`

### 2.2.9 ThermalGenerator._add_UT_DT_constraints.min_up_time_rule

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 230

**Signature** `def min_up_time_rule(b, h):`

**Summary** No docstring.

**Calls**

- `value`
- `sum`
- `range`

### 2.2.10 ThermalGenerator.populate_model

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 253

**Signature** `def populate_model(self, b, horizon):`

**Summary** This function builds the model for a thermal generator.

**Calls**

- `Set`
- `Param`
- `Var`
- `Constraint`
- `_add_UT_DT_constraints`
- `Expression`
- `list`
- `int`
- `sum`
- `range`

### 2.2.11 ThermalGenerator.populate_model.init_pre_pow_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 319

**Signature** `def init_pre_pow_fun(b):`

**Summary** No docstring.

### 2.2.12 ThermalGenerator.populate_model.lhs_bnd_gen_pow_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 350

**Signature** `def lhs_bnd_gen_pow_fun(b, h):`

**Summary** No docstring.

### 2.2.13 ThermalGenerator.populate_model.rhs_bnd_gen_pow_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 355

**Signature** `def rhs_bnd_gen_pow_fun(b, h):`

**Summary** No docstring.

### 2.2.14 ThermalGenerator.populate_model.linear_power_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 361

**Signature** `def linear_power_fun(b, h):`

**Summary** No docstring.

**Calls**

- `sum`

### 2.2.15 ThermalGenerator.populate_model.seg_pow_bnd_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 370

**Signature** `def seg_pow_bnd_fun(b, h, l):`

**Summary** No docstring.

### 2.2.16 ThermalGenerator.populate_model.start_up_shut_dw_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 386

**Signature** `def start_up_shut_dw_fun(b, h):`

**Summary** No docstring.

### 2.2.17 ThermalGenerator.populate_model.start_up_or_shut_dw_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 395

**Signature** `def start_up_or_shut_dw_fun(b, h):`

**Summary** No docstring.

### 2.2.18 ThermalGenerator.populate_model.ramp_up_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 401

**Signature** `def ramp_up_fun(b, h):`

**Summary** h stand for hour

### 2.2.19 ThermalGenerator.populate_model.ramp_shut_dw_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 423

**Signature** `def ramp_shut_dw_fun(b, h):`

**Summary** h stand for hour.

### 2.2.20 ThermalGenerator.populate_model.ramp_dw_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 437

**Signature** `def ramp_dw_fun(b, h):`

**Summary** h stand for hour.

### 2.2.21 ThermalGenerator.populate_model.prod_cost_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 458

**Signature** `def prod_cost_fun(b, h):`

**Summary** No docstring.

**Calls**

- `sum`

### 2.2.22 ThermalGenerator.populate_model.start_cost_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 466

**Signature** `def start_cost_fun(b, h):`

**Summary** No docstring.

### 2.2.23 ThermalGenerator.populate_model.tot_cost_fun

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 472

**Signature** `def tot_cost_fun(b, h):`

**Summary** No docstring.

### 2.2.24 ThermalGenerator._update_UT_DT

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 480

**Signature** `def _update_UT_DT(b, implemented_shut_down, implemented_start_up):`

**Summary** This method updates the parameters in the minimum up/down time

**Calls**

- `deque`
- `append`
- `len`
- `popleft`
- `round`
- `value`

### 2.2.25 ThermalGenerator._update_power

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 527

**Signature** `def _update_power(b, implemented_power_output):`

**Summary** This method updates the parameters in the ramping constraints based on

**Calls**

- `round`
- `int`

### 2.2.26 ThermalGenerator.update_model

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 545

**Signature** `def update_model(`

**Summary** This method updates the parameters in the model based on

**Calls**

- `_update_UT_DT`
- `_update_power`

### 2.2.27 ThermalGenerator.get_implemented_profile

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 568

**Signature** `def get_implemented_profile(b, last_implemented_time_step):`

**Summary** This method gets the implemented variable profiles in the last optimization

**Calls**

- `deque`
- `value`
- `range`

### 2.2.28 ThermalGenerator.get_last_delivered_power

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 600

**Signature** `def get_last_delivered_power(b, last_implemented_time_step):`

**Summary** Returns the last delivered power output.

**Calls**

- `value`

### 2.2.29 ThermalGenerator.record_results

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 613

**Signature** `def record_results(self, b, date=None, hour=None, **kwargs):`

**Summary** Record the operations stats for the model.

**Calls**

- `append`
- `int`
- `float`
- `from_dict`
- `concat`
- `round`
- `value`
- `abs`

### 2.2.30 ThermalGenerator.write_results

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 678

**Signature** `def write_results(self, path):`

**Summary** This methods writes the saved operation stats into an csv file.

**Calls**

- `to_csv`
- `concat`

### 2.2.31 ThermalGenerator.power_output

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 692

**Signature** `def power_output(self):`

**Summary** No docstring.

### 2.2.32 ThermalGenerator.total_cost

**Kind** method

**Defined in file** `examples/thermal_generator.py` at line 696

**Signature** `def total_cost(self):`

**Summary** No docstring.

## 2.3 thermal_generator_prescient_plugin.py

**File path** `examples/thermal_generator_prescient_plugin.py`

**Called functions/methods in this file**

- line 29: `dirname`
- line 39: `PlaceHolderForecaster`
- line 47: `SolverFactory`
- line 52: `ThermalGenerator`
- line 57: `Tracker`
- line 83: `Bidder`
- line 93: `DoubleLoopCoordinator`

## 2.4 utils.py

**File path** `examples/utils.py`

**Called functions/methods in this file**

- line 20: `as_file`
- line 21: `joinpath`
- line 21: `files`
- line 23: `Path`
- line 26: `read_csv`
- line 26: `join`

# Chapter 3

# multiperiod

## 3.1 __init__.py

**File path** `multiperiod/__init__.py`

## 3.2 multiperiod.py

**File path** `multiperiod/multiperiod.py`

**Definitions Index**

- **class**: `MultiPeriodModel` (line 24)
- **method**: `MultiPeriodModel.__init__` (line 53)
- **method**: `MultiPeriodModel.build_multi_period_model` (line 135)
- **method**: `MultiPeriodModel.advance_time` (line 225)
- **method**: `MultiPeriodModel.pyomo_model` (line 280)
- **method**: `MultiPeriodModel.current_time` (line 287)
- **method**: `MultiPeriodModel.get_active_process_blocks` (line 293)
- **method**: `MultiPeriodModel._create_linking_constraints` (line 299)
- **method**: `MultiPeriodModel._create_periodic_constraints` (line 307)
- **method**: `MultiPeriodModel._construct_flowsheet_instance` (line 315)
- **method**: `MultiPeriodModel.build_stochastic_multi_period` (line 351)
- **method**: `MultiPeriodModel.build_stochastic_multi_period._build_scenario_model` (line 385)
- **method**: `MultiPeriodModel.plot_lmp_signal` (line 493)
- **method**: `MultiPeriodModel.plot_lmp_and_schedule` (line 597)

### Called functions/methods in this file

- line 19: `getLogger`
- line 71: `__init__`
- line 71: `super`
- line 94: `RangeSet`
- line 97: `Set`
- line 118: `get_solver`
- line 121: `setLevel`
- line 124: `build_stochastic_multi_period`
- line 174: `range`
- line 175: `Block`
- line 178: `_construct_flowsheet_instance`
- line 185: `info`
- line 186: `clone`
- line 189: `len`
- line 190: `error`
- line 197: `warning`
- line 204: `create_process_model`
- line 207: `data`
- line 208: `get_linking_variable_pairs`
- line 211: `_create_linking_constraints`
- line 215: `get_periodic_variable_pairs`
- line 218: `_create_periodic_constraints`
- line 222: `first`
- line 235: `next`
- line 238: `deactivate`
- line 244: `last`
- line 246: `add`
- line 297: `values`

- line 303: `Constraint`
- line 304: `enumerate`
- line 323: `ConcreteModel`
- line 330: `initialization_func`
- line 331: `solve`
- line 333: `check_optimal_termination`
- line 334: `InitializationError`
- line 347: `unfix_dof_func`
- line 399: `transfer_attributes_from`
- line 483: `_build_scenario_model`
- line 511: `type`
- line 528: `Exception`
- line 569: `str`
- line 578: `figure`
- line 581: `add_subplot`
- line 582: `plot`
- line 583: `set_xlabel`
- line 584: `set_ylabel`
- line 585: `set_title`
- line 588: `set_xlim`
- line 591: `set_ylim`
- line 593: `tight_layout`
- line 594: `show`
- line 659: `tick_params`
- line 667: `twinx`

**Definition Notes**

### 3.2.1 MultiPeriodModel

**Kind** class

**Defined in file** `multiperiod/multiperiod.py` at line 24

**Signature** `class MultiPeriodModel(pyo.ConcreteModel):`

**Summary** The ‘MultiPeriodModel‘ class helps transfer existing steady-state

**Calls**

- `__init__`
- `Set`
- `Block`
- `first`
- `next`
- `deactivate`
- `last`
- `add`
- `create_process_model`
- `get_linking_variable_pairs`
- `_create_linking_constraints`
- `Constraint`
- `enumerate`
- `ConcreteModel`
- `_construct_flowsheet_instance`
- `figure`
- `range`
- `tight_layout`
- `show`
- `RangeSet`
- `getLogger`
- `setLevel`
- `build_stochastic_multi_period`
- `get_solver`
- `warning`

### 3.2.2 MultiPeriodModel.__init__

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 53

**Signature** `def __init__(`

**Summary** No docstring.

**Calls**

- `__init__`
- `RangeSet`
- `getLogger`
- `setLevel`
- `build_stochastic_multi_period`
- `super`
- `Set`
- `get_solver`

### 3.2.3 MultiPeriodModel.build_multi_period_model

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 135

**Signature** `def build_multi_period_model(`

**Summary** Build a multi-period capable model using user-provided functions

**Calls**

- `Set`
- `Block`
- `first`
- `get_solver`
- `_construct_flowsheet_instance`
- `warning`
- `data`

- get_linking_variable_pairs
- _create_linking_constraints
- len
- get_periodic_variable_pairs
- _create_periodic_constraints
- range
- info
- clone
- error
- create_process_model

### 3.2.4 MultiPeriodModel.advance_time

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 225

**Signature** `def advance_time(self, **model_data_kwargs):`

**Summary** Advance the current model instance to the next time period

**Calls**

- next
- deactivate
- last
- add
- create_process_model
- get_linking_variable_pairs
- _create_linking_constraints
- get_periodic_variable_pairs
- _create_periodic_constraints

### 3.2.5 MultiPeriodModel.pyomo_model

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 280

**Signature** `def pyomo_model(self):`

**Summary** Retrieve the underlying pyomo model

### 3.2.6 MultiPeriodModel.current_time

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 287

**Signature** `def current_time(self):`

**Summary** Retrieve the current multiperiod model time

### 3.2.7 MultiPeriodModel.get_active_process_blocks

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 293

**Signature** `def get_active_process_blocks(self):`

**Summary** Retrieve the active time blocks of the pyomo model

**Calls**

- `values`

### 3.2.8 MultiPeriodModel._create_linking_constraints

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 299

**Signature** `def _create_linking_constraints(self, b1, variable_pairs):`

**Summary** Create linking constraint on ‘b1‘ using ‘variable_pairs‘

**Calls**

- `Constraint`
- `enumerate`
- `range`
- `len`

### 3.2.9 MultiPeriodModel._create_periodic_constraints

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 307

**Signature** `def _create_periodic_constraints(self, b1, variable_pairs):`

**Summary** Create periodic linking constraint on ‘b1‘ using ‘variable_pairs‘

**Calls**

- `Constraint`
- `enumerate`
- `range`
- `len`

### 3.2.10 MultiPeriodModel._construct_flowsheet_instance

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 315

**Signature** `def _construct_flowsheet_instance(`

**Summary** No docstring.

**Calls**

- `ConcreteModel`
- `create_process_model`
- `warning`
- `initialization_func`
- `solve`
- `unfix_dof_func`
- `check_optimal_termination`
- `InitializationError`

### 3.2.11 MultiPeriodModel.build_stochastic_multi_period

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 351

**Signature** `def build_stochastic_multi_period(`

**Summary** This function constructs the stochastic multiperiod optimization problem

**Calls**

- `Set`
- `_construct_flowsheet_instance`
- `Block`
- `ConcreteModel`
- `_build_scenario_model`
- `info`
- `transfer_attributes_from`
- `warning`
- `clone`
- `data`
- `get_linking_variable_pairs`
- `_create_linking_constraints`

### 3.2.12 MultiPeriodModel.build_stochastic_multi_period._build_scenario_model

**Kind** method

**Defined in file** `multiperiod/multiperiod.py` at line 385

**Signature** `def _build_scenario_model(m, fs_blk):`

**Summary** Construct a multiperiod model for one scenario

**Calls**

- `Block`
- `info`
- `transfer_attributes_from`
- `warning`
- `clone`

- data
- get_linking_variable_pairs
- _create_linking_constraints

### 3.2.13 MultiPeriodModel.plot_lmp_signal

**Kind** method

**Defined in file** multiperiod/multiperiod.py at line 493

**Signature** def plot_lmp_signal(

**Summary** This function plots LMP signals as a function of time.

**Calls**

- figure
- range
- tight_layout
- show
- type
- add_subplot
- plot
- set_xlabel
- set_ylabel
- set_title
- len
- set_xlim
- set_ylim
- Exception
- str

### 3.2.14 MultiPeriodModel.plot_lmp_and_schedule

**Kind** method

**Defined in file** multiperiod/multiperiod.py at line 597

**Signature** `def plot_lmp_and_schedule(`

**Summary** The function plots optimal operation schedule as a function of time.

**Calls**

- `figure`
- `range`
- `tight_layout`
- `show`
- `len`
- `Exception`
- `add_subplot`
- `set_xlabel`
- `set_ylabel`
- `plot`
- `tick_params`
- `twinx`
- `enumerate`
- `set_xlim`
- `set_ylim`
- `str`

# Chapter 4

# multiperiod/tests

## 4.1 __init__.py

**File path** `multiperiod/tests/__init__.py`

## 4.2 test_multiperiod.py

**File path** `multiperiod/tests/test_multiperiod.py`

**Module docstring** Tests for ‘MultiPeriodModel‘ class

**Definitions Index**

- **function**: `build_flowsheet` (line 28)
- **function**: `fix_dof_and_initialize` (line 42)
- **function**: `fix_dof_and_initialize_2` (line 48)
- **function**: `unfix_dof` (line 53)
- **function**: `get_linking_variable_pairs` (line 58)
- **function**: `build_multi_period_model` (line 64)
- **function**: `build_multi_period_stochastic_model` (line 78)
- **function**: `build_multi_day_model` (line 93)
- **function**: `build_multi_day_stochastic_model` (line 108)
- **function**: `build_multi_year_model` (line 124)
- **function**: `build_multi_year_stochastic_model` (line 139)
- **function**: `build_multi_day_year_model` (line 155)
- **function**: `build_multi_day_year_stochastic_model` (line 171)
- **function**: `build_multi_period_model_manual` (line 188)

- **function**: `test_lmp_plot` (line 201)
- **function**: `test_lmp_and_schedule_plot` (line 234)
- **function**: `test_multi_period_model` (line 277)
- **function**: `test_multi_period_stochastic_model` (line 290)
- **function**: `test_multi_day_model` (line 310)
- **function**: `test_multi_day_stochastic_model` (line 327)
- **function**: `test_multi_year_model` (line 349)
- **function**: `test_multi_year_stochastic_model` (line 366)
- **function**: `test_multi_day_year_model` (line 388)
- **function**: `test_multi_day_year_stochastic_model` (line 406)
- **function**: `test_no_initialization` (line 429)
- **function**: `test_data_kwargs` (line 496)
- **function**: `test_data_kwargs_fail` (line 532)
- **function**: `test_initialization_fail` (line 569)
- **function**: `test_multi_period_model_manual` (line 590)

### Called functions/methods in this file

- line 31: `ConcreteModel`
- line 33: `FlowsheetBlock`
- line 35: `Var`
- line 37: `Constraint`
- line 44: `fix`
- line 45: `set_value`
- line 55: `unfix`
- line 65: `MultiPeriodModel`
- line 63: `fixture`
- line 196: `build_multi_period_model`
- line 213: `range`
- line 213: `len`
- line 216: `setattr`

- line 218: `plot_lmp_signal`
- line 223: `raises`
- line 260: `plot_lmp_and_schedule`
- line 280: `hasattr`
- line 286: `degrees_of_freedom`
- line 431: `at_level`
- line 446: `clear`

**Definition Notes**

### 4.2.1 build_flowsheet

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 28

**Signature** `def build_flowsheet(m=None, a=None, b=None):`

**Summary** This function builds a dummy flowsheet

**Calls**

- `FlowsheetBlock`
- `Var`
- `Constraint`
- `ConcreteModel`

### 4.2.2 fix_dof_and_initialize

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 42

**Signature** `def fix_dof_and_initialize(m):`

**Summary** This function fixes dof and initializes the dummy flowsheet

**Calls**

- `fix`
- `set_value`

### 4.2.3 fix_dof_and_initialize_2

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 48

**Signature** `def fix_dof_and_initialize_2(m):`

**Summary** This function fixes dof and initializes the dummy flowsheet

**Calls**

- `Constraint`

### 4.2.4 unfix_dof

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 53

**Signature** `def unfix_dof(m):`

**Summary** This function unfixes dof for optimization

**Calls**

- `unfix`

### 4.2.5 get_linking_variable_pairs

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 58

**Signature** `def get_linking_variable_pairs(m1, m2):`

**Summary** This function returns pairs of linking variables

### 4.2.6 build_multi_period_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 64

**Signature** `def build_multi_period_model():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`

### 4.2.7 build_multi_period_stochastic_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 78

**Signature** `def build_multi_period_stochastic_model():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`

### 4.2.8 build_multi_day_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 93

**Signature** `def build_multi_day_model():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`

### 4.2.9 build_multi_day_stochastic_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 108

**Signature** `def build_multi_day_stochastic_model():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`

### 4.2.10 build_multi_year_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 124

**Signature** `def build_multi_year_model():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`

### 4.2.11 build_multi_year_stochastic_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 139

**Signature** `def build_multi_year_stochastic_model():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`

### 4.2.12 build_multi_day_year_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 155

**Signature** `def build_multi_day_year_model():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`

### 4.2.13 build_multi_day_year_stochastic_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 171

**Signature** `def build_multi_day_year_stochastic_model():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`

### 4.2.14 build_multi_period_model_manual

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 188

**Signature** `def build_multi_period_model_manual():`

**Summary** No docstring.

**Calls**

- `fixture`
- `MultiPeriodModel`
- `build_multi_period_model`

### 4.2.15 test_lmp_plot

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 201

**Signature** `def test_lmp_plot(monkeypatch):`

**Summary** No docstring.

**Calls**

- `setattr`
- `plot_lmp_signal`
- `raises`
- `range`
- `len`

### 4.2.16 test_lmp_and_schedule_plot

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 234

**Signature** `def test_lmp_and_schedule_plot(monkeypatch):`

**Summary** No docstring.

**Calls**

- `setattr`
- `plot_lmp_and_schedule`
- `raises`
- `range`
- `len`

### 4.2.17 test_multi_period_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 277

**Signature** `def test_multi_period_model(build_multi_period_model):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `len`
- `degrees_of_freedom`

### 4.2.18 test_multi_period_stochastic_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 290

**Signature** `def test_multi_period_stochastic_model(build_multi_period_stochastic_model):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `len`
- `degrees_of_freedom`

### 4.2.19 test_multi_day_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 310

**Signature** `def test_multi_day_model(build_multi_day_model):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `degrees_of_freedom`

### 4.2.20 test_multi_day_stochastic_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 327

**Signature** `def test_multi_day_stochastic_model(build_multi_day_stochastic_model):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `degrees_of_freedom`

### 4.2.21 test_multi_year_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 349

**Signature** `def test_multi_year_model(build_multi_year_model):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `degrees_of_freedom`

### 4.2.22 test_multi_year_stochastic_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 366

**Signature** `def test_multi_year_stochastic_model(build_multi_year_stochastic_model):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `degrees_of_freedom`

### 4.2.23 test_multi_day_year_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 388

**Signature** `def test_multi_day_year_model(build_multi_day_year_model):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `degrees_of_freedom`

### 4.2.24 test_multi_day_year_stochastic_model

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 406

**Signature** `def test_multi_day_year_stochastic_model(build_multi_day_year_stochastic_model):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `degrees_of_freedom`

### 4.2.25 test_no_initialization

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 429

**Signature** `def test_no_initialization(caplog):`

**Summary** No docstring.

**Calls**

- `clear`
- `at_level`
- `MultiPeriodModel`

### 4.2.26 test_data_kwargs

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 496

**Signature** `def test_data_kwargs(caplog, n_time_points=3):`

**Summary** No docstring.

**Calls**

- `at_level`
- `MultiPeriodModel`
- `build_multi_period_model`

### 4.2.27 test_data_kwargs_fail

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 532

**Signature** `def test_data_kwargs_fail(caplog, n_time_points=2):`

**Summary** No docstring.

**Calls**

- `at_level`
- `MultiPeriodModel`
- `build_multi_period_model`
- `len`

### 4.2.28 test_initialization_fail

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 569

**Signature** `def test_initialization_fail():`

**Summary** No docstring.

**Calls**

- `raises`
- `MultiPeriodModel`

### 4.2.29 test_multi_period_model_manual

**Kind** function

**Defined in file** `multiperiod/tests/test_multiperiod.py` at line 590

**Signature** `def test_multi_period_model_manual(build_multi_period_model_manual):`

**Summary** No docstring.

**Calls**

- `hasattr`
- `len`
- `degrees_of_freedom`

# Chapter 5

# pricetaker

## 5.1 __init__.py

**File path** `pricetaker/__init__.py`

## 5.2 clustering.py

**File path** `pricetaker/clustering.py`

**Module docstring** Contains functions for clustering the price signal into representative days/periods.

**Definitions Index**

- **function**: `generate_daily_data` (line 35)
- **function**: `cluster_lmp_data` (line 76)
- **function**: `get_optimal_num_clusters` (line 142)
- **function**: `_elbow_method` (line 238)
- **function**: `_normalize_values` (line 332)

**Called functions/methods in this file**

- line 26: `attempt_import`
- line 32: `getLogger`
- line 52: `list`
- line 54: `len`
- line 55: `ValueError`
- line 61: `warning`
- line 68: `range`
- line 73: `transpose`

- line 73: `DataFrame`
- line 119: `generate_daily_data`
- line 122: `fit`
- line 122: `KMeans`
- line 127: `abs`
- line 134: `float`
- line 137: `int`
- line 137: `sum`
- line 199: `append`
- line 200: `silhouette_score`
- line 204: `index`
- line 204: `max`
- line 208: `_elbow_method`
- line 218: `info`
- line 227: `plot`
- line 228: `axvline`
- line 229: `xlabel`
- line 230: `ylabel`
- line 231: `title`
- line 232: `xlim`
- line 233: `show`
- line 248: `splrep`
- line 249: `arange`
- line 250: `splev`
- line 258: `_normalize_values`
- line 278: `RuntimeError`
- line 334: `min`

### Definition Notes

### 5.2.1 generate_daily_data

**Kind** function

**Defined in file** `pricetaker/clustering.py` at line 35

**Signature** `def generate_daily_data(raw_data: list, horizon_length: int):`

**Summary** Function used to generate the daily data in a usable format

**Calls**

- `list`
- `transpose`
- `len`
- `ValueError`
- `warning`
- `range`
- `DataFrame`

### 5.2.2 cluster_lmp_data

**Kind** function

**Defined in file** `pricetaker/clustering.py` at line 76

**Signature** `def cluster_lmp_data(`

**Summary** Clusters the given price signal into n_clusters using the k-means clustering

**Calls**

- `generate_daily_data`
- `fit`
- `int`
- `KMeans`
- `abs`
- `float`
- `range`
- `sum`

### 5.2.3 get_optimal_num_clusters

**Kind** function

**Defined in file** `pricetaker/clustering.py` at line 142

**Signature** `def get_optimal_num_clusters(`

**Summary** Determines the appropriate number of clusters needed for a

**Calls**

- `list`
- `info`
- `ValueError`
- `len`
- `range`
- `fit`
- `append`
- `index`
- `warning`
- `plot`
- `axvline`
- `xlabel`
- `ylabel`
- `title`
- `xlim`
- `show`
- `silhouette_score`
- `max`
- `_elbow_method`
- `KMeans`

### 5.2.4 _elbow_method

**Kind** function

**Defined in file** `pricetaker/clustering.py` at line 238

**Signature** `def _elbow_method(k_values, inertia_values, kmin, kmax, sensitivity: int = 1):`

**Summary** No docstring.

**Calls**

- `splrep`
- `arange`
- `splev`
- `_normalize_values`
- `len`
- `RuntimeError`
- `range`
- `index`
- `append`
- `warning`

### 5.2.5 _normalize_values

**Kind** function

**Defined in file** `pricetaker/clustering.py` at line 332

**Signature** `def _normalize_values(values):`

**Summary** No docstring.

**Calls**

- `max`
- `min`

## 5.3 design_and_operation_models.py

**File path** `pricetaker/design_and_operation_models.py`

### Definitions Index

- **function**: `_format_data` (line 30)
- **function**: `is_valid_variable_design_data` (line 48)
- **function**: `is_valid_polynomial_surrogate_data` (line 70)
- **class**: `DesignModelData` (line 91)
- **method**: `DesignModelData.build` (line 163)
- **method**: `DesignModelData._build_variable_design_model` (line 209)
- **class**: `OperationModelData` (line 243)
- **method**: `OperationModelData.build` (line 399)
- **method**: `OperationModelData.build_polynomial_surrogates` (line 449)
- **method**: `OperationModelData.build_expressions` (line 460)
- **function**: `_is_valid_data_type_for_storage_model` (line 477)
- **class**: `StorageModelData` (line 502)
- **method**: `StorageModelData.build` (line 562)
- **method**: `StorageModelData.build._add_upper_bound` (line 575)

### Called functions/methods in this file

- line 27: `getLogger`
- line 34: `isinstance`
- line 43: `ConfigurationError`
- line 44: `type`
- line 58: `pop`
- line 64: `items`
- line 65: `_format_data`
- line 134: `ConfigDict`
- line 135: `declare`
- line 137: `ConfigValue`
- line 164: `build`
- line 164: `super`
- line 166: `Var`

- line 174: `setattr`
- line 174: `Param`
- line 178: `_build_variable_design_model`
- line 182: `model_func`
- line 186: `warning`
- line 195: `hasattr`
- line 224: `getattr`
- line 225: `Constraint`
- line 236: `sum`
- line 237: `range`
- line 237: `len`
- line 239: `Expression`
- line 90: `declare_process_block_class`
- line 343: `is_in_range`
- line 429: `build_polynomial_surrogates`
- line 458: `build_expressions`
- line 487: `any`
- line 488: `is_expression_type`
- line 488: `is_variable_type`
- line 488: `is_parameter_type`
- line 580: `setub`
- line 592: `_add_upper_bound`
- line 602: `setlb`

**Definition Notes**

### 5.3.1 _format_data

**Kind** function

**Defined in file** `pricetaker/design_and_operation_models.py` at line 30

**Signature** `def _format_data(coeffs):`

**Summary** Helper function to correctly format the surrogate model coefficients

**Calls**

- `isinstance`
- `ConfigurationError`
- `type`

### 5.3.2 is_valid_variable_design_data

**Kind** function

**Defined in file** `pricetaker/design_and_operation_models.py` at line 48

**Signature** `def is_valid_variable_design_data(data: dict):`

**Summary** Validates the arguments received for the variable design case

**Calls**

- `items`
- `ConfigurationError`
- `pop`
- `_format_data`

### 5.3.3 is_valid_polynomial_surrogate_data

**Kind** function

**Defined in file** `pricetaker/design_and_operation_models.py` at line 70

**Signature** `def is_valid_polynomial_surrogate_data(data: dict):`

**Summary** Validates the arguments received for the variable design case

**Calls**

- `items`
- `ConfigurationError`
- `pop`
- `_format_data`

### 5.3.4 DesignModelData

**Kind** class

**Defined in file** `pricetaker/design_and_operation_models.py` at line 91

**Signature** `class DesignModelData(ProcessBlockData):`

**Summary** Builds the 'design model' for a unit/process.

**Calls**

- `declare_process_block_class`
- `ConfigDict`
- `declare`
- `ConfigValue`
- `build`
- `Var`
- `setattr`
- `getattr`
- `Constraint`
- `items`
- `hasattr`
- `warning`
- `super`
- `_build_variable_design_model`
- `sum`
- `Expression`
- `Param`
- `model_func`
- `range`
- `len`

### 5.3.5 DesignModelData.build

**Kind** method

**Defined in file** `pricetaker/design_and_operation_models.py` at line 163

**Signature** `def build(self):`

**Summary** No docstring.

**Calls**

- `build`
- `Var`
- `items`
- `hasattr`
- `warning`
- `super`
- `setattr`
- `_build_variable_design_model`
- `Param`
- `model_func`

### 5.3.6 DesignModelData._build_variable_design_model

**Kind** method

**Defined in file** `pricetaker/design_and_operation_models.py` at line 209

**Signature** `def _build_variable_design_model(self):`

**Summary** No docstring.

**Calls**

- `setattr`
- `getattr`
- `Constraint`
- `items`
- `Var`
- `sum`
- `Expression`
- `range`
- `len`

### 5.3.7 OperationModelData

**Kind** class

**Defined in file** `pricetaker/design_and_operation_models.py` at line 243

**Signature** `class OperationModelData(ProcessBlockData):`

**Summary** Builds the 'operation model' for a unit/process.

**Calls**

- `declare_process_block_class`
- `ConfigDict`
- `declare`
- `ConfigValue`
- `build`
- `build_expressions`
- `items`
- `Var`
- `Param`
- `setattr`
- `build_polynomial_surrogates`
- `is_in_range`
- `super`
- `model_func`
- `warning`
- `sum`
- `getattr`
- `Constraint`
- `Expression`
- `range`
- `len`

### 5.3.8 OperationModelData.build

**Kind** method

**Defined in file** `pricetaker/design_and_operation_models.py` at line 399

**Signature** `def build(self):`

**Summary** No docstring.

**Calls**

- `build`
- `Var`
- `Param`
- `setattr`
- `build_polynomial_surrogates`
- `super`
- `model_func`
- `warning`
- `getattr`

### 5.3.9 OperationModelData.build_polynomial_surrogates

**Kind** method

**Defined in file** `pricetaker/design_and_operation_models.py` at line 449

**Signature** `def build_polynomial_surrogates(`

**Summary** Builds polynomial-type surrgogate models

**Calls**

- `build_expressions`
- `sum`
- `items`
- `range`
- `len`

### 5.3.10 OperationModelData.build_expressions

**Kind** method

**Defined in file** `pricetaker/design_and_operation_models.py` at line 460

**Signature** `def build_expressions(self, expressions: dict, declare_variables: bool = False):`

**Summary** Declares user-defined expressions

**Calls**

- `items`
- `setattr`
- `Var`
- `Constraint`
- `Expression`
- `getattr`

### 5.3.11 _is_valid_data_type_for_storage_model

**Kind** function

**Defined in file** `pricetaker/design_and_operation_models.py` at line 477

**Signature** `def _is_valid_data_type_for_storage_model(obj):`

**Summary** Domain validator for config arguments of the storage model

**Calls**

- `isinstance`
- `any`
- `ConfigurationError`
- `is_expression_type`
- `is_variable_type`
- `is_parameter_type`

### 5.3.12 StorageModelData

**Kind** class

**Defined in file** `pricetaker/design_and_operation_models.py` at line 502

**Signature** `class StorageModelData(ProcessBlockData):`

**Summary** Builds the 'storage model' for a unit/process.

**Calls**

- `declare_process_block_class`
- `ConfigDict`
- `declare`
- `ConfigValue`
- `build`
- `Var`
- `_add_upper_bound`
- `Constraint`
- `isinstance`
- `setlb`
- `is_in_range`
- `super`
- `setub`
- `setattr`
- `getattr`

### 5.3.13 StorageModelData.build

**Kind** method

**Defined in file** `pricetaker/design_and_operation_models.py` at line 562

**Signature** `def build(self):`

**Summary** No docstring.

**Calls**

- `build`
- `Var`
- `_add_upper_bound`
- `Constraint`
- `isinstance`
- `setlb`
- `super`
- `setub`
- `setattr`
- `getattr`

### 5.3.14 StorageModelData.build._add_upper_bound

**Kind** method

**Defined in file** `pricetaker/design_and_operation_models.py` at line 575

**Signature** `def _add_upper_bound(var_name, bound):`

**Summary** No docstring.

**Calls**

- `isinstance`
- `setub`
- `setattr`
- `Constraint`
- `getattr`

## 5.4 price_taker_model.py

**File path** `pricetaker/price_taker_model.py`

**Definitions Index**


- **class**: PriceTakerModel (line 161)
- **method**: PriceTakerModel.__init__ (line 164)
- **method**: PriceTakerModel.num_representative_days (line 174)
- **method**: PriceTakerModel.num_representative_days (line 179)
- **method**: PriceTakerModel.horizon_length (line 191)
- **method**: PriceTakerModel.horizon_length (line 203)
- **method**: PriceTakerModel._assert_lmp_data_exists (line 213)
- **method**: PriceTakerModel.append_lmp_data (line 221)
- **method**: PriceTakerModel.get_optimal_representative_days (line 262)
- **method**: PriceTakerModel._assert_mp_model_exists (line 283)
- **method**: PriceTakerModel.build_multiperiod_model (line 291)
- **method**: PriceTakerModel._get_operation_blocks (line 445)
- **method**: PriceTakerModel._get_operation_vars (line 482)
- **method**: PriceTakerModel.fix_operation_var (line 504)
- **method**: PriceTakerModel.unfix_operation_var (line 518)
- **method**: PriceTakerModel.update_operation_params (line 530)
- **method**: PriceTakerModel.add_linking_constraints (line 564)
- **method**: PriceTakerModel.add_linking_constraints._rule_linking_constraints (line 582)
- **method**: PriceTakerModel.add_periodic_constraints (line 598)
- **method**: PriceTakerModel.add_periodic_constraints._rule_linking_constraints (line 614)
- **method**: PriceTakerModel._retrieve_uc_data (line 628)
- **method**: PriceTakerModel._get_valid_block_name (line 645)
- **method**: PriceTakerModel.add_capacity_limits (line 654)
- **method**: PriceTakerModel.add_ramping_limits (line 723)
- **method**: PriceTakerModel.add_startup_shutdown (line 813)
- **method**: PriceTakerModel.add_hourly_cashflows (line 897)
- **method**: PriceTakerModel.add_overall_cashflows (line 997)
- **method**: PriceTakerModel.add_objective_function (line 1123)

- **method**: PriceTakerModel._get_num (line 1155)
- **method**: PriceTakerModel.get_num_startups (line 1182)
- **method**: PriceTakerModel.get_num_shutdowns (line 1186)
- **method**: PriceTakerModel._get_pyomo_obj_value (line 1191)
- **method**: PriceTakerModel.get_operation_var_values (line 1202)
- **method**: PriceTakerModel.get_design_var_values (line 1245)
- **method**: PriceTakerModel.plot_operation_profile (line 1281)
- **method**: PriceTakerModel.plot_lmp_histogram (line 1352)

**Called functions/methods in this file**

- line 60: getLogger
- line 65: ConfigDict
- line 68: declare
- line 70: ConfigValue
- line 87: ListOf
- line 133: is_in_range
- line 165: __init__
- line 165: super
- line 166: CONFIG
- line 182: ConfigurationError
- line 196: warning
- line 249: len
- line 273: _assert_lmp_data_exists
- line 277: generate_daily_data
- line 279: get_optimal_num_clusters
- line 285: hasattr
- line 330: cluster_lmp_data
- line 341: enumerate
- line 344: RangeSet
- line 349: ConcreteModel

- line 352: `flowsheet_func`
- line 358: `Block`
- line 360: `transfer_attributes_from`
- line 360: `clone`
- line 367: `component_data_objects`
- line 374: `isinstance`
- line 377: `info`
- line 380: `add_linking_constraints`
- line 386: `add_periodic_constraints`
- line 402: `add_startup_shutdown`
- line 419: `add_capacity_limits`
- line 435: `add_ramping_limits`
- line 456: `_assert_mp_model_exists`
- line 460: `find_component`
- line 470: `AttributeError`
- line 514: `_get_operation_vars`
- line 516: `fix`
- line 528: `unfix`
- line 543: `NotImplementedError`
- line 549: `items`
- line 557: `list`
- line 562: `set_value`
- line 587: `setattr`
- line 589: `str`
- line 590: `Constraint`
- line 635: `update`
- line 638: `UnitCommitmentData`
- line 649: `sub`
- line 649: `split`

- line 651: `rstrip`
- line 685: `_get_operation_blocks`
- line 689: `_retrieve_uc_data`
- line 698: `_get_valid_block_name`
- line 706: `getattr`
- line 710: `capacity_limits`
- line 784: `assert_ramping_args_present`
- line 800: `ramping_limits`
- line 876: `startup_shutdown_constraints`
- line 979: `Expression`
- line 1071: `Var`

**Definition Notes**

### 5.4.1 PriceTakerModel

**Kind** class

**Defined in file** `pricetaker/price_taker_model.py` at line 161

**Signature** `class PriceTakerModel(ConcreteModel):`

**Summary** Builds a price-taker model for a given system

**Calls**

- `__init__`
- `CONFIG`
- `warning`
- `_assert_lmp_data_exists`
- `generate_daily_data`
- `get_optimal_num_clusters`
- `hasattr`
- `RangeSet`
- `ConcreteModel`
- `flowsheet_func`

- Block
- component_data_objects
- _assert_mp_model_exists
- _get_operation_vars
- items
- setattr
- info
- sub
- rstrip
- _get_operation_blocks
- _retrieve_uc_data
- getattr
- assert_ramping_args_present
- Var
- Constraint

### 5.4.2 PriceTakerModel.__init__

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 164

**Signature** `def __init__(self, *args, **kwds):`

**Summary** No docstring.

**Calls**

- __init__
- CONFIG
- super

### 5.4.3 PriceTakerModel.num_representative_days

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 174

**Signature** `def num_representative_days(self):`

**Summary** Returns the number of representative days

### 5.4.4 PriceTakerModel.num_representative_days

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 179

**Signature** `def num_representative_days(self, value):`

**Summary** Setter for the num_representative_days property

**Calls**

- `ConfigurationError`

### 5.4.5 PriceTakerModel.horizon_length

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 191

**Signature** `def horizon_length(self):`

**Summary** Returns the length of each representative day

**Calls**

- `warning`

### 5.4.6 PriceTakerModel.horizon_length

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 203

**Signature** `def horizon_length(self, value):`

**Summary** Setter for the horizon_length property

**Calls**

- `ConfigurationError`

### 5.4.7 PriceTakerModel._assert_lmp_data_exists

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 213

**Signature** `def _assert_lmp_data_exists(self):`

**Summary** Raise an error if LMP data does not exist

**Calls**

- `ConfigurationError`

### 5.4.8 PriceTakerModel.append_lmp_data

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 221

**Signature** `def append_lmp_data(`

**Summary** Appends the locational marginal price (LMP) data to the

**Calls**

- `ConfigurationError`
- `len`

### 5.4.9 PriceTakerModel.get_optimal_representative_days

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 262

**Signature** `def get_optimal_representative_days(`

**Summary** Returns the optimal number of representative days for the

**Calls**

- `_assert_lmp_data_exists`
- `generate_daily_data`
- `get_optimal_num_clusters`

### 5.4.10 PriceTakerModel._assert_mp_model_exists

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 283

**Signature** `def _assert_mp_model_exists(self):`

**Summary** Raise an error if the multiperiod model does not exist

**Calls**

- `hasattr`
- `ConfigurationError`

### 5.4.11 PriceTakerModel.build_multiperiod_model

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 291

**Signature** `def build_multiperiod_model(`

**Summary** Builds the multiperiod model.

**Calls**

- `_assert_lmp_data_exists`
- `hasattr`
- `RangeSet`
- `ConcreteModel`
- `flowsheet_func`
- `Block`
- `component_data_objects`
- `ConfigurationError`
- `cluster_lmp_data`
- `len`
- `transfer_attributes_from`
- `info`
- `clone`
- `isinstance`
- `add_linking_constraints`
- `add_periodic_constraints`

- add_startup_shutdown
- add_capacity_limits
- add_ramping_limits
- enumerate

### 5.4.12 PriceTakerModel._get_operation_blocks

**Kind** method

**Defined in file** pricetaker/price_taker_model.py at line 445

**Signature** def _get_operation_blocks(

**Summary** Returns a dictionary of operational blocks named 'blk_name'.

**Calls**

- _assert_mp_model_exists
- AttributeError
- find_component
- hasattr

### 5.4.13 PriceTakerModel._get_operation_vars

**Kind** method

**Defined in file** pricetaker/price_taker_model.py at line 482

**Signature** def _get_operation_vars(self, var_name: str):

**Summary** Returns a dictionary of pointers to the var_name variable located in each flowsheet

**Calls**

- _assert_mp_model_exists
- AttributeError
- find_component

### 5.4.14 PriceTakerModel.fix_operation_var

**Kind** method

**Defined in file** pricetaker/price_taker_model.py at line 504

**Signature** `def fix_operation_var(self, var_name: str, value: Union[int, float]):`

**Summary** Fixes the specified variable to the given value in all operation blocks

**Calls**

- `_get_operation_vars`
- `fix`

### 5.4.15 PriceTakerModel.unfix_operation_var

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 518

**Signature** `def unfix_operation_var(self, var_name: str):`

**Summary** Unfixes the specified variable in all operation blocks

**Calls**

- `_get_operation_vars`
- `unfix`

### 5.4.16 PriceTakerModel.update_operation_params

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 530

**Signature** `def update_operation_params(self, params: dict):`

**Summary** This method updates the time-varying parameter values. This

**Calls**

- `_assert_mp_model_exists`
- `items`
- `NotImplementedError`
- `list`
- `_get_operation_vars`
- `len`
- `ConfigurationError`
- `set_value`

### 5.4.17 PriceTakerModel.add_linking_constraints

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 564

**Signature** `def add_linking_constraints(self, previous_time_var: str, current_time_var: str):`

**Summary** Adds constraints to relate variables across two consecutive time periods. This method is

**Calls**

- `_get_operation_vars`
- `setattr`
- `info`
- `Constraint`
- `str`

### 5.4.18 PriceTakerModel.add_linking_constraints._rule_linking_constraints

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 582

**Signature** `def _rule_linking_constraints(_, d, t):`

**Summary** No docstring.

### 5.4.19 PriceTakerModel.add_periodic_constraints

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 598

**Signature** `def add_periodic_constraints(self, initial_time_var: str, final_time_var: str):`

**Summary** Adds constraints to relate variables across the entire time horizon. This method is

**Calls**

- _get_operation_vars
- setattr
- info
- Constraint
- str

### 5.4.20 PriceTakerModel.add_periodic_constraints._rule_linking_constraints

**Kind** method

**Defined in file** pricetaker/price_taker_model.py at line 614

**Signature** def _rule_linking_constraints(_, d):

**Summary** No docstring.

### 5.4.21 PriceTakerModel._retrieve_uc_data

**Kind** method

**Defined in file** pricetaker/price_taker_model.py at line 628

**Signature** def _retrieve_uc_data(self, op_blk: str, commodity: str, **kwargs):

**Summary** Retrieves unit commitment data for the given operation

**Calls**

- update
- UnitCommitmentData

### 5.4.22 PriceTakerModel._get_valid_block_name

**Kind** method

**Defined in file** pricetaker/price_taker_model.py at line 645

**Signature** def _get_valid_block_name(blk_name: str):

**Summary** Returns a valid python variable name for the given block

**Calls**

- `sub`
- `rstrip`
- `split`

### 5.4.23 PriceTakerModel.add_capacity_limits

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 654

**Signature** `def add_capacity_limits(`

**Summary** Adds capacity limit constraints of the form:

**Calls**

- `_get_operation_blocks`
- `_retrieve_uc_data`
- `hasattr`
- `setattr`
- `getattr`
- `info`
- `_get_valid_block_name`
- `ConfigurationError`
- `Block`
- `capacity_limits`

### 5.4.24 PriceTakerModel.add_ramping_limits

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 723

**Signature** `def add_ramping_limits(`

**Summary** Adds ramping constraints of the form:

**Calls**

- _get_operation_blocks
- _retrieve_uc_data
- assert_ramping_args_present
- hasattr
- setattr
- getattr
- info
- _get_valid_block_name
- ConfigurationError
- Block
- ramping_limits

### 5.4.25 PriceTakerModel.add_startup_shutdown

**Kind** method

**Defined in file** pricetaker/price_taker_model.py at line 813

**Signature** def add_startup_shutdown(

**Summary** Adds minimum uptime/downtime constraints for a given unit/process

**Calls**

- _get_operation_blocks
- hasattr
- setattr
- getattr
- info
- find_component
- _get_valid_block_name
- ConfigurationError
- Block
- startup_shutdown_constraints
- AttributeError

### 5.4.26 PriceTakerModel.add_hourly_cashflows

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 897

**Signature** `def add_hourly_cashflows(`

**Summary** Adds an expression for the net cash inflow for each flowsheet

**Calls**

- `_assert_mp_model_exists`
- `info`
- `warning`
- `component_data_objects`
- `Expression`
- `find_component`

### 5.4.27 PriceTakerModel.add_overall_cashflows

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 997

**Signature** `def add_overall_cashflows(`

**Summary** Builds overall cashflow expressions.

**Calls**

- `_assert_mp_model_exists`
- `component_data_objects`
- `Block`
- `Var`
- `Constraint`
- `Expression`
- `info`
- `ConfigurationError`
- `isinstance`
- `warning`
- `sum`

### 5.4.28 PriceTakerModel.add_objective_function

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1123

**Signature** `def add_objective_function(self, objective_type="npv"):`

**Summary** Appends the objective function to the model.

**Calls**

- `ConfigurationError`
- `Objective`
- `getattr`

### 5.4.29 PriceTakerModel._get_num

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1155

**Signature** `def _get_num(self, op_block_name: str, attribute: str):`

**Summary** Returns the number of times the given operation block undergoes an event,

**Calls**

- `_get_operation_blocks`
- `sum`
- `AttributeError`
- `getattr`
- `values`

### 5.4.30 PriceTakerModel.get_num_startups

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1182

**Signature** `def get_num_startups(self, op_block_name: str):`

**Summary** Returns the number of times the given operation block undergoes startup

**Calls**

- _get_num

### 5.4.31 PriceTakerModel.get_num_shutdowns

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1186

**Signature** `def get_num_shutdowns(self, op_block_name: str):`

**Summary** Returns the number of times the given operation block undergoes shutdown

**Calls**

- _get_num

### 5.4.32 PriceTakerModel._get_pyomo_obj_value

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1191

**Signature** `def _get_pyomo_obj_value(var_or_expr: Union[Var, Expression]):`

**Summary** Returns the value of a given variable/expression

**Calls**

- pyo_value

### 5.4.33 PriceTakerModel.get_operation_var_values

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1202

**Signature** `def get_operation_var_values(self, var_list: Optional[list] = None):`

**Summary** Returns a DataFrame of all operation values

**Calls**

- DataFrame
- component_data_objects
- _get_pyomo_obj_value
- split
- find_component

### 5.4.34 PriceTakerModel.get_design_var_values

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1245

**Signature** `def get_design_var_values(self, var_list: Optional[list] = None):`

**Summary** Returns the values of non-time-varying variables and expressions.

**Calls**

- `component_data_objects`
- `hasattr`
- `_get_pyomo_obj_value`
- `isinstance`
- `find_component`

### 5.4.35 PriceTakerModel.plot_operation_profile

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1281

**Signature** `def plot_operation_profile(`

**Summary** Returns a plot object containing operational profiles for a specific

**Calls**

- `get_operation_var_values`
- `subplots`
- `tight_layout`
- `list`
- `len`
- `plot`
- `set_title`
- `twinx`
- `tick_params`
- `pop`

### 5.4.36 PriceTakerModel.plot_lmp_histogram

**Kind** method

**Defined in file** `pricetaker/price_taker_model.py` at line 1352

**Signature** `def plot_lmp_histogram(self):`

**Summary** Returns a histogram of the specified LMP signal

**Calls**

- `_assert_lmp_data_exists`
- `NotImplementedError`

## 5.5 unit_commitment.py

**File path** `pricetaker/unit_commitment.py`

**Module docstring** This module contains functions that build unit commitment-type

**Definitions Index**

- **class**: `UnitCommitmentData` (line 26)
- **method**: `UnitCommitmentData.__init__` (line 29)
- **method**: `UnitCommitmentData._get_config` (line 35)
- **method**: `UnitCommitmentData.update` (line 84)
- **method**: `UnitCommitmentData.assert_startup_rate_validity` (line 90)
- **method**: `UnitCommitmentData.assert_shutdown_rate_validity` (line 103)
- **method**: `UnitCommitmentData.assert_ramping_args_present` (line 116)
- **function**: `startup_shutdown_constraints` (line 137)
- **method**: `startup_shutdown_constraints.binary_relationship_con` (line 150)
- **method**: `startup_shutdown_constraints.minimum_up_time_con` (line 160)
- **method**: `startup_shutdown_constraints.minimum_down_time_con` (line 170)
- **function**: `capacity_limits` (line 180)
- **method**: `capacity_limits.capacity_low_limit_con` (line 197)
- **method**: `capacity_limits.capacity_high_limit_con` (line 201)
- **function**: `ramping_limits` (line 205)
- **method**: `ramping_limits.ramp_up_con` (line 222)
- **method**: `ramping_limits.ramp_down_con` (line 233)

**Called functions/methods in this file**

- line 32: `_get_config`
- line 33: `update`
- line 36: `ConfigDict`
- line 37: `declare`
- line 39: `ConfigValue`
- line 40: `is_in_range`
- line 86: `set_value`
- line 87: `assert_startup_rate_validity`
- line 88: `assert_shutdown_rate_validity`
- line 97: `ConfigurationError`
- line 149: `Constraint`
- line 165: `sum`
- line 165: `range`
- line 194: `getattr`
- line 194: `items`

**Definition Notes**

### 5.5.1 UnitCommitmentData

**Kind** class

**Defined in file** `pricetaker/unit_commitment.py` at line 26

**Signature** `class UnitCommitmentData:`

**Summary** Dataclass to store startup, shutdown, and ramp rates

**Calls**

- `_get_config`
- `update`
- `ConfigDict`
- `declare`
- `set_value`

- assert_startup_rate_validity
- assert_shutdown_rate_validity
- ConfigValue
- ConfigurationError
- is_in_range

### 5.5.2 UnitCommitmentData.__init__

**Kind** method

**Defined in file** pricetaker/unit_commitment.py at line 29

**Signature** `def __init__(self, blk_name: str, commodity_name: str, **kwargs):`

**Summary** No docstring.

**Calls**

- _get_config
- update

### 5.5.3 UnitCommitmentData._get_config

**Kind** method

**Defined in file** pricetaker/unit_commitment.py at line 35

**Signature** `def _get_config(self):`

**Summary** No docstring.

**Calls**

- ConfigDict
- declare
- ConfigValue
- is_in_range

### 5.5.4 UnitCommitmentData.update

**Kind** method

**Defined in file** pricetaker/unit_commitment.py at line 84

**Signature** `def update(self, **kwargs):`

**Summary** Updates the attribute values

**Calls**

- `set_value`
- `assert_startup_rate_validity`
- `assert_shutdown_rate_validity`

### 5.5.5 UnitCommitmentData.assert_startup_rate_validity

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 90

**Signature** `def assert_startup_rate_validity(self):`

**Summary** Raises an error if startup rate value is not valid.

**Calls**

- `ConfigurationError`

### 5.5.6 UnitCommitmentData.assert_shutdown_rate_validity

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 103

**Signature** `def assert_shutdown_rate_validity(self):`

**Summary** Raises an error if shutdown rate value is not valid

**Calls**

- `ConfigurationError`

### 5.5.7 UnitCommitmentData.assert_ramping_args_present

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 116

**Signature** `def assert_ramping_args_present(self):`

**Summary** Raises an error if any of the arguments needed for the ramping

**Calls**

- `ConfigurationError`

### 5.5.8 startup_shutdown_constraints

**Kind** function

**Defined in file** `pricetaker/unit_commitment.py` at line 137

**Signature** `def startup_shutdown_constraints(`

**Summary** Appends startup and shutdown constraints for a given unit/process

**Calls**

- `Constraint`
- `sum`
- `range`

### 5.5.9 startup_shutdown_constraints.binary_relationship_con

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 150

**Signature** `def binary_relationship_con(_, t):`

**Summary** No docstring.

**Calls**

- `Constraint`

### 5.5.10 startup_shutdown_constraints.minimum_up_time_con

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 160

**Signature** `def minimum_up_time_con(_, t):`

**Summary** No docstring.

**Calls**

- `Constraint`
- `sum`
- `range`

### 5.5.11 startup_shutdown_constraints.minimum_down_time_con

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 170

**Signature** `def minimum_down_time_con(_, t):`

**Summary** No docstring.

**Calls**

- `Constraint`
- `sum`
- `range`

### 5.5.12 capacity_limits

**Kind** function

**Defined in file** `pricetaker/unit_commitment.py` at line 180

**Signature** `def capacity_limits(`

**Summary** Appends capacity limit constraints

**Calls**

- `Constraint`
- `getattr`
- `items`

### 5.5.13 capacity_limits.capacity_low_limit_con

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 197

**Signature** `def capacity_low_limit_con(_, t):`

**Summary** No docstring.

**Calls**

- `Constraint`

### 5.5.14 capacity_limits.capacity_high_limit_con

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 201

**Signature** `def capacity_high_limit_con(_, t):`

**Summary** No docstring.

**Calls**

- `Constraint`

### 5.5.15 ramping_limits

**Kind** function

**Defined in file** `pricetaker/unit_commitment.py` at line 205

**Signature** `def ramping_limits(`

**Summary** Appends ramping constraints

**Calls**

- `Constraint`
- `getattr`
- `items`

### 5.5.16 ramping_limits.ramp_up_con

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 222

**Signature** `def ramp_up_con(_, t):`

**Summary** No docstring.

**Calls**

- `Constraint`

### 5.5.17 ramping_limits.ramp_down_con

**Kind** method

**Defined in file** `pricetaker/unit_commitment.py` at line 233

**Signature** `def ramp_down_con(_, t):`

**Summary** No docstring.

**Calls**

- `Constraint`

# Chapter 6

# pricetaker/tests

## 6.1 __init__.py

**File path** `pricetaker/tests/__init__.py`

## 6.2 lmp_data.csv

**File path** `pricetaker/tests/lmp_data.csv`

**File type** .csv

**Preview**

```
time,BaseCase_2030
1,17.6239982
2,21.22599985
3,26.69300001
4,26.693
5,26.69300001
6,26.693
7,26.69300001
8,26.693
9,26.693
10,26.69300001
11,26.69300002
12,26.69300004
13,26.69300009
14,26.69300047
15,26.69300003
16,26.69300001
17,26.693
18,26.693
19,26.693
```

## 6.3 test_clustering.py

**File path** `pricetaker/tests/test_clustering.py`

### Definitions Index

- **function**: `sample_data_fixture` (line 33)
- **function**: `dummy_data_fixture` (line 41)
- **function**: `test_daily_data_logger_message1` (line 76)
- **function**: `test_daily_data_logger_message2` (line 86)
- **function**: `test_generate_daily_data` (line 101)
- **function**: `test_cluster_lmp_data` (line 116)
- **function**: `test_optimal_num_clusters_elbow` (line 134)
- **function**: `test_optimal_num_clusters_silhouette` (line 162)
- **function**: `test_optimal_clusters_logger_message1` (line 187)
- **function**: `test_optimal_clusters_logger_message2` (line 206)
- **function**: `test_optimal_clusters_logger_message3` (line 225)
- **function**: `test_optimal_clusters_logger_message4` (line 244)
- **function**: `test_optimal_clusters_logger_message5` (line 267)
- **function**: `test_normalize_values` (line 280)

### Called functions/methods in this file

- line 29: `importorskip`
- line 36: `read_csv`
- line 32: `fixture`
- line 78: `raises`
- line 82: `generate_daily_data`
- line 89: `at_level`
- line 90: `list`
- line 90: `range`
- line 96: `isinstance`
- line 107: `equals`
- line 108: `transpose`

- line 108: `DataFrame`
- line 119: `cluster_lmp_data`
- line 113: `skipif`
- line 137: `tolist`
- line 139: `get_optimal_num_clusters`
- line 151: `gcf`
- line 153: `gca`
- line 155: `close`
- line 284: `_normalize_values`

**Definition Notes**

### 6.3.1 sample_data_fixture

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 33

**Signature** `def sample_data_fixture():`

**Summary** Returns a sample price signal for testing

**Calls**

- `fixture`
- `read_csv`

### 6.3.2 dummy_data_fixture

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 41

**Signature** `def dummy_data_fixture():`

**Summary** Returns dummy data for clustering

**Calls**

- `fixture`

### 6.3.3 test_daily_data_logger_message1

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 76

**Signature** `def test_daily_data_logger_message1():`

**Summary** Tests ValueError in generate_daily_data function

**Calls**

- `raises`
- `generate_daily_data`

### 6.3.4 test_daily_data_logger_message2

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 86

**Signature** `def test_daily_data_logger_message2(caplog):`

**Summary** Tests non-integer-multiple length warning in generate_daily_data function

**Calls**

- `isinstance`
- `at_level`
- `generate_daily_data`
- `list`
- `range`

### 6.3.5 test_generate_daily_data

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 101

**Signature** `def test_generate_daily_data():`

**Summary** Tests the generate_daily_data function

**Calls**

- `list`
- `generate_daily_data`
- `equals`
- `range`
- `transpose`
- `DataFrame`

### 6.3.6 test_cluster_lmp_data

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 116

**Signature** `def test_cluster_lmp_data(dummy_data):`

**Summary** Tests the cluster_lmp_data function

**Calls**

- `skipif`
- `cluster_lmp_data`

### 6.3.7 test_optimal_num_clusters_elbow

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 134

**Signature** `def test_optimal_num_clusters_elbow(sample_data):`

**Summary** Tests the get_optimal_num_clusters function

**Calls**

- `skipif`
- `generate_daily_data`
- `get_optimal_num_clusters`
- `close`
- `tolist`
- `gcf`
- `gca`

### 6.3.8 test_optimal_num_clusters_silhouette

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 162

**Signature** `def test_optimal_num_clusters_silhouette(dummy_data):`

**Summary** Tests the get_optimal_num_clusters function

**Calls**

- `skipif`
- `generate_daily_data`
- `get_optimal_num_clusters`
- `close`
- `gcf`
- `gca`

### 6.3.9 test_optimal_clusters_logger_message1

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 187

**Signature** `def test_optimal_clusters_logger_message1(dummy_data):`

**Summary** Tests the error message associated with kmin >= kmax

**Calls**

- `skipif`
- `generate_daily_data`
- `raises`
- `get_optimal_num_clusters`

### 6.3.10 test_optimal_clusters_logger_message2

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 206

**Signature** `def test_optimal_clusters_logger_message2(dummy_data):`

**Summary** Tests the error message associated with kmin $< 2$

**Calls**

- `skipif`
- `generate_daily_data`
- `raises`
- `get_optimal_num_clusters`

### 6.3.11 test_optimal_clusters_logger_message3

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 225

**Signature** `def test_optimal_clusters_logger_message3(dummy_data):`

**Summary** Tests the error message associated with kmax >= len(samples)

**Calls**

- `skipif`
- `generate_daily_data`
- `raises`
- `get_optimal_num_clusters`

### 6.3.12 test_optimal_clusters_logger_message4

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 244

**Signature** `def test_optimal_clusters_logger_message4(dummy_data):`

**Summary** Tests the error message associated with unrecognized method

**Calls**

- `skipif`
- `generate_daily_data`
- `raises`
- `get_optimal_num_clusters`

### 6.3.13 test_optimal_clusters_logger_message5

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 267

**Signature** `def test_optimal_clusters_logger_message5(dummy_data, caplog):`

**Summary** Tests the logger warning associated with n_clusters close to kmax

**Calls**

- `skipif`
- `generate_daily_data`
- `at_level`
- `get_optimal_num_clusters`

### 6.3.14 test_normalize_values

**Kind** function

**Defined in file** `pricetaker/tests/test_clustering.py` at line 280

**Signature** `def test_normalize_values():`

**Summary** Tests the normalize_values function

**Calls**

- `_normalize_values`

## 6.4 test_design_and_operation_models.py

**File path** `pricetaker/tests/test_design_and_operation_models.py`

**Definitions Index**

- **function**: `test_format_data` (line 31)
- **function**: `test_is_valid_variable_design_data` (line 56)
- **function**: `test_is_valid_polynomial_surrogate_data` (line 87)
- **function**: `test_design_model_class` (line 111)
- **method**: `test_design_model_class._my_design_model` (line 114)
- **method**: `test_design_model_class.dummy_func` (line 152)

- **function**: test_design_model_class_logger_message1 (line 170)
- **function**: test_design_model_with_fixed_design (line 195)
- **function**: test_design_model_with_variable_design (line 208)
- **function**: test_operation_model_class (line 241)
- **method**: test_operation_model_class.des_model (line 244)
- **method**: test_operation_model_class.op_model (line 249)
- **function**: test_operation_model_class_logger_message1 (line 275)
- **method**: test_operation_model_class_logger_message1.des_model (line 282)
- **function**: test_operation_model_with_surrogates (line 300)
- **function**: test_is_valid_data_type_for_storage_model (line 359)
- **function**: test_storage_model_fixed_design (line 401)
- **function**: test_storage_model_variable_design (line 438)

### Called functions/methods in this file

- line 33: ConcreteModel
- line 34: Var
- line 35: Param
- line 38: _format_data
- line 48: raises
- line 61: is_valid_variable_design_data
- line 91: is_valid_polynomial_surrogate_data
- line 116: Constraint
- line 120: Expression
- line 124: DesignModel
- line 135: hasattr
- line 137: isinstance
- line 175: clear
- line 231: str
- line 258: OperationModel
- line 326: build_polynomial_surrogates

- line 337: `build_expressions`
- line 375: `_is_valid_data_type_for_storage_model`
- line 404: `StorageModel`

**Definition Notes**

### 6.4.1 test_format_data

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 31

**Signature** `def test_format_data():`

**Summary** Tests the _format_data function

**Calls**

- `ConcreteModel`
- `Var`
- `Param`
- `_format_data`
- `raises`

### 6.4.2 test_is_valid_variable_design_data

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 56

**Signature** `def test_is_valid_variable_design_data():`

**Summary** Tests the is_valid_variable_design_data function

**Calls**

- `is_valid_variable_design_data`
- `raises`

### 6.4.3 test_is_valid_polynomial_surrogate_data

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 87

**Signature** `def test_is_valid_polynomial_surrogate_data():`

**Summary** Tests the is_valid_surrogate_data function

**Calls**

- `is_valid_polynomial_surrogate_data`
- `raises`

### 6.4.4 test_design_model_class

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 111

**Signature** `def test_design_model_class(caplog):`

**Summary** Tests the DesignModel class

**Calls**

- `ConcreteModel`
- `DesignModel`
- `isinstance`
- `hasattr`
- `Var`
- `Constraint`
- `Expression`

### 6.4.5 test_design_model_class._my_design_model

**Kind** method

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 114

**Signature** `def _my_design_model(m, p_min, p_max, cost):`

**Summary** No docstring.

**Calls**

- `Var`
- `Constraint`
- `Expression`

### 6.4.6 test_design_model_class.dummy_func

**Kind** method

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 152

**Signature** `def dummy_func(_):`

**Summary** No docstring.

### 6.4.7 test_design_model_class_logger_message1

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 170

**Signature** `def test_design_model_class_logger_message1(caplog):`

**Summary** Tests the DesignModel class logger message when no

**Calls**

- `clear`
- `ConcreteModel`
- `DesignModel`

### 6.4.8 test_design_model_with_fixed_design

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 195

**Signature** `def test_design_model_with_fixed_design():`

**Summary** Tests the DesignModel class with fixed_design_data argument

**Calls**

- `ConcreteModel`
- `DesignModel`
- `isinstance`

### 6.4.9 test_design_model_with_variable_design

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 208

**Signature** `def test_design_model_with_variable_design():`

**Summary** Tests the DesignModel class with variable_design_data argument

**Calls**

- `ConcreteModel`
- `DesignModel`
- `isinstance`
- `str`

### 6.4.10 test_operation_model_class

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 241

**Signature** `def test_operation_model_class():`

**Summary** Tests the OperationModel class

**Calls**

- `ConcreteModel`
- `DesignModel`
- `OperationModel`
- `isinstance`
- `Var`
- `Expression`
- `hasattr`

### 6.4.11 test_operation_model_class.des_model

**Kind** method

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 244

**Signature** `def des_model(m):`

**Summary** No docstring.

**Calls**

- `Var`

### 6.4.12 test_operation_model_class.op_model

**Kind** method

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 249

**Signature** `def op_model(m, des_blk):`

**Summary** No docstring.

**Calls**

- `Var`
- `Expression`

### 6.4.13 test_operation_model_class_logger_message1

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 275

**Signature** `def test_operation_model_class_logger_message1(caplog):`

**Summary** Testst the OperationModel class when no input

**Calls**

- `clear`
- `ConcreteModel`
- `DesignModel`
- `OperationModel`
- `Var`

### 6.4.14 test_operation_model_class_logger_message1.des_model

**Kind** method

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 282

**Signature** `def des_model(m):`

**Summary** No docstring.

**Calls**

- `Var`

### 6.4.15 test_operation_model_with_surrogates

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 300

**Signature** `def test_operation_model_with_surrogates():`

**Summary** Tests the OperationModel with additional surrogate models

**Calls**

- `ConcreteModel`
- `DesignModel`
- `OperationModel`
- `isinstance`
- `build_polynomial_surrogates`
- `build_expressions`
- `str`

### 6.4.16 test_is_valid_data_type_for_storage_model

**Kind** function

**Defined in file** `pricetaker/tests/test_design_and_operation_models.py` at line 359

**Signature** `def test_is_valid_data_type_for_storage_model():`

**Summary** Tests the _is_valid_data_type_for_storage_model function

**Calls**

- `ConcreteModel`
- `DesignModel`
- `Var`
- `Param`
- `Expression`

- raises
- _is_valid_data_type_for_storage_model
- str

### 6.4.17 test_storage_model_fixed_design

**Kind** function

**Defined in file** pricetaker/tests/test_design_and_operation_models.py at line 401

**Signature** `def test_storage_model_fixed_design():`

**Summary** Tests the StorageModel class with int/float-type arguments

**Calls**

- ConcreteModel
- StorageModel
- hasattr
- str

### 6.4.18 test_storage_model_variable_design

**Kind** function

**Defined in file** pricetaker/tests/test_design_and_operation_models.py at line 438

**Signature** `def test_storage_model_variable_design():`

**Summary** Tests the storage model with a variable design

**Calls**

- ConcreteModel
- Var
- StorageModel
- str

## 6.5 test_price_taker_model.py

**File path** pricetaker/tests/test_price_taker_model.py

**Definitions Index**


- **function**: simple_flowsheet_func (line 36)
- **function**: foo_design_model (line 49)
- **function**: foo_operation_model (line 66)
- **function**: build_foo_flowsheet (line 79)
- **function**: build_power_cycle_with_storage (line 90)
- **function**: initialized_foo_flowsheet_fixture (line 125)
- **function**: test_num_representative_days (line 153)
- **function**: test_horizon_length (line 185)
- **function**: test_append_lmp_data (line 226)
- **function**: test_get_optimal_representative_days (line 259)
- **function**: test_mp_model_full_year (line 287)
- **function**: test_build_mp_model_w_period (line 332)
- **function**: test_mp_model_rep_days (line 351)
- **function**: test_get_operation_blocks (line 386)
- **function**: test_get_operation_blocks_rep_days (line 435)
- **function**: test_get_operation_vars (line 453)
- **function**: test_get_operation_vars_rep_days (line 501)
- **function**: test_update_operation_params (line 527)
- **function**: test_update_operation_params_errors (line 551)
- **function**: test_add_linking_constraints (line 567)
- **function**: test_add_linking_constraints_rep_days (line 599)
- **function**: test_add_periodic_constraints (line 618)
- **function**: test_add_periodic_constraints_rep_days (line 635)
- **function**: test_retrieve_uc_data (line 656)
- **function**: test_add_capacity_limits (line 713)
- **function**: test_add_capacity_limits_rep_days (line 756)
- **function**: test_add_ramping_limits (line 774)
- **function**: test_add_ramping_limits_rep_days (line 834)

- **function**: `test_start_shut_no_install` (line 862)
- **function**: `test_start_shut_with_install` (line 910)
- **function**: `test_start_shut_with_rep_days` (line 961)
- **function**: `test_automatic_addition_operational_constr` (line 1004)
- **function**: `test_automatic_addition_operational_rep_days` (line 1029)
- **function**: `test_add_hourly_cashflows_warnings` (line 1051)
- **function**: `test_add_hourly_cashflows` (line 1085)
- **function**: `test_add_overall_cashflows_warnings` (line 1127)
- **function**: `test_add_overall_cashflows` (line 1189)
- **function**: `test_add_objective_function` (line 1215)
- **function**: `test_get_valid_block_name` (line 1250)
- **function**: `test_get_num_startups_shutdowns` (line 1269)
- **function**: `test_get_num_startups_shutdowns_rep_days` (line 1297)
- **function**: `test_get_num_startups_shutdowns_exception` (line 1327)
- **function**: `test_get_pyomo_obj_value` (line 1359)
- **function**: `test_get_operation_var_values` (line 1381)
- **function**: `test_get_design_var_values` (line 1413)
- **function**: `test_plot_operation_profile` (line 1448)
- **function**: `test_plot_lmp_histogram` (line 1467)

### Called functions/methods in this file

- line 33: `importorskip`
- line 38: `Var`
- line 40: `Constraint`
- line 41: `Param`
- line 43: `Block`
- line 74: `Expression`
- line 81: `OperationModel`
- line 121: `StorageModel`
- line 127: `PriceTakerModel`

- line 128: `append_lmp_data`
- line 129: `DesignModel`
- line 134: `build_multiperiod_model`
- line 140: `list`
- line 140: `range`
- line 124: `fixture`
- line 165: `raises`
- line 190: `at_level`
- line 232: `Series`
- line 271: `get_optimal_representative_days`
- line 256: `skipif`
- line 308: `len`
- line 311: `isinstance`
- line 316: `enumerate`
- line 323: `approx`
- line 326: `component_data_objects`
- line 327: `hasattr`
- line 400: `_get_operation_blocks`
- line 467: `_get_operation_vars`
- line 488: `fix_operation_var`
- line 490: `value`
- line 492: `unfix_operation_var`
- line 494: `is_fixed`
- line 535: `update_operation_params`
- line 572: `add_linking_constraints`
- line 577: `str`
- line 623: `add_periodic_constraints`
- line 643: `pprint`
- line 662: `_retrieve_uc_data`

- line 705: `keys`
- line 733: `copy`
- line 735: `add_capacity_limits`
- line 805: `add_ramping_limits`
- line 877: `add_startup_shutdown`
- line 931: `del_component`
- line 932: `find_component`
- line 1066: `add_hourly_cashflows`
- line 1141: `add_overall_cashflows`
- line 1228: `add_objective_function`
- line 1262: `items`
- line 1264: `_get_valid_block_name`
- line 1286: `fix`
- line 1289: `get_num_startups`
- line 1290: `get_num_shutdowns`
- line 1371: `_get_pyomo_obj_value`
- line 1375: `values`
- line 1385: `get_operation_var_values`
- line 1387: `set`
- line 1402: `to_list`
- line 1404: `sum`
- line 1404: `isna`

**Definition Notes**

### 6.5.1 simple_flowsheet_func

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 36

**Signature** `def simple_flowsheet_func(m):`

**Summary** Dummy flowsheet function for testing

**Calls**

- `Var`
- `Constraint`
- `Param`
- `Block`

### 6.5.2 foo_design_model

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 49

**Signature** `def foo_design_model(m, max_power, min_power):`

**Summary** Dummy design model

**Calls**

- `Param`
- `Var`
- `Constraint`

### 6.5.3 foo_operation_model

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 66

**Signature** `def foo_operation_model(m, design_blk):`

**Summary** Dummy operation model

**Calls**

- `Var`
- `Expression`

### 6.5.4 build_foo_flowsheet

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 79

**Signature** `def build_foo_flowsheet(m, design_blk):`

**Summary** Dummy flowsheet model for testing

**Calls**

- `OperationModel`
- `Expression`

### 6.5.5 build_power_cycle_with_storage

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 90

**Signature** `def build_power_cycle_with_storage(m):`

**Summary** Builds a simple model for a cycle integrated with storage system

**Calls**

- `OperationModel`
- `StorageModel`

### 6.5.6 initialized_foo_flowsheet_fixture

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 125

**Signature** `def initialized_foo_flowsheet_fixture():`

**Summary** Creates a multiperiod model and initializes it

**Calls**

- `fixture`
- `PriceTakerModel`
- `append_lmp_data`
- `DesignModel`
- `build_multiperiod_model`
- `list`
- `range`

### 6.5.7 test_num_representative_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 153

**Signature** `def test_num_representative_days():`

**Summary** Tests the num_representative_days property

**Calls**

- `PriceTakerModel`
- `raises`

### 6.5.8 test_horizon_length

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 185

**Signature** `def test_horizon_length(caplog):`

**Summary** Tests the horizon_length property

**Calls**

- `PriceTakerModel`
- `at_level`
- `raises`

### 6.5.9 test_append_lmp_data

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 226

**Signature** `def test_append_lmp_data():`

**Summary** Tests the append_lmp_data method

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `raises`
- `Series`

### 6.5.10 test_get_optimal_representative_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 259

**Signature** `def test_get_optimal_representative_days(dummy_data):`

**Summary** Tests the get_optimal_representative_days method

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `raises`
- `get_optimal_representative_days`

### 6.5.11 test_mp_model_full_year

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 287

**Signature** `def test_mp_model_full_year(dummy_data):`

**Summary** Tests the build_multiperiod_model using the full year price signal

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `isinstance`
- `raises`
- `len`
- `component_data_objects`
- `approx`
- `hasattr`
- `enumerate`

### 6.5.12 test_build_mp_model_w_period

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 332

**Signature** `def test_build_mp_model_w_period(dummy_data):`

**Summary** Tests the ConfigurationError in build_multiperiod_model method

**Calls**

- `PriceTakerModel`
- `raises`
- `append_lmp_data`
- `build_multiperiod_model`

### 6.5.13 test_mp_model_rep_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 351

**Signature** `def test_mp_model_rep_days(dummy_data):`

**Summary** Tests the build_multiperiod_model using representative days

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `len`
- `isinstance`
- `approx`

### 6.5.14 test_get_operation_blocks

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 386

**Signature** `def test_get_operation_blocks(dummy_data):`

**Summary** Tests the _get_operation_blocks method

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `_get_operation_blocks`
- `raises`
- `len`

### 6.5.15 test_get_operation_blocks_rep_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 435

**Signature** `def test_get_operation_blocks_rep_days(dummy_data):`

**Summary** Tests the get_operation_blocks method with representative days

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `_get_operation_blocks`
- `len`

### 6.5.16 test_get_operation_vars

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 453

**Signature** `def test_get_operation_vars(dummy_data):`

**Summary** Tests the _get_operation_vars method

**Calls**

- PriceTakerModel
- append_lmp_data
- build_multiperiod_model
- _get_operation_vars
- fix_operation_var
- unfix_operation_var
- raises
- len
- value
- is_fixed

### 6.5.17 test_get_operation_vars_rep_days

**Kind** function

**Defined in file** pricetaker/tests/test_price_taker_model.py at line 501

**Signature** `def test_get_operation_vars_rep_days(dummy_data):`

**Summary** Tests the get_operation_vars method with representative days

**Calls**

- skipif
- PriceTakerModel
- append_lmp_data
- build_multiperiod_model
- _get_operation_vars
- fix_operation_var
- unfix_operation_var
- len
- value
- is_fixed

### 6.5.18 test_update_operation_params

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 527

**Signature** `def test_update_operation_params(dummy_data):`

**Summary** Tests the update_operation_params method

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `update_operation_params`
- `value`
- `raises`

### 6.5.19 test_update_operation_params_errors

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 551

**Signature** `def test_update_operation_params_errors(dummy_data):`

**Summary** Tests the update_operation_params error logs

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `raises`
- `update_operation_params`

### 6.5.20 test_add_linking_constraints

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 567

**Signature** `def test_add_linking_constraints(dummy_data):`

**Summary** Tests the add_linking_constraints method

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `add_linking_constraints`
- `len`
- `str`

### 6.5.21 test_add_linking_constraints_rep_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 599

**Signature** `def test_add_linking_constraints_rep_days(dummy_data):`

**Summary** Tests the add_linking_constraints method with representative days

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `add_linking_constraints`
- `len`
- `str`

### 6.5.22 test_add_periodic_constraints

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 618

**Signature** `def test_add_periodic_constraints(dummy_data):`

**Summary** Tests the add_linking_constraints method

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `add_periodic_constraints`
- `len`
- `str`

### 6.5.23 test_add_periodic_constraints_rep_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 635

**Signature** `def test_add_periodic_constraints_rep_days(dummy_data):`

**Summary** Tests the add_periodic_constraints method with representative days

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `add_periodic_constraints`
- `pprint`
- `len`
- `str`

### 6.5.24 test_retrieve_uc_data

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 656

**Signature** `def test_retrieve_uc_data():`

**Summary** Tests the _retrieve_uc_data method

**Calls**

- PriceTakerModel
- _retrieve_uc_data
- isinstance
- len
- approx
- raises
- list
- keys

### 6.5.25 test_add_capacity_limits

**Kind** function

**Defined in file** pricetaker/tests/test_price_taker_model.py at line 713

**Signature** def test_add_capacity_limits(dummy_data):

**Summary** Tests the add_capacity_limits method

**Calls**

- PriceTakerModel
- append_lmp_data
- build_multiperiod_model
- add_capacity_limits
- raises
- copy
- len

### 6.5.26 test_add_capacity_limits_rep_days

**Kind** function

**Defined in file** pricetaker/tests/test_price_taker_model.py at line 756

**Signature** def test_add_capacity_limits_rep_days(dummy_data):

**Summary** Tests the add_capacity_limits method with representative days

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `add_capacity_limits`
- `len`

### 6.5.27 test_add_ramping_limits

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 774

**Signature** `def test_add_ramping_limits(dummy_data):`

**Summary** Tests the add_ramping_limits method

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `DesignModel`
- `build_multiperiod_model`
- `add_ramping_limits`
- `raises`
- `copy`
- `approx`
- `len`

### 6.5.28 test_add_ramping_limits_rep_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 834

**Signature** `def test_add_ramping_limits_rep_days(dummy_data):`

**Summary** Tests the add_ramping_limits method with representative days

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `DesignModel`
- `build_multiperiod_model`
- `add_ramping_limits`
- `len`

### 6.5.29 test_start_shut_no_install

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 862

**Signature** `def test_start_shut_no_install(dummy_data):`

**Summary** Tests the add_startup_shutdown method without install_unit variable.

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `DesignModel`
- `build_multiperiod_model`
- `add_startup_shutdown`
- `hasattr`
- `len`
- `str`
- `raises`

### 6.5.30 test_start_shut_with_install

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 910

**Signature** `def test_start_shut_with_install(dummy_data):`

**Summary** Tests the add_startup_shutdown method with install_unit variable.

**Calls**

- PriceTakerModel
- append_lmp_data
- DesignModel
- build_multiperiod_model
- del_component
- Var
- add_startup_shutdown
- find_component
- raises
- str

### 6.5.31 test_start_shut_with_rep_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 961

**Signature** `def test_start_shut_with_rep_days(dummy_data):`

**Summary** Tests the add_startup_shutdown method with install_unit variable,

**Calls**

- skipif
- PriceTakerModel
- append_lmp_data
- DesignModel
- build_multiperiod_model
- add_startup_shutdown
- len
- str

### 6.5.32 test_automatic_addition_operational_constr

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1004

**Signature** `def test_automatic_addition_operational_constr(dummy_data):`

**Summary** Tests automatic addition of operational constraints if the required data is present

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `hasattr`
- `len`

### 6.5.33 test_automatic_addition_operational_rep_days

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1029

**Signature** `def test_automatic_addition_operational_rep_days(dummy_data):`

**Summary** Tests automatic addition of operational constraints if the required data is present

**Calls**

- `skipif`
- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `hasattr`
- `len`

### 6.5.34 test_add_hourly_cashflows_warnings

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1051

**Signature** `def test_add_hourly_cashflows_warnings(dummy_data, caplog):`

**Summary** Tests the add_hourly_cashflows method with empty args

**Calls**

- PriceTakerModel
- append_lmp_data
- DesignModel
- build_multiperiod_model
- at_level
- add_hourly_cashflows
- str

### 6.5.35 test_add_hourly_cashflows

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1085

**Signature** `def test_add_hourly_cashflows(dummy_data, caplog):`

**Summary** Tests the add_hourly_cashflows method

**Calls**

- PriceTakerModel
- append_lmp_data
- DesignModel
- build_multiperiod_model
- at_level
- add_hourly_cashflows
- str

### 6.5.36 test_add_overall_cashflows_warnings

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1127

**Signature** `def test_add_overall_cashflows_warnings(dummy_data, caplog):`

**Summary** Tests the warning/error messages in add_overall_cashflows

**Calls**

- PriceTakerModel
- append_lmp_data
- build_multiperiod_model
- hasattr
- raises
- add_overall_cashflows
- at_level
- add_hourly_cashflows
- str

### 6.5.37 test_add_overall_cashflows

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1189

**Signature** `def test_add_overall_cashflows(dummy_data):`

**Summary** Tests the add_overall_cashflows method

**Calls**

- PriceTakerModel
- append_lmp_data
- DesignModel
- build_multiperiod_model
- add_hourly_cashflows
- add_overall_cashflows
- str

### 6.5.38 test_add_objective_function

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1215

**Signature** `def test_add_objective_function(dummy_data):`

**Summary** Tests the add_objective_function method

**Calls**

- `PriceTakerModel`
- `append_lmp_data`
- `build_multiperiod_model`
- `add_hourly_cashflows`
- `add_overall_cashflows`
- `add_objective_function`
- `hasattr`
- `raises`

### 6.5.39 test_get_valid_block_name

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1250

**Signature** `def test_get_valid_block_name():`

**Summary** Tests the get_valid_block_name static method

**Calls**

- `PriceTakerModel`
- `items`
- `_get_valid_block_name`

### 6.5.40 test_get_num_startups_shutdowns

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1269

**Signature** `def test_get_num_startups_shutdowns(dummy_data):`

**Summary** Tests the get_num_startups and get_num_shutdowns methods

**Calls**

- PriceTakerModel
- append_lmp_data
- DesignModel
- build_multiperiod_model
- fix
- get_num_startups
- get_num_shutdowns

### 6.5.41 test_get_num_startups_shutdowns_rep_days

**Kind** function

**Defined in file** pricetaker/tests/test_price_taker_model.py at line 1297

**Signature** `def test_get_num_startups_shutdowns_rep_days(dummy_data):`

**Summary** Tests the get_num_startups and get_num_shutdowns methods

**Calls**

- skipif
- PriceTakerModel
- append_lmp_data
- DesignModel
- build_multiperiod_model
- fix
- get_num_startups
- get_num_shutdowns

### 6.5.42 test_get_num_startups_shutdowns_exception

**Kind** function

**Defined in file** pricetaker/tests/test_price_taker_model.py at line 1327

**Signature** `def test_get_num_startups_shutdowns_exception(dummy_data):`

**Summary** Tests the get_num_startups and get_num_shutdowns methods

**Calls**

- PriceTakerModel
- append_lmp_data
- DesignModel
- build_multiperiod_model
- raises
- get_num_startups
- get_num_shutdowns

### 6.5.43 test_get_pyomo_obj_value

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1359

**Signature** `def test_get_pyomo_obj_value():`

**Summary** Tests the _get_pyomo_obj_value function

**Calls**

- PriceTakerModel
- Var
- Expression
- _get_pyomo_obj_value
- values

### 6.5.44 test_get_operation_var_values

**Kind** function

**Defined in file** `pricetaker/tests/test_price_taker_model.py` at line 1381

**Signature** `def test_get_operation_var_values(initialized_foo_flowsheet):`

**Summary** Tests the get_operation_var_values method

**Calls**

- get_operation_var_values
- isinstance
- set
- to_list
- sum
- isna

### 6.5.45 test_get_design_var_values

**Kind** function

**Defined in file** pricetaker/tests/test_price_taker_model.py at line 1413

**Signature** `def test_get_design_var_values(initialized_foo_flowsheet):`

**Summary** Tests the get_design_var_values method

**Calls**

- Var
- Expression
- add_hourly_cashflows
- add_overall_cashflows
- get_design_var_values
- isinstance
- len

### 6.5.46 test_plot_operation_profile

**Kind** function

**Defined in file** pricetaker/tests/test_price_taker_model.py at line 1448

**Signature** `def test_plot_operation_profile(initialized_foo_flowsheet):`

**Summary** Tests the plot_operation_profile method

**Calls**

- plot_operation_profile
- isinstance

### 6.5.47 test_plot_lmp_histogram

**Kind** function

**Defined in file** pricetaker/tests/test_price_taker_model.py at line 1467

**Signature** `def test_plot_lmp_histogram(dummy_data):`

**Summary** Tests the plot_lmp_histogram method

**Calls**

- PriceTakerModel
- append_lmp_data
- raises
- plot_lmp_histogram

## 6.6 test_unit_commitment.py

**File path** pricetaker/tests/test_unit_commitment.py

**Definitions Index**

- **function**: test_unit_commitment_data (line 21)
- **function**: test_startup_shutdown_constraints (line 84)
- **method**: test_startup_shutdown_constraints.op_blk (line 90)
- **function**: test_capacity_limits_with_float (line 162)
- **method**: test_capacity_limits_with_float.op_blk (line 171)
- **function**: test_capacity_limits_with_var (line 205)
- **method**: test_capacity_limits_with_var.op_blk (line 215)
- **function**: test_ramping_limits (line 245)
- **method**: test_ramping_limits.op_blk (line 252)

**Called functions/methods in this file**

- line 23: UnitCommitmentData
- line 32: approx
- line 37: ConcreteModel
- line 38: Var
- line 40: update
- line 47: isinstance
- line 51: raises
- line 80: assert_ramping_args_present
- line 87: RangeSet
- line 89: Block
- line 98: startup_shutdown_constraints
- line 112: len
- line 133: str
- line 179: capacity_limits
- line 268: ramping_limits

**Definition Notes**

### 6.6.1 test_unit_commitment_data

**Kind** function

**Defined in file** pricetaker/tests/test_unit_commitment.py at line 21

**Signature** def test_unit_commitment_data():

**Summary** Tests the UnitCommitmentData class

**Calls**

- UnitCommitmentData
- ConcreteModel
- Var
- update
- isinstance

- `approx`
- `raises`
- `assert_ramping_args_present`

### 6.6.2 test_startup_shutdown_constraints

**Kind** function

**Defined in file** `pricetaker/tests/test_unit_commitment.py` at line 84

**Signature** `def test_startup_shutdown_constraints():`

**Summary** Tests the startup_shutdown_constraints function

**Calls**

- `ConcreteModel`
- `RangeSet`
- `Block`
- `startup_shutdown_constraints`
- `Var`
- `len`
- `str`

### 6.6.3 test_startup_shutdown_constraints.op_blk

**Kind** method

**Defined in file** `pricetaker/tests/test_unit_commitment.py` at line 90

**Signature** `def op_blk(b, _):`

**Summary** No docstring.

**Calls**

- `Block`
- `Var`

### 6.6.4 test_capacity_limits_with_float

**Kind** function

**Defined in file** `pricetaker/tests/test_unit_commitment.py` at line 162

**Signature** `def test_capacity_limits_with_float():`

**Summary** Tests the capacity_limits function with a constant

**Calls**

- `ConcreteModel`
- `RangeSet`
- `Block`
- `UnitCommitmentData`
- `capacity_limits`
- `Var`
- `len`
- `str`

### 6.6.5 test_capacity_limits_with_float.op_blk

**Kind** method

**Defined in file** `pricetaker/tests/test_unit_commitment.py` at line 171

**Signature** `def op_blk(b, _):`

**Summary** No docstring.

**Calls**

- `Block`
- `Var`

### 6.6.6 test_capacity_limits_with_var

**Kind** function

**Defined in file** `pricetaker/tests/test_unit_commitment.py` at line 205

**Signature** `def test_capacity_limits_with_var():`

**Summary** Tests the capacity_limits function with a Pyomo Var

**Calls**

- `ConcreteModel`
- `RangeSet`
- `Var`
- `Block`
- `UnitCommitmentData`
- `capacity_limits`
- `str`

### 6.6.7 test_capacity_limits_with_var.op_blk

**Kind** method

**Defined in file** `pricetaker/tests/test_unit_commitment.py` at line 215

**Signature** `def op_blk(b, _):`

**Summary** No docstring.

**Calls**

- `Block`
- `Var`

### 6.6.8 test_ramping_limits

**Kind** function

**Defined in file** `pricetaker/tests/test_unit_commitment.py` at line 245

**Signature** `def test_ramping_limits():`

**Summary** Tests the ramping_limits constraint

**Calls**

- `ConcreteModel`
- `RangeSet`
- `Var`
- `Block`

- UnitCommitmentData
- ramping_limits
- len
- str

### 6.6.9 test_ramping_limits.op_blk

**Kind** method

**Defined in file** `pricetaker/tests/test_unit_commitment.py` at line 252

**Signature** `def op_blk(b, _):`

**Summary** No docstring.

**Calls**

- Block
- Var

# Chapter 7

# tests

## 7.1 __init__.py

**File path** `tests/__init__.py`

## 7.2 self_scheduler_integration_test_plugin.py

**File path** `tests/self_scheduler_integration_test_plugin.py`

### Called functions/methods in this file

- line 20: `make_testing_tracker`
- line 22: `make_testing_selfscheduler`
- line 25: `DoubleLoopCoordinator`

## 7.3 test_PEM_Parameterized_bidder.py

**File path** `tests/test_PEM_Parameterized_bidder.py`

### Definitions Index

- **function**: `test_creat_PEMParametrizedBidder_with_wrong_PEM_power` (line 34)
- **function**: `wind_df` (line 59)
- **function**: `bidder_object` (line 86)
- **function**: `test_compute_DA_bids` (line 110)
- **function**: `test_compute_RT_bids` (line 152)

**Called functions/methods in this file**

- line 30: `OptSolver`
- line 39: `ExampleModel`
- line 40: `ExampleForecaster`
- line 44: `raises`
- line 47: `PEMParametrizedBidder`
- line 67: `Timestamp`
- line 68: `date_range`
- line 71: `DateOffset`
- line 75: `DataFrame`
- line 90: `wind_df`
- line 91: `PerfectForecaster`
- line 122: `compute_day_ahead_bids`
- line 125: `range`
- line 135: `convert_marginal_costs_to_actual_costs`
- line 145: `assertStructuredAlmostEqual`
- line 107: `skipif`
- line 170: `compute_real_time_bids`
- line 198: `max`

**Definition Notes**

### 7.3.1 test_creat_PEMParametrizedBidder_with_wrong_PEM_power

**Kind** function

**Defined in file** `tests/test_PEM_Parameterized_bidder.py` at line 34

**Signature** `def test_creat_PEMParametrizedBidder_with_wrong_PEM_power():`

**Summary** This is to test when we creat the PEM bidder with a PEM power greater than the renewable power,

**Calls**

- ExampleModel
- ExampleForecaster
- raises
- PEMParametrizedBidder

### 7.3.2 wind_df

**Kind** function

**Defined in file** `tests/test_PEM_Parameterized_bidder.py` at line 59

**Signature** `def wind_df():`

**Summary** This is to define a dataframe fed to the PerfectForecaster with example DA/RT

**Calls**

- Timestamp
- date_range
- DataFrame
- DateOffset

### 7.3.3 bidder_object

**Kind** function

**Defined in file** `tests/test_PEM_Parameterized_bidder.py` at line 86

**Signature** `def bidder_object():`

**Summary** This is to define a bidder object.

**Calls**

- wind_df
- PerfectForecaster
- ExampleModel
- PEMParametrizedBidder

### 7.3.4 test_compute_DA_bids

**Kind** function

**Defined in file** `tests/test_PEM_Parameterized_bidder.py` at line 110

**Signature** `def test_compute_DA_bids(bidder_object):`

**Summary** This is to test the if the compute_DA_bids function works correctly.

**Calls**

- `skipif`
- `compute_day_ahead_bids`
- `range`
- `assertStructuredAlmostEqual`
- `convert_marginal_costs_to_actual_costs`

### 7.3.5 test_compute_RT_bids

**Kind** function

**Defined in file** `tests/test_PEM_Parameterized_bidder.py` at line 152

**Signature** `def test_compute_RT_bids(bidder_object):`

**Summary** This is to test the if the compute_DA_bids function works correctly.

**Calls**

- `skipif`
- `compute_real_time_bids`
- `range`
- `assertStructuredAlmostEqual`
- `convert_marginal_costs_to_actual_costs`
- `max`

## 7.4 test_backcaster.py

**File path** `tests/test_backcaster.py`

**Definitions Index**


- **function**: historical_da_prices (line 21)
- **function**: historical_rt_prices (line 26)
- **function**: base_backcaster (line 31)
- **function**: test_create_backcaster (line 36)
- **function**: test_create_backcaster_with_small_max_historical_days (line 43)
- **function**: test_invalid_max_historical_days_type (line 72)
- **function**: test_invalid_max_historical_days_value (line 83)
- **function**: test_create_backcaster_with_non_dict (line 93)
- **function**: test_create_backcaster_with_empty_dict (line 102)
- **function**: test_create_backcaster_with_non_list (line 109)
- **function**: test_create_backcaster_with_not_enough_entries (line 116)
- **function**: test_create_backcaster_with_non_24_entries (line 127)
- **function**: test_forecast_real_time_prices (line 136)
- **function**: test_forecast_day_ahead_prices (line 154)
- **function**: test_forecast_day_ahead_and_real_time_prices (line 170)
- **function**: test_forecast_nonexistent_bus_prices (line 198)
- **class**: MockPrescientHourlyStats (line 231)
- **method**: MockPrescientHourlyStats.__init__ (line 232)
- **function**: test_fetch_hourly_stats_from_prescient (line 237)
- **function**: test_fetch_hourly_stats_from_prescient_greater_than_max_historical_days (line 270)
- **class**: DAPrices (line 290)
- **method**: DAPrices.__init__ (line 291)
- **method**: DAPrices.get (line 294)
- **class**: MockRucMarket (line 299)
- **method**: MockRucMarket.__init__ (line 300)
- **class**: MockPrescientRucPlan (line 304)
- **method**: MockPrescientRucPlan.__init__ (line 305)
- **function**: test_fetch_day_ahead_stats_from_prescient (line 310)

**Called functions/methods in this file**

- line 32: `Backcaster`
- line 47: `at_level`
- line 61: `assertStructuredAlmostEqual`
- line 75: `raises`
- line 71: `parametrize`
- line 141: `forecast_real_time_prices`
- line 145: `range`
- line 159: `forecast_day_ahead_prices`
- line 175: `forecast_day_ahead_and_real_time_prices`
- line 239: `MockPrescientHourlyStats`
- line 240: `fetch_hourly_stats_from_prescient`
- line 280: `append`
- line 301: `DAPrices`
- line 306: `MockRucMarket`
- line 314: `MockPrescientRucPlan`
- line 315: `fetch_day_ahead_stats_from_prescient`

**Definition Notes**

### 7.4.1 historical_da_prices

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 21

**Signature** `def historical_da_prices():`

**Summary** No docstring.

### 7.4.2 historical_rt_prices

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 26

**Signature** `def historical_rt_prices():`

**Summary** No docstring.

### 7.4.3 base_backcaster

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 31

**Signature** `def base_backcaster(historical_da_prices, historical_rt_prices):`

**Summary** No docstring.

**Calls**

- `Backcaster`

### 7.4.4 test_create_backcaster

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 36

**Signature** `def test_create_backcaster(historical_da_prices, historical_rt_prices):`

**Summary** No docstring.

**Calls**

- `Backcaster`

### 7.4.5 test_create_backcaster_with_small_max_historical_days

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 43

**Signature** `def test_create_backcaster_with_small_max_historical_days(`

**Summary** No docstring.

**Calls**

- `assertStructuredAlmostEqual`
- `at_level`
- `Backcaster`

### 7.4.6 test_invalid_max_historical_days_type

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 72

**Signature** `def test_invalid_max_historical_days_type(`

**Summary** No docstring.

**Calls**

- `parametrize`
- `raises`
- `Backcaster`

### 7.4.7 test_invalid_max_historical_days_value

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 83

**Signature** `def test_invalid_max_historical_days_value(`

**Summary** No docstring.

**Calls**

- `parametrize`
- `raises`
- `Backcaster`

### 7.4.8 test_create_backcaster_with_non_dict

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 93

**Signature** `def test_create_backcaster_with_non_dict(historical_da_prices, historical_rt_prices):`

**Summary** No docstring.

**Calls**

- `raises`
- `Backcaster`

### 7.4.9 test_create_backcaster_with_empty_dict

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 102

**Signature** `def test_create_backcaster_with_empty_dict(historical_da_prices, historical_rt_prices`

**Summary** No docstring.

**Calls**

- `raises`
- `Backcaster`

### 7.4.10 test_create_backcaster_with_non_list

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 109

**Signature** `def test_create_backcaster_with_non_list(historical_da_prices, historical_rt_prices):`

**Summary** No docstring.

**Calls**

- `raises`
- `Backcaster`

### 7.4.11 test_create_backcaster_with_not_enough_entries

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 116

**Signature** `def test_create_backcaster_with_not_enough_entries(`

**Summary** No docstring.

**Calls**

- `raises`
- `Backcaster`

### 7.4.12 test_create_backcaster_with_non_24_entries

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 127

**Signature** `def test_create_backcaster_with_non_24_entries(`

**Summary** No docstring.

**Calls**

- `raises`
- `Backcaster`

### 7.4.13 test_forecast_real_time_prices

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 136

**Signature** `def test_forecast_real_time_prices(base_backcaster):`

**Summary** No docstring.

**Calls**

- `forecast_real_time_prices`
- `assertStructuredAlmostEqual`
- `range`

### 7.4.14 test_forecast_day_ahead_prices

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 154

**Signature** `def test_forecast_day_ahead_prices(base_backcaster):`

**Summary** No docstring.

**Calls**

- `forecast_day_ahead_prices`
- `assertStructuredAlmostEqual`

### 7.4.15 test_forecast_day_ahead_and_real_time_prices

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 170

**Signature** `def test_forecast_day_ahead_and_real_time_prices(base_backcaster):`

**Summary** No docstring.

**Calls**

- `forecast_day_ahead_and_real_time_prices`
- `assertStructuredAlmostEqual`
- `range`

### 7.4.16 test_forecast_nonexistent_bus_prices

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 198

**Signature** `def test_forecast_nonexistent_bus_prices(base_backcaster):`

**Summary** No docstring.

**Calls**

- `raises`
- `forecast_real_time_prices`
- `forecast_day_ahead_prices`

### 7.4.17 MockPrescientHourlyStats

**Kind** class

**Defined in file** `tests/test_backcaster.py` at line 231

**Signature** `class MockPrescientHourlyStats:`

**Summary** No docstring.

### 7.4.18 MockPrescientHourlyStats.__init__

**Kind** method

**Defined in file** `tests/test_backcaster.py` at line 232

**Signature** `def __init__(self, bus_to_prices_dict) -> None:`

**Summary** No docstring.

### 7.4.19 test_fetch_hourly_stats_from_prescient

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 237

**Signature** `def test_fetch_hourly_stats_from_prescient(base_backcaster, historical_rt_prices):`

**Summary** No docstring.

**Calls**

- `MockPrescientHourlyStats`
- `fetch_hourly_stats_from_prescient`
- `assertStructuredAlmostEqual`
- `range`

### 7.4.20 test_fetch_hourly_stats_from_prescient_greater_than_max_historical_days

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 270

**Signature** `def test_fetch_hourly_stats_from_prescient_greater_than_max_historical_days(`

**Summary** No docstring.

**Calls**

- `range`
- `assertStructuredAlmostEqual`
- `MockPrescientHourlyStats`
- `fetch_hourly_stats_from_prescient`
- `append`

### 7.4.21 DAPrices

**Kind** class

**Defined in file** `tests/test_backcaster.py` at line 290

**Signature** `class DAPrices:`

**Summary** No docstring.

### 7.4.22 DAPrices.__init__

**Kind** method

**Defined in file** `tests/test_backcaster.py` at line 291

**Signature** `def __init__(self, da_prices) -> None:`

**Summary** No docstring.

### 7.4.23 DAPrices.get

**Kind** method

**Defined in file** `tests/test_backcaster.py` at line 294

**Signature** `def get(self, info):`

**Summary** No docstring.

### 7.4.24 MockRucMarket

**Kind** class

**Defined in file** `tests/test_backcaster.py` at line 299

**Signature** `class MockRucMarket:`

**Summary** No docstring.

**Calls**

- `DAPrices`

### 7.4.25 MockRucMarket.__init__

**Kind** method

**Defined in file** `tests/test_backcaster.py` at line 300

**Signature** `def __init__(self, da_prices) -> None:`

**Summary** No docstring.

**Calls**

- DAPrices

### 7.4.26 MockPrescientRucPlan

**Kind** class

**Defined in file** `tests/test_backcaster.py` at line 304

**Signature** `class MockPrescientRucPlan:`

**Summary** No docstring.

**Calls**

- MockRucMarket

### 7.4.27 MockPrescientRucPlan.__init__

**Kind** method

**Defined in file** `tests/test_backcaster.py` at line 305

**Signature** `def __init__(self, da_prices) -> None:`

**Summary** No docstring.

**Calls**

- MockRucMarket

### 7.4.28 test_fetch_day_ahead_stats_from_prescient

**Kind** function

**Defined in file** `tests/test_backcaster.py` at line 310

**Signature** `def test_fetch_day_ahead_stats_from_prescient(base_backcaster, historical_da_prices):`

**Summary** No docstring.

### Calls

- `range`
- `assertStructuredAlmostEqual`
- `MockPrescientRucPlan`
- `fetch_day_ahead_stats_from_prescient`

## 7.5 test_bidder.py

**File path** `tests/test_bidder.py`

### Definitions Index

- **class**: `MissingModel` (line 27)
- **method**: `MissingModel.__init__` (line 41)
- **function**: `test_model_object_missing_methods` (line 64)
- **function**: `test_model_object_missing_attr` (line 87)
- **function**: `test_n_scenario_checker` (line 110)
- **function**: `test_solver_checker` (line 140)
- **function**: `bidder_object` (line 160)
- **function**: `test_compute_DA_bids` (line 182)
- **function**: `test_compute_RT_bids` (line 260)
- **function**: `bidder_object_pcost` (line 301)
- **function**: `test_compute_RT_bids_pcost` (line 324)

### Called functions/methods in this file

- line 49: `setattr`
- line 66: `SolverFactory`
- line 67: `ExampleForecaster`
- line 74: `MissingModel`
- line 75: `raises`
- line 75: `format`
- line 76: `Bidder`
- line 114: `ExampleModel`

- line 196: `compute_day_ahead_bids`
- line 212: `range`
- line 215: `assertStructuredAlmostEqual`
- line 245: `append`
- line 179: `skipif`
- line 275: `compute_real_time_bids`

**Definition Notes**

### 7.5.1 MissingModel

**Kind** class

**Defined in file** `tests/test_bidder.py` at line 27

**Signature** `class MissingModel:`

**Summary** A class for testing missing methods and attributes.

**Calls**

- `setattr`

### 7.5.2 MissingModel.__init__

**Kind** method

**Defined in file** `tests/test_bidder.py` at line 41

**Signature** `def __init__(self, missing_method=None, missing_attr=None):`

**Summary** Constructs a model class without the specified missing methods and/or

**Calls**

- `setattr`

### 7.5.3 test_model_object_missing_methods

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 64

**Signature** `def test_model_object_missing_methods():`

**Summary** No docstring.

**Calls**

- `SolverFactory`
- `ExampleForecaster`
- `MissingModel`
- `raises`
- `Bidder`
- `format`

### 7.5.4 test_model_object_missing_attr

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 87

**Signature** `def test_model_object_missing_attr():`

**Summary** No docstring.

**Calls**

- `SolverFactory`
- `ExampleForecaster`
- `MissingModel`
- `raises`
- `Bidder`
- `format`

### 7.5.5 test_n_scenario_checker

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 110

**Signature** `def test_n_scenario_checker():`

**Summary** No docstring.

**Calls**

- `SolverFactory`
- `ExampleForecaster`
- `ExampleModel`
- `raises`
- `Bidder`

### 7.5.6 test_solver_checker

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 140

**Signature** `def test_solver_checker():`

**Summary** No docstring.

**Calls**

- `ExampleForecaster`
- `ExampleModel`
- `raises`
- `Bidder`

### 7.5.7 bidder_object

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 160

**Signature** `def bidder_object():`

**Summary** No docstring.

**Calls**

- `SolverFactory`
- `ExampleForecaster`
- `ExampleModel`
- `Bidder`

### 7.5.8 test_compute_DA_bids

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 182

**Signature** `def test_compute_DA_bids(bidder_object):`

**Summary** No docstring.

**Calls**

- `skipif`
- `compute_day_ahead_bids`
- `assertStructuredAlmostEqual`
- `range`
- `append`

### 7.5.9 test_compute_RT_bids

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 260

**Signature** `def test_compute_RT_bids(bidder_object):`

**Summary** No docstring.

**Calls**

- `skipif`
- `compute_real_time_bids`
- `assertStructuredAlmostEqual`
- `range`

### 7.5.10 bidder_object_pcost

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 301

**Signature** `def bidder_object_pcost():`

**Summary** No docstring.

**Calls**

- `SolverFactory`
- `ExampleForecaster`
- `ExampleModel`
- `Bidder`

### 7.5.11 test_compute_RT_bids_pcost

**Kind** function

**Defined in file** `tests/test_bidder.py` at line 324

**Signature** `def test_compute_RT_bids_pcost(bidder_object_pcost):`

**Summary** No docstring.

**Calls**

- `skipif`
- `compute_real_time_bids`
- `assertStructuredAlmostEqual`
- `range`

## 7.6 test_coordinator.py

**File path** `tests/test_coordinator.py`

**Definitions Index**

- **function**: `coordinator_object` (line 38)
- **function**: `test_static_params` (line 90)
- **function**: `test_assemble_sced_tracking_market_signals` (line 102)

**Called functions/methods in this file**

- line 41: `SolverFactory`
- line 50: `ExampleModel`
- line 51: `Tracker`
- line 68: `ExampleForecaster`
- line 70: `Bidder`

- line 80: DoubleLoopCoordinator
- line 37: fixture
- line 98: _update_static_params
- line 112: range
- line 120: _assemble_sced_tracking_market_signals
- line 125: assertStructuredAlmostEqual

**Definition Notes**

### 7.6.1 coordinator_object

**Kind** function

**Defined in file** tests/test_coordinator.py at line 38

**Signature** `def coordinator_object(request):`

**Summary** No docstring.

**Calls**

- fixture
- SolverFactory
- ExampleModel
- Tracker
- ExampleForecaster
- Bidder
- DoubleLoopCoordinator

### 7.6.2 test_static_params

**Kind** function

**Defined in file** tests/test_coordinator.py at line 90

**Signature** `def test_static_params(coordinator_object):`

**Summary** No docstring.

**Calls**

- _update_static_params

### 7.6.3 test_assemble_sced_tracking_market_signals

**Kind** function

**Defined in file** `tests/test_coordinator.py` at line 102

**Signature** `def test_assemble_sced_tracking_market_signals(coordinator_object):`

**Summary** No docstring.

**Calls**

- `_assemble_sced_tracking_market_signals`
- `assertStructuredAlmostEqual`
- `range`

## 7.7 test_integration.py

**File path** `tests/test_integration.py`

**Definitions Index**

- **class**: `TestDoubleLoopIntegration` (line 44)
- **method**: `TestDoubleLoopIntegration.data_path` (line 48)
- **method**: `TestDoubleLoopIntegration.test_data_path_available` (line 58)
- **method**: `TestDoubleLoopIntegration.output_dir` (line 62)
- **method**: `TestDoubleLoopIntegration.self_scheduler_output_dir` (line 68)
- **method**: `TestDoubleLoopIntegration.self_scheduler_plugin_path` (line 74)
- **method**: `TestDoubleLoopIntegration.test_self_scheduler_plugin_path_is_existing_file` (line 83)
- **method**: `TestDoubleLoopIntegration.prescient_options` (line 89)
- **method**: `TestDoubleLoopIntegration.bidder_sim_options` (line 119)
- **method**: `TestDoubleLoopIntegration.self_scheduler_sim_options` (line 134)
- **method**: `TestDoubleLoopIntegration.run_bidder_simulator` (line 150)
- **method**: `TestDoubleLoopIntegration.run_self_scheduler_simulator` (line 159)
- **method**: `TestDoubleLoopIntegration.simulation_results_dir` (line 170)
- **method**: `TestDoubleLoopIntegration.self_scheduler_simulation_results_dir` (line 174)
- **method**: `TestDoubleLoopIntegration.test_prescient_outputs_exist` (line 180)

**Called functions/methods in this file**

- line 29: `make_testing_tracker`
- line 31: `make_testing_bidder`
- line 34: `DoubleLoopCoordinator`
- line 52: `as_file`
- line 53: `joinpath`
- line 53: `files`
- line 59: `is_dir`
- line 64: `mkdir`
- line 80: `Path`
- line 86: `is_file`
- line 91: `str`
- line 151: `importorskip`
- line 156: `simulate`
- line 156: `Prescient`
- line 183: `isdir`
- line 205: `join`
- line 206: `isfile`

**Definition Notes**

### 7.7.1 TestDoubleLoopIntegration

**Kind** class

**Defined in file** `tests/test_integration.py` at line 44

**Signature** `class TestDoubleLoopIntegration:`

**Summary** Integration test for the double loop using 5bus use case.

**Calls**

- `is_dir`
- `mkdir`
- `is_file`
- `str`
- `importorskip`
- `simulate`
- `isdir`
- `as_file`
- `Path`
- `join`
- `isfile`
- `joinpath`
- `Prescient`
- `files`

### 7.7.2 TestDoubleLoopIntegration.data_path

**Kind** method

**Defined in file** `tests/test_integration.py` at line 48

**Signature** `def data_path(self) -> Path:`

**Summary** No docstring.

**Calls**

- `as_file`
- `joinpath`
- `files`

### 7.7.3 TestDoubleLoopIntegration.test_data_path_available

**Kind** method

**Defined in file** `tests/test_integration.py` at line 58

**Signature** `def test_data_path_available(self, data_path: Path):`

**Summary** No docstring.

**Calls**

- `is_dir`

### 7.7.4 TestDoubleLoopIntegration.output_dir

**Kind** method

**Defined in file** `tests/test_integration.py` at line 62

**Signature** `def output_dir(self, tmp_path: Path) -> Path:`

**Summary** No docstring.

**Calls**

- `mkdir`

### 7.7.5 TestDoubleLoopIntegration.self_scheduler_output_dir

**Kind** method

**Defined in file** `tests/test_integration.py` at line 68

**Signature** `def self_scheduler_output_dir(self, tmp_path: Path) -> Path:`

**Summary** No docstring.

**Calls**

- `mkdir`

### 7.7.6 TestDoubleLoopIntegration.self_scheduler_plugin_path

**Kind** method

**Defined in file** `tests/test_integration.py` at line 74

**Signature** `def self_scheduler_plugin_path(self) -> Path:`

**Summary** No docstring.

**Calls**

- `as_file`
- `Path`
- `joinpath`
- `files`

### 7.7.7 TestDoubleLoopIntegration.test_self_scheduler_plugin_path_is_existing_file

**Kind** method

**Defined in file** `tests/test_integration.py` at line 83

**Signature** `def test_self_scheduler_plugin_path_is_existing_file(`

**Summary** No docstring.

**Calls**

- `is_file`

### 7.7.8 TestDoubleLoopIntegration.prescient_options

**Kind** method

**Defined in file** `tests/test_integration.py` at line 89

**Signature** `def prescient_options(self, data_path: Path) -> PrescientOptions:`

**Summary** No docstring.

**Calls**

- `str`

### 7.7.9 TestDoubleLoopIntegration.bidder_sim_options

**Kind** method

**Defined in file** `tests/test_integration.py` at line 119

**Signature** `def bidder_sim_options(`

**Summary** No docstring.

**Calls**

- `str`

### 7.7.10 TestDoubleLoopIntegration.self_scheduler_sim_options

**Kind** method

**Defined in file** `tests/test_integration.py` at line 134

**Signature** `def self_scheduler_sim_options(`

**Summary** No docstring.

**Calls**

- `str`

### 7.7.11 TestDoubleLoopIntegration.run_bidder_simulator

**Kind** method

**Defined in file** `tests/test_integration.py` at line 150

**Signature** `def run_bidder_simulator(self, bidder_sim_options: PrescientOptions) -> None:`

**Summary** No docstring.

**Calls**

- `importorskip`
- `simulate`
- `Prescient`

### 7.7.12 TestDoubleLoopIntegration.run_self_scheduler_simulator

**Kind** method

**Defined in file** `tests/test_integration.py` at line 159

**Signature** `def run_self_scheduler_simulator(`

**Summary** No docstring.

**Calls**

- `importorskip`
- `simulate`
- `Prescient`

### 7.7.13 TestDoubleLoopIntegration.simulation_results_dir

**Kind** method

**Defined in file** `tests/test_integration.py` at line 170

**Signature** `def simulation_results_dir(self, run_bidder_simulator, output_dir):`

**Summary** No docstring.

### 7.7.14 TestDoubleLoopIntegration.self_scheduler_simulation_results_dir

**Kind** method

**Defined in file** `tests/test_integration.py` at line 174

**Signature** `def self_scheduler_simulation_results_dir(`

**Summary** No docstring.

### 7.7.15 TestDoubleLoopIntegration.test_prescient_outputs_exist

**Kind** method

**Defined in file** `tests/test_integration.py` at line 180

**Signature** `def test_prescient_outputs_exist(`

**Summary** No docstring.

**Calls**

- `isdir`
- `join`
- `isfile`

## 7.8 test_model_data.py

**File path** `tests/test_model_data.py`

**Definitions Index**

- **function**: `generator_params` (line 23)
- **function**: `renewable_generator_params` (line 44)
- **function**: `generator_data_object` (line 56)
- **function**: `renewable_generator_data_object` (line 61)
- **function**: `test_create_thermal_model_data_object` (line 66)
- **function**: `test_create_renewable_model_data_object` (line 85)
- **function**: `test_create_model_data_with_non_real_numbers` (line 110)
- **function**: `test_create_model_data_with_negative_values` (line 129)
- **function**: `test_create_model_data_with_less_than_pmin_data` (line 140)
- **function**: `test_invalid_fixed_commitment_value` (line 151)
- **function**: `test_invalid_generator_name` (line 160)
- **function**: `test_invalid_bus_name` (line 167)
- **function**: `test_empty_bid_pairs` (line 177)
- **function**: `test_bid_missing_pmin` (line 186)
- **function**: `test_bid_missing_pmax` (line 197)
- **function**: `test_invalid_start_up_bid` (line 206)
- **function**: `test_invalid_pcost` (line 216)
- **function**: `test_model_data_iterator` (line 226)
- **function**: `test_invalid_initial_status` (line 265)
- **function**: `test_invalid_initial_p_output` (line 286)

**Called functions/methods in this file**

- line 57: `ThermalGeneratorModelData`
- line 62: `RenewableGeneratorModelData`
- line 71: `getattr`
- line 78: `zip`
- line 81: `assertStructuredAlmostEqual`
- line 112: `raises`
- line 97: `parametrize`
- line 249: `sort`

**Definition Notes**

### 7.8.1 generator_params

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 23

**Signature** `def generator_params():`

**Summary** No docstring.

### 7.8.2 renewable_generator_params

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 44

**Signature** `def renewable_generator_params():`

**Summary** No docstring.

### 7.8.3 generator_data_object

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 56

**Signature** `def generator_data_object(generator_params):`

**Summary** No docstring.

**Calls**

- `ThermalGeneratorModelData`

### 7.8.4 renewable_generator_data_object

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 61

**Signature** `def renewable_generator_data_object(renewable_generator_params):`

**Summary** No docstring.

**Calls**

- RenewableGeneratorModelData

### 7.8.5 test_create_thermal_model_data_object

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 66

**Signature** `def test_create_thermal_model_data_object(generator_params, generator_data_object):`

**Summary** No docstring.

**Calls**

- `zip`
- `getattr`
- `assertStructuredAlmostEqual`

### 7.8.6 test_create_renewable_model_data_object

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 85

**Signature** `def test_create_renewable_model_data_object(`

**Summary** No docstring.

### 7.8.7 test_create_model_data_with_non_real_numbers

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 110

**Signature** `def test_create_model_data_with_non_real_numbers(param_name, value, generator_params)`

**Summary** No docstring.

**Calls**

- `parametrize`
- `raises`
- ThermalGeneratorModelData

### 7.8.8 test_create_model_data_with_negative_values

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 129

**Signature** `def test_create_model_data_with_negative_values(param_name, value, generator_params):`

**Summary** No docstring.

**Calls**

- `parametrize`
- `raises`
- `ThermalGeneratorModelData`

### 7.8.9 test_create_model_data_with_less_than_pmin_data

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 140

**Signature** `def test_create_model_data_with_less_than_pmin_data(`

**Summary** No docstring.

**Calls**

- `parametrize`
- `raises`
- `ThermalGeneratorModelData`

### 7.8.10 test_invalid_fixed_commitment_value

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 151

**Signature** `def test_invalid_fixed_commitment_value(generator_params):`

**Summary** No docstring.

**Calls**

- `raises`
- `ThermalGeneratorModelData`

### 7.8.11 test_invalid_generator_name

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 160

**Signature** `def test_invalid_generator_name(generator_params):`

**Summary** No docstring.

**Calls**

- `raises`
- `ThermalGeneratorModelData`

### 7.8.12 test_invalid_bus_name

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 167

**Signature** `def test_invalid_bus_name(generator_params):`

**Summary** No docstring.

**Calls**

- `raises`
- `ThermalGeneratorModelData`

### 7.8.13 test_empty_bid_pairs

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 177

**Signature** `def test_empty_bid_pairs(param_name, generator_params):`

**Summary** No docstring.

**Calls**

- `parametrize`
- `raises`
- `ThermalGeneratorModelData`

### 7.8.14 test_bid_missing_pmin

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 186

**Signature** `def test_bid_missing_pmin(generator_params):`

**Summary** No docstring.

**Calls**

- `raises`
- `ThermalGeneratorModelData`

### 7.8.15 test_bid_missing_pmax

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 197

**Signature** `def test_bid_missing_pmax(generator_params):`

**Summary** No docstring.

**Calls**

- `raises`
- `ThermalGeneratorModelData`

### 7.8.16 test_invalid_start_up_bid

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 206

**Signature** `def test_invalid_start_up_bid(generator_params):`

**Summary** No docstring.

**Calls**

- `raises`
- `ThermalGeneratorModelData`

### 7.8.17 test_invalid_pcost

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 216

**Signature** `def test_invalid_pcost(generator_params):`

**Summary** No docstring.

**Calls**

- `raises`
- `ThermalGeneratorModelData`

### 7.8.18 test_model_data_iterator

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 226

**Signature** `def test_model_data_iterator(generator_data_object):`

**Summary** No docstring.

**Calls**

- `sort`
- `assertStructuredAlmostEqual`

### 7.8.19 test_invalid_initial_status

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 265

**Signature** `def test_invalid_initial_status(value, error, msg, generator_params):`

**Summary** No docstring.

**Calls**

- `parametrize`
- `raises`
- `ThermalGeneratorModelData`

### 7.8.20 test_invalid_initial_p_output

**Kind** function

**Defined in file** `tests/test_model_data.py` at line 286

**Signature** `def test_invalid_initial_p_output(`

**Summary** No docstring.

**Calls**

- `parametrize`
- `raises`
- `ThermalGeneratorModelData`

## 7.9 test_perfectforecaster.py

**File path** `tests/test_perfectforecaster.py`

**Definitions Index**

- **function**: `wind_df` (line 23)
- **function**: `base_perfectforecaster` (line 46)
- **function**: `test_create_perfectforecaster` (line 51)
- **function**: `test_create_perfectforecaster_with_ndarray_and_list` (line 58)
- **function**: `test_get_column_from_data` (line 70)
- **function**: `test_forecast_day_ahead_prices` (line 89)
- **function**: `test_forecast_real_time_prices` (line 108)
- **function**: `test_forecast_day_ahead_capacity_factor` (line 127)
- **function**: `test_forecast_real_time_capacity_factor` (line 145)

**Called functions/methods in this file**

- line 31: `Timestamp`
- line 32: `date_range`
- line 34: `DateOffset`
- line 37: `DataFrame`
- line 38: `list`

- line 38: `range`
- line 47: `PerfectForecaster`
- line 62: `raises`
- line 57: `parametrize`
- line 57: `array`
- line 79: `get_column_from_data`
- line 83: `assertStructuredAlmostEqual`
- line 84: `tolist`
- line 98: `forecast_day_ahead_prices`
- line 117: `forecast_real_time_prices`
- line 135: `forecast_day_ahead_capacity_factor`
- line 153: `forecast_real_time_capacity_factor`

**Definition Notes**

### 7.9.1 wind_df

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 23

**Signature** `def wind_df():`

**Summary** This is to define a dataframe fed to the PerfectForecaster with example DA/RT

**Calls**

- `Timestamp`
- `date_range`
- `DataFrame`
- `list`
- `DateOffset`
- `range`

### 7.9.2 base_perfectforecaster

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 46

**Signature** `def base_perfectforecaster(wind_df):`

**Summary** No docstring.

**Calls**

- `PerfectForecaster`

### 7.9.3 test_create_perfectforecaster

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 51

**Signature** `def test_create_perfectforecaster(wind_df):`

**Summary** No docstring.

**Calls**

- `PerfectForecaster`

### 7.9.4 test_create_perfectforecaster_with_ndarray_and_list

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 58

**Signature** `def test_create_perfectforecaster_with_ndarray_and_list(value):`

**Summary** This is to test when we creat the PerfectForecaster with array or list, there will be an error.

**Calls**

- `parametrize`
- `raises`
- `PerfectForecaster`
- `array`

### 7.9.5 test_get_column_from_data

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 70

**Signature** `def test_get_column_from_data(base_perfectforecaster):`

**Summary** This is to test the if the dataframe can be read correctly.

**Calls**

- `get_column_from_data`
- `assertStructuredAlmostEqual`
- `range`
- `tolist`

### 7.9.6 test_forecast_day_ahead_prices

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 89

**Signature** `def test_forecast_day_ahead_prices(base_perfectforecaster):`

**Summary** This is to test if the forecast_day_ahead_price function works correctly.

**Calls**

- `forecast_day_ahead_prices`
- `assertStructuredAlmostEqual`
- `range`
- `tolist`

### 7.9.7 test_forecast_real_time_prices

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 108

**Signature** `def test_forecast_real_time_prices(base_perfectforecaster):`

**Summary** This is to test if the forecast_real_time_price function works correctly.

**Calls**

- `forecast_real_time_prices`
- `assertStructuredAlmostEqual`
- `range`
- `tolist`

### 7.9.8 test_forecast_day_ahead_capacity_factor

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 127

**Signature** `def test_forecast_day_ahead_capacity_factor(base_perfectforecaster):`

**Summary** This is to test if the forecast_day_ahead_capacity_factor function works correctly.

**Calls**

- `forecast_day_ahead_capacity_factor`
- `assertStructuredAlmostEqual`
- `range`
- `tolist`

### 7.9.9 test_forecast_real_time_capacity_factor

**Kind** function

**Defined in file** `tests/test_perfectforecaster.py` at line 145

**Signature** `def test_forecast_real_time_capacity_factor(base_perfectforecaster):`

**Summary** This is to test if the forecast_real_time_capacity_factor function works correctly.

**Calls**

- `forecast_real_time_capacity_factor`
- `assertStructuredAlmostEqual`
- `range`
- `tolist`

## 7.10 test_tracker.py

**File path** `tests/test_tracker.py`

**Definitions Index**

- **class**: `MissingModel` (line 19)
- **method**: `MissingModel.__init__` (line 36)
- **function**: `test_model_object_missing_methods` (line 55)
- **function**: `test_model_object_missing_attr` (line 82)
- **function**: `test_n_tracking_hour_checker` (line 101)
- **function**: `test_solver_checker` (line 127)
- **function**: `tracker_object` (line 144)
- **function**: `test_track_market_dispatch` (line 163)
- **function**: `test_track_deviation_penalty` (line 183)

**Called functions/methods in this file**

- line 44: `setattr`
- line 57: `SolverFactory`
- line 71: `MissingModel`
- line 72: `raises`
- line 72: `format`
- line 73: `Tracker`
- line 103: `ExampleModel`
- line 165: `track_market_dispatch`
- line 169: `zip`
- line 169: `range`
- line 171: `approx`
- line 171: `value`
- line 177: `get_last_delivered_power`
- line 160: `skipif`
- line 161: `available`

**Definition Notes**

### 7.10.1 MissingModel

**Kind** class

**Defined in file** `tests/test_tracker.py` at line 19

**Signature** `class MissingModel:`

**Summary** A class for testing missing methods and attributes.

**Calls**

- `setattr`

### 7.10.2 MissingModel.__init__

**Kind** method

**Defined in file** `tests/test_tracker.py` at line 36

**Signature** `def __init__(self, missing_method=None, missing_attr=None):`

**Summary** Constructs a model class without the specified missing methods and/or

**Calls**

- `setattr`

### 7.10.3 test_model_object_missing_methods

**Kind** function

**Defined in file** `tests/test_tracker.py` at line 55

**Signature** `def test_model_object_missing_methods():`

**Summary** No docstring.

**Calls**

- `SolverFactory`
- `MissingModel`
- `raises`
- `Tracker`
- `format`

### 7.10.4 test_model_object_missing_attr

**Kind** function

**Defined in file** `tests/test_tracker.py` at line 82

**Signature** `def test_model_object_missing_attr():`

**Summary** No docstring.

**Calls**

- `SolverFactory`
- `MissingModel`
- `raises`
- `Tracker`
- `format`

### 7.10.5 test_n_tracking_hour_checker

**Kind** function

**Defined in file** `tests/test_tracker.py` at line 101

**Signature** `def test_n_tracking_hour_checker():`

**Summary** No docstring.

**Calls**

- `SolverFactory`
- `ExampleModel`
- `raises`
- `Tracker`

### 7.10.6 test_solver_checker

**Kind** function

**Defined in file** `tests/test_tracker.py` at line 127

**Signature** `def test_solver_checker():`

**Summary** No docstring.

**Calls**

- ExampleModel
- raises
- Tracker

### 7.10.7 tracker_object

**Kind** function

**Defined in file** tests/test_tracker.py at line 144

**Signature** def tracker_object():

**Summary** No docstring.

**Calls**

- SolverFactory
- ExampleModel
- Tracker

### 7.10.8 test_track_market_dispatch

**Kind** function

**Defined in file** tests/test_tracker.py at line 163

**Signature** def test_track_market_dispatch(tracker_object):

**Summary** No docstring.

**Calls**

- skipif
- track_market_dispatch
- zip
- range
- approx
- available
- get_last_delivered_power
- value
- SolverFactory

### 7.10.9 test_track_deviation_penalty

**Kind** function

**Defined in file** `tests/test_tracker.py` at line 183

**Signature** `def test_track_deviation_penalty(tracker_object):`

**Summary** No docstring.

**Calls**

- `range`
- `approx`
- `value`

## 7.11 util.py

**File path** `tests/util.py`

**Definitions Index**

- **class**: `ExampleModel` (line 25)
- **method**: `ExampleModel.__init__` (line 30)
- **method**: `ExampleModel.model_data` (line 58)
- **method**: `ExampleModel.populate_model` (line 61)
- **method**: `ExampleModel.populate_model.prod_cost_fun` (line 93)
- **method**: `ExampleModel.populate_model.tot_cost_fun` (line 99)
- **method**: `ExampleModel._update_power` (line 107)
- **method**: `ExampleModel.update_model` (line 124)
- **method**: `ExampleModel.get_implemented_profile` (line 142)
- **method**: `ExampleModel.get_last_delivered_power` (line 165)
- **method**: `ExampleModel.record_results` (line 178)
- **method**: `ExampleModel.write_results` (line 237)
- **method**: `ExampleModel.power_output` (line 251)
- **method**: `ExampleModel.total_cost` (line 255)
- **class**: `ExampleForecaster` (line 259)

- **method**: `ExampleForecaster.__init__` (line 264)
- **method**: `ExampleForecaster.forecast_day_ahead_and_real_time_prices` (line 267)
- **method**: `ExampleForecaster.forecast_real_time_prices` (line 279)
- **method**: `ExampleForecaster.forecast_day_ahead_prices` (line 282)
- **method**: `ExampleForecaster._forecast` (line 285)
- **method**: `ExampleForecaster.fetch_hourly_stats_from_prescient` (line 288)
- **method**: `ExampleForecaster.fetch_day_ahead_stats_from_prescient` (line 291)
- **function**: `make_testing_forecaster` (line 337)
- **function**: `make_testing_tracker` (line 351)
- **function**: `make_testing_bidder` (line 373)
- **function**: `make_testing_selfscheduler` (line 399)

**Called functions/methods in this file**

- line 75: `Set`
- line 75: `range`
- line 78: `Param`
- line 90: `Var`
- line 96: `Expression`
- line 120: `round`
- line 137: `_update_power`
- line 158: `deque`
- line 159: `value`
- line 203: `int`
- line 206: `float`
- line 218: `abs`
- line 228: `from_dict`
- line 229: `append`
- line 233: `concat`
- line 248: `to_csv`
- line 270: `forecast_real_time_prices`

- line 273: `forecast_day_ahead_prices`
- line 280: `_forecast`
- line 318: `ThermalGeneratorModelData`
- line 328: `RenewableGeneratorModelData`
- line 334: `SolverFactory`
- line 348: `ExampleForecaster`
- line 362: `ExampleModel`
- line 363: `Tracker`
- line 384: `make_testing_forecaster`
- line 387: `Bidder`
- line 413: `SelfScheduler`

**Definition Notes**

### 7.11.1 ExampleModel

**Kind** class

**Defined in file** `tests/util.py` at line 25

**Signature** `class ExampleModel:`

**Summary** Simple model object for testing.

**Calls**

- `Set`
- `Param`
- `Var`
- `Expression`
- `round`
- `_update_power`
- `deque`
- `value`
- `append`
- `to_csv`

- `int`
- `float`
- `from_dict`
- `concat`
- `range`
- `abs`

### 7.11.2 ExampleModel.__init__

**Kind** method

**Defined in file** `tests/util.py` at line 30

**Signature** `def __init__(self, model_data):`

**Summary** Initializes the class object by building the thermal generator model.

### 7.11.3 ExampleModel.model_data

**Kind** method

**Defined in file** `tests/util.py` at line 58

**Signature** `def model_data(self):`

**Summary** No docstring.

### 7.11.4 ExampleModel.populate_model

**Kind** method

**Defined in file** `tests/util.py` at line 61

**Signature** `def populate_model(self, b, horizon):`

**Summary** This function builds the model for a thermal generator.

**Calls**

- `Set`
- `Param`
- `Var`
- `Expression`
- `range`

### 7.11.5 ExampleModel.populate_model.prod_cost_fun

**Kind** method

**Defined in file** `tests/util.py` at line 93

**Signature** `def prod_cost_fun(b, h):`

**Summary** No docstring.

### 7.11.6 ExampleModel.populate_model.tot_cost_fun

**Kind** method

**Defined in file** `tests/util.py` at line 99

**Signature** `def tot_cost_fun(b, h):`

**Summary** No docstring.

### 7.11.7 ExampleModel._update_power

**Kind** method

**Defined in file** `tests/util.py` at line 107

**Signature** `def _update_power(b, implemented_power_output):`

**Summary** This method updates the parameters in the ramping constraints based on

**Calls**

- `round`

### 7.11.8 ExampleModel.update_model

**Kind** method

**Defined in file** `tests/util.py` at line 124

**Signature** `def update_model(self, b, implemented_power_output):`

**Summary** This method updates the parameters in the model based on

**Calls**

- `_update_power`

### 7.11.9 ExampleModel.get_implemented_profile

**Kind** method

**Defined in file** `tests/util.py` at line 142

**Signature** `def get_implemented_profile(b, last_implemented_time_step):`

**Summary** This method gets the implemented variable profiles in the last optimization

**Calls**

- `deque`
- `value`
- `range`

### 7.11.10 ExampleModel.get_last_delivered_power

**Kind** method

**Defined in file** `tests/util.py` at line 165

**Signature** `def get_last_delivered_power(b, last_implemented_time_step):`

**Summary** Returns the last delivered power output.

**Calls**

- `value`

### 7.11.11 ExampleModel.record_results

**Kind** method

**Defined in file** `tests/util.py` at line 178

**Signature** `def record_results(self, b, date=None, hour=None, **kwargs):`

**Summary** Record the operations stats for the model.

**Calls**

- `append`
- `int`
- `float`
- `from_dict`
- `concat`
- `round`
- `value`
- `abs`

### 7.11.12 ExampleModel.write_results

**Kind** method

**Defined in file** `tests/util.py` at line 237

**Signature** `def write_results(self, path):`

**Summary** This methods writes the saved operation stats into an csv file.

**Calls**

- `to_csv`
- `concat`

### 7.11.13 ExampleModel.power_output

**Kind** method

**Defined in file** `tests/util.py` at line 251

**Signature** `def power_output(self):`

**Summary** No docstring.

### 7.11.14 ExampleModel.total_cost

**Kind** method

**Defined in file** `tests/util.py` at line 255

**Signature** `def total_cost(self):`

**Summary** No docstring.

### 7.11.15 ExampleForecaster

**Kind** class

**Defined in file** `tests/util.py` at line 259

**Signature** `class ExampleForecaster(AbstractPrescientPriceForecaster):`

**Summary** A fake forecaster class for testing.

**Calls**

- `forecast_real_time_prices`
- `forecast_day_ahead_prices`
- `_forecast`
- `range`

### 7.11.16 ExampleForecaster.__init__

**Kind** method

**Defined in file** `tests/util.py` at line 264

**Signature** `def __init__(self, prediction):`

**Summary** No docstring.

### 7.11.17 ExampleForecaster.forecast_day_ahead_and_real_time_prices

**Kind** method

**Defined in file** `tests/util.py` at line 267

**Signature** `def forecast_day_ahead_and_real_time_prices(`

**Summary** No docstring.

**Calls**

- `forecast_real_time_prices`
- `forecast_day_ahead_prices`

### 7.11.18 ExampleForecaster.forecast_real_time_prices

**Kind** method

**Defined in file** `tests/util.py` at line 279

**Signature** `def forecast_real_time_prices(self, date, hour, bus, horizon, n_samples):`

**Summary** No docstring.

**Calls**

- `_forecast`

### 7.11.19 ExampleForecaster.forecast_day_ahead_prices

**Kind** method

**Defined in file** `tests/util.py` at line 282

**Signature** `def forecast_day_ahead_prices(self, date, hour, bus, horizon, n_samples):`

**Summary** No docstring.

**Calls**

- `_forecast`

### 7.11.20 ExampleForecaster._forecast

**Kind** method

**Defined in file** `tests/util.py` at line 285

**Signature** `def _forecast(self, horizon, n_samples, prediction):`

**Summary** No docstring.

**Calls**

- `range`

### 7.11.21 ExampleForecaster.fetch_hourly_stats_from_prescient

**Kind** method

**Defined in file** `tests/util.py` at line 288

**Signature** `def fetch_hourly_stats_from_prescient(self, prescient_hourly_stats):`

**Summary** No docstring.

### 7.11.22 ExampleForecaster.fetch_day_ahead_stats_from_prescient

**Kind** method

**Defined in file** `tests/util.py` at line 291

**Signature** `def fetch_day_ahead_stats_from_prescient(self, uc_date, uc_hour, day_ahead_result):`

**Summary** No docstring.

### 7.11.23 make_testing_forecaster

**Kind** function

**Defined in file** `tests/util.py` at line 337

**Signature** `def make_testing_forecaster():`

**Summary** Create a forecaster for testing.

**Calls**

- `ExampleForecaster`

### 7.11.24 make_testing_tracker

**Kind** function

**Defined in file** `tests/util.py` at line 351

**Signature** `def make_testing_tracker():`

**Summary** Create a tracker for testing.

**Calls**

- `ExampleModel`
- `Tracker`

### 7.11.25 make_testing_bidder

**Kind** function

**Defined in file** `tests/util.py` at line 373

**Signature** `def make_testing_bidder():`

**Summary** Create a bidder for testing.

**Calls**

- `make_testing_forecaster`
- `ExampleModel`
- `Bidder`

### 7.11.26 make_testing_selfscheduler

**Kind** function

**Defined in file** `tests/util.py` at line 399

**Signature** `def make_testing_selfscheduler():`

**Summary** Create a self-scheduler for testing.

**Calls**

- `make_testing_forecaster`
- `ExampleModel`
- `SelfScheduler`